

When Institutions Interact: How the Effects of Unemployment Insurance are Shaped by Retirement Policies*

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Abstract

We show that the non-employment effects of unemployment insurance (UI) for older workers depend critically on retirement policy. Using German data, we document large bunching in UI inflows at the age that allows workers to claim their pension following UI expiration. Inflows respond strongly to several UI and pension reforms. We probe the implications of these behavioral responses using a dynamic model and find that Germany's UI and retirement policy changes had substantial effects on the unemployment rate of older workers. Furthermore, we calculate large fiscal externalities from extending UI for older workers, especially under generous retirement policies.

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Understanding how unemployment insurance (UI) generosity affects non-employment duration is central to UI design. A large literature, spanning a range of contexts, estimates the elasticity of non-employment with respect to UI benefit levels or potential benefit durations for prime-age workers (Schmieder and von Wachter, 2016). However, it is difficult to know how well these estimates generalize to older workers approaching retirement age. Even when estimates of UI expansions at older ages are available, their effects may differ depending on the surrounding retirement-policy environment. Although it is well understood that reduced-form estimates are inherently context-specific, less is known about whether interactions between UI and pension policies substantially shape the effects of UI extensions for older workers and their associated policy takeaways.

Germany offers an intriguing context to study how UI extensions interact with pension policies. During the 1990s, the unemployment rate of German men aged 55–59 rose sharply, diverging from that of slightly younger workers (Figure 1 (a)).¹ UI payments to this age group also ballooned (Figure 1 (b)). In subsequent years, just as remarkably, the unemployment rate of older workers declined rapidly and converged with that of younger workers. These shifts coincided with major policy changes. Germany first substantially extended UI potential benefit durations (PBD), later partially reversed those extensions, and, beginning in the late 1990s, implemented pension reforms that gradually reduced the attractiveness of early retirement. To what extent did these UI and retirement reforms shape aggregate unemployment trends of older workers? Would we reach similar conclusions about the effects of these policies based on existing evidence focused on prime-age workers? This paper addresses these questions using a combination of reduced-form evidence and a structural model estimated on German social security data between 1975 and 2017.

We begin our empirical analysis by documenting substantial bunching of UI inflows at the age that allows workers to claim a pension immediately after UI exhaustion (bridge to retirement age). For example, birth cohorts nearing retirement age in the early 1980s faced a maximum UI PBD of 12 months and could retire as early as age 60 following a UI spell.² Accordingly, we see large and precise bunching of UI inflows for these birth cohorts at exactly age 59.³ Most of these workers remain on UI until exhaustion.

Importantly, these UI inflows respond to a series of UI extensions and pension rule changes. Our study period encompasses numerous reforms to both Germany’s UI and retirement system that altered the payoffs to entering UI and the search incentives of the unemployed, first through a series of PBD extensions and later

¹This paper focuses on men, although we also estimate and present all key results for women in a self-contained [Online Companion Appendix](#). We focus on men because retirement rules and bridge ages differ by gender due to the existence of a specific women’s pension (Section I). Over our sample period, the more generous terms of the women’s pension allowed women, at different points in time, to enter early retirement without an intervening UI spell, retire at younger ages, or face lower financial penalties for early retirement relative to men. Consequently, the UI system played a less important role as a pathway to early effective retirement for women. Nonetheless, the results are qualitatively similar for women.

²Entering UI voluntarily in Germany is also feasible and at most lightly penalized. A worker may be sanctioned if they quit a job voluntarily. These sanctions take the form of losing the first few weeks of benefits, but these sanctions do not always seem to be applied, could be offset by severance payments from firms, and are small relative to the length of maximum PBD.

³UI replacement rates are relatively generous in our context, with at most limited penalties for voluntary quits, and no job search obligations for workers above age 58. Firms also contributed to generating worker inflow responses to UI extensions by negotiating collective labor agreements (CLAs) and ‘social plans’ during downturns with their workforce that often took UI and retirement-based incentives into account (Trampusch, 2005).

through retirement rule changes such as implementing a financial penalty on pension benefits for retiring early. As maximum PBD was extended from 12 to 24 and then 32 months, the large UI inflow bunching mass moves from age 59 to age 58, and then to 57 and 4 months. For the 1935 birth cohort, which faced the most generous UI and retirement rules in worsening macroeconomic conditions, almost 12% of the sampled cohort bunched into UI at the bridge-to-retirement age, and almost 25% of the cohort was receiving UI benefits at age 59. Later, as penalties for retiring early were gradually introduced but PBD remained fixed, UI inflow spikes diminished. And eventually, as the earliest possible retirement age also increased, the age at which UI inflow bunching occurred increased as well. Thus, UI inflows at the bridge-to-retirement age respond to UI policy changes, but also to retirement policy changes when holding UI policy fixed.

These large, adaptable UI inflow responses suggest that Germany’s historical UI extensions, introduced during a period of relatively generous retirement rules, may have substantially affected older workers’ aggregate unemployment and generated larger fiscal externalities than one might expect from prior estimates for prime-age workers.⁴ However, quantifying these aggregate and fiscal effects requires additional modeling assumptions. A direct extrapolation from the bunching evidence, for example, would require taking a stance on how the inflow mass shifts under non-marginal PBD changes, even though its size and shape vary with the distance to retirement, macroeconomic conditions, and early-retirement penalties. It would also need to account jointly for effects on UI entry and non-employment duration conditional on entry.⁵

We therefore specify a dynamic life-cycle model in which workers transition among employment, unemployment, and an absorbing out-of-the-labor-force state. Employment ends following a sufficiently adverse productivity shock or when UI and retirement institutions raise the value of non-employment above the value of continued work. As workers approach retirement, the value of non-employment rises. Heterogeneity in the disutility of work generates a smooth distribution of exit ages, while the UI bridge creates a kink that produces bunching at the bridge age. Productivity shocks spread this bunching across nearby ages and generate exits at younger ages. Endogenous search allows workers to return to employment, so the model captures both UI entry responses and changes in non-employment duration conditional on entry.

We estimate the model by matching employment-to-unemployment transitions, non-employment durations, and a regression discontinuity (RD) estimate of the intensive-margin response to PBD for three cohorts facing different UI and retirement institutions. We then simulate all birth cohorts from 1924 to 1963 to evaluate out-of-sample fit and construct age-specific unemployment rates in calendar time.

Our fairly standard model, built on clear economic incentives and their interactions, is able to capture UI entries and non-employment duration trends in the empirical data well. Our simulated unemployment rates mirror both their empirical counterparts from the social security data and the OECD unemployment rates. Furthermore, simple model-based comparative statistics are consistent with what we would expect from the reduced-form data. For example, extending the maximum PBD by one year increases the simulated unem-

⁴For example, the regression-discontinuity evidence from [Schmieder et al. \(2012\)](#) implies that non-employment duration rises by about three days for each additional month of PBD.

⁵Simpler bunching approaches are also difficult to reconcile with voluntary and involuntary unemployment inflows that vary over the business cycle, and their predictions can be sensitive to restrictions on the counterfactual distribution ([Saez, 2010](#); [Kleven, 2016](#); [Blomquist et al., 2021](#)).

ployment rates of slightly younger (aged 52-55) versus slightly older workers (aged 56-59) very differently. Under the generous retirement rules prevailing in the early 1990s, it increases the simulated unemployment rates of those aged 52-55 by 0.6pp. There is close to no effect on these workers' inflows into UI. In stark contrast, this same change increases the unemployment rate of those aged 56-59 by a comparatively high 2.5pp, in large part due to UI inflow effects around bridge-to-retirement ages.

With our model estimated, we return to the motivating questions surrounding Germany's older workers' unique unemployment trends. We ask to what extent Germany's PBD extensions explain the large 10pp increase in the unemployment rate of workers aged 56-59 between the early 1980s and the mid 1990s. When we simulate an environment in which maximum PBD had never been extended past 12 months, we find that the rise in unemployment rates of workers aged 56-59 would have been 58% lower. We also show that changes in UI and retirement policies played a key role in explaining the improved labor market performance of older workers in the late 1990s and 2000s.

These large effects of PBD extensions on older workers are partially a consequence of the relatively generous retirement institutions of the time. These PBD effects would not have been nearly as large under different retirement rules: when simulating an environment with less generous retirement policies, our model predicts significantly lower effects of the PBD extensions on unemployment rates. We reinforce this point by showing how the effect of the *same UI extension* varies for same-aged workers across different (historically observed) retirement regimes (but with all else held constant). Model simulations show that the non-employment effects of an identical one-year extension vary considerably between actual, current institutions and a setting that re-institutes Germany's historically more generous retirement rules.

We conclude by showing that the behavioral costs of UI extensions for older workers can, under the right conditions, be sufficiently different from what has been previously estimated for younger workers to alter the welfare assessment of UI extensions. Specifically, building on [Hendren and Sprung-Keyser \(2020\)](#)'s marginal value of public funds (MVPF) framework, we calculate the fiscal externality of an extra dollar of spending on UI PBD extensions. Using intensive-margin estimates for prime-age workers in Germany, [Schmieder et al. \(2016\)](#) show that the net cost of an extra dollar towards PBD extensions is around 1.41. Using a willingness to pay for an extra dollar of spending on PBD of 1.36 ([Le Barbanchon et al., 2024](#)), this yields an MVPF of 0.97.⁶ Using our model, we calculate the fiscal externality associated with a PBD extension for older workers from each separate birth cohort. Using the 1935 cohort as an example, a cohort that faced generous retirement and PBD rules and a poor labor market, we find that the net cost of an extra dollar of spending on PBD extensions for workers aged 50 and up is \$5.62 (due to a large \$4.62 fiscal externality). If we use the same willingness to pay for a \$1 of PBD extensions for older workers as for younger workers (1.36), this implies an MVPF of 0.24 – considerably lower than 0.97. While one may still choose to nevertheless expand PBDs for older workers, it is clear that the inputs into the welfare assessment

⁶The willingness to pay for an extra dollar of spending on UI PBD, transferred to UI exhaustees, is given by the marginal rate of substitution between the unemployed and employed states, which depends on the difference in consumption across these states. Following [Le Barbanchon et al. \(2024\)](#) and the limited available evidence, we set $WTP^{PBD} = 1.36$, based on a 9.1% average consumption drop for the long-term unemployed from [Kolsrud et al. \(2018\)](#).

for such a policy can differ meaningfully from these better-established baselines.

Related Literature. Much of the empirical UI literature has focused on intensive-margin (i.e., conditional on entry) effects of UI extensions for prime-age workers (Schmieder et al., 2016). This is evident in the many regression discontinuity papers that show no inflow bunching at policy thresholds as a standard RD check (e.g., Card et al., 2007; Lalive, 2007). More recently, however, there has been added emphasis on UI-induced inflows into non-employment with a growing number of papers documenting the importance of UI policies on job separation and UI inflow responses (Anderson and Meyer, 1997; Green and Riddell, 1997; Albanese et al., 2020; Leung and O’leary, 2020; Van Doornik et al., 2023; Khoury, 2023; Hartung et al., 2025; Jäger et al., 2023; Jessen et al., 2025).⁷ Our work reinforces the importance of considering the UI inflow margin, which may be especially relevant for workers nearing retirement.

Several papers have also directly estimated UI effects for older workers across different countries (Winter-Ebmer, 2003; Kyrrä and Wilke, 2007; Kyrrä and Ollikainen, 2008; Lalive, 2008; Tuit and van Ours, 2010; Baguelin and Remillon, 2014; Dlugosz et al., 2014; Inderbitzin et al., 2016; Kyrrä and Pesola, 2020).⁸ Many of these papers either directly or indirectly emphasize the importance of early retirement incentives, and several have found relatively large UI effects for older workers. For example, Inderbitzin et al. (2016) consider interactions between UI, retirement and disability insurance, and apply the Baily-Chetty framework (Chetty, 2008) to Austria, finding that UI for older workers was too generous. A handful of these papers also provide direct evidence of UI inflow bunching (Kyrrä and Wilke, 2007; Tuit and van Ours, 2010; Baguelin and Remillon, 2014; Kyrrä and Pesola, 2020). Our reduced-form evidence complements and expands on this prior work. Relative to these papers, which typically study a single PBD change in isolation, we document decades of inflow responses and show how UI inflow bunching changes following both multiple PBD reforms and, importantly, also pension reforms that made bridging into early retirement via UI less attractive by introducing a financial penalty for early pension claiming. The scope of our work illustrates how the effects of UI extensions can vary as pension rules change, even within the same broader context: at their most generous, UI inflow effects at the bridge-to-retirement ages were comparable in magnitude to direct bunching at the statutory retirement ages (Seibold, 2021), emphasizing how common the UI pathway to retirement was. However, once retirement ages increase and penalties are introduced, this pathway, while still relevant, became less important.

Crucially, we also add to prior work by using our model to speak directly to how UI and retirement rules interact to impact market-level aggregates (e.g. old age unemployment) and alter policy takeaways.⁹

⁷Anderson and Meyer (1997); Green and Riddell (1997) show early empirical evidence of UI effects on separations linked to firms’ experience requirement. Albanese et al. (2020); Leung and O’leary (2020); Van Doornik et al. (2023) use quasi-experimental evidence to estimate the impact of UI eligibility on separation. More recent literature, such as Khoury (2023); Hartung et al. (2025); Jäger et al. (2023); Jessen et al. (2025) show that among prime-age workers (up to 50 years old), UI policies (such as eligibility, replacement rate and PBD) affect UI inflows, formal employment probability and job separations.

⁸There is also related work for Germany, including papers that study at least one of the reforms covered in our long time horizon (see, for example, among others, Fitzenberger and Wilke (2010); Dlugosz et al. (2014); Riphahn and Schrader (2023)).

⁹Our focus on the challenging question of how nation-wide policy changes affect market aggregates is shared with Autor and Dugan (2003), who are interested in how programmatic changes to disability insurance affected market-level labor force participation of low-skilled workers.

By quantifying how UI and retirement policies affected Germany’s unemployment trends, we relate to a literature focused on understanding the drivers of Germany’s stark labor market improvements since the mid to late 90s. Germany’s labor market ‘miracle’ has been the subject of many studies with authors highlighting factors ranging from the Hartz reforms to Germany’s governance structure (e.g. [Dustmann et al., 2014](#); [Hochmuth et al., 2021](#); [Hartung et al., 2025](#)). While these studies focus on workers of all ages, there is perhaps no more striking example of both Germany’s initial sluggishness and subsequent improvements than the massive rise and later decline in the unemployment rates among workers in their late 50s. Our model simulations provide novel evidence that changes in UI and retirement policies jointly played a substantial role in driving the unemployment rate trends of workers in their late 50s. By using our model to explore the fiscal externalities of various UI policies, we join studies like [Inderbitzin et al. \(2016\)](#) in highlighting that the welfare implications of UI extensions for older workers can look less attractive than at younger ages. We further emphasize that when considering the welfare implications of UI policy for older workers it is critical to consider interactions between UI and retirement policies: the effects and policy implications of any UI extension can change within a country over time as retirement policies adapt.

I Institutional Background and Data

I.A Unemployment Insurance

The German unemployment insurance system provides income replacement to eligible workers who lose their jobs. Before 1985, eligible workers were entitled to at most 12 months of benefits. Net replacement rates (i.e., benefits divided by post-tax earnings) for UI are 67-68% for an individual with children and 60-63% for an individual without children and remained relatively stable over our study period (1980–present). Beginning in 1985, numerous reforms changed the maximum UI potential benefit duration (PBD) in a manner that tied the maximum PBD to recipients’ exact age at the beginning of their UI spell.¹⁰

Reforms in 1985 and 1987 increased maximum PBDs for workers aged 42 and older. The most generous PBD, up to 32 months, became available to workers aged 54 and up following the 1987 reform. Reforms in 1999 and 2006 gradually decreased the generosity of the system. In 1999, age thresholds were increased, and then, beginning in 2006, maximum PBD was reduced from 32 to 18 months for workers above age 55, while everyone else could only receive 12 months. There was a modest reversal of this trend in 2008 when PBD for workers above age 50 was extended again to between 15 and 24 months (depending on age). Figure [H.1](#) plots the maximum PBD by age for older workers in each different institutional regime.¹¹

Individuals who exhausted UI benefits before 2005 and whose net liquid wealth fell below a certain threshold were eligible for unemployment assistance (UA). In principle, UA replacement rates were between 50% and 58% of net wages (in the presence of dependent children) but lower in practice due to deductions

¹⁰See [Hunt \(1995\)](#); [Lange \(2003\)](#); [Fitzenberger and Wilke \(2010\)](#) for an analysis and discussion of these reforms.

¹¹We omit the short 1985 regime in the interest of brevity and because it appears that some individuals who entered UI in 1985 retroactively benefited from the UI extensions in later years. We only plot changes in maximum PBD from age 48 to 62 in Figure [H.1](#) to focus on the changes in PBD at older ages. Table [I.1](#) provides the full details about each reform.

like spousal income (see [Schmieder et al. \(2012\)](#)). From 2005 on, UA was replaced by unemployment insurance benefits 2 (UIB II), an entirely means-tested program. Both UA and UIB II are unlimited in duration but they are, especially due to the means-testing, a very imperfect substitute for UI for older workers.

I.B Pension System and Early Retirement Via Unemployment

Germany has a pay-as-you-go public pension system. Participation is mandatory, except for civil servants and the self-employed. Pension benefits depend on workers' earnings, years of contributions, an adjustment factor, and the type of pension claimed. In 2017, pension benefits averaged approximately 50% of post-tax earnings in the year prior to retirement ([Deutsche Rentenversicherung \(2017\)](#)).

For most of our study period, the statutory retirement age (SRA) for a regular old-age pension remained at 65, with the only prerequisite being 5 contribution years. Beginning with the 1947 birth cohort in 2012, the statutory retirement age was gradually raised, reaching age 67 for cohorts born 1964 or later. Early retirement was possible under several alternate pathways, each with its own eligibility conditions: a normal retirement age (NRA), the age at which unpenalized pension payments can begin, and an early retirement age (ERA), the earliest age at which pension payments can begin.¹² Most relevantly, the pension due to the unemployment pathway (UI pathway) allowed for retirement after an unemployment spell.

The UI pathway provided eligible workers with an option to retire early at the age of 60.¹³ The eligibility requirements for this pathway were: 1) at least 15 years of contributions, at least 8 of which must have occurred in the past 10 years, and 2) being unemployed for at least one year after the age of 58 and a half. The generosity of UI benefits, combined with lenient job search requirements for older workers and the ability to retire earlier than other pathways, made the UI pathway attractive.¹⁴

This system incentivizes workers considering early retirement to time their entry into UI around the age that allows them to transition directly from UI to pensions, without any uncovered period. Put differently, the possibility of using UI as a bridge-to-retirement introduces a kink in a lifetime budget constraint relating lifetime income to the year of exit into UI. This kink occurs at the bridge-to-retirement age: $ERA - P$, with P being the maximum PBD. We show that UI entries react to the location and size of the kink in [Section II](#).

The NRA and ERA via the UI pathway remained at 60 until a 1992 reform. Cohorts born between January 1937 and December 1941 saw their NRA increase in steps by birth month from 60 to 65. While the ERA remained fixed at 60 meaning they could continue to retire at 60 via the UI pathway, they now

¹²The full list of alternative pathways to retirement can be found in [Table I.2](#) with associated discussion in [Appendix D.1](#). These pathways are old-age pensions for long-term insured (which required 35 years of contributions, had an ERA of 63 throughout our study period) and especially long-term insured, old-age pensions for women, old-age pensions due to unemployment (and, later, part-time work), and old-age pensions for severely disabled persons ([Boersch-Supan and Wilke, 2006](#)). We note that while early retirement due to disability is quantitatively important, [Riphahn \(1997\)](#) argues that in practice this is not a close substitute to retirement via unemployment and that retirement due to disability is usually associated with a health shock.

¹³For our first three focal cohorts (1924, 1929, and 1935), the unpenalized NRA and ERA via the UI pathway were age 60. Persons satisfying the requirements could retire at 60 with no penalty, missing out only on the marginal benefit gains from a few additional years of pension contributions. For later cohorts, the NRA and ERA increase.

¹⁴After the late 1980s, unemployed individuals aged 58 and older were exempt from actively looking for a job or other obligations. This so-called "58er-Regelung" was formally introduced at the end of 1985 and remained in place until the end of 2007.

faced an actuarial adjustment in the form of a 0.3% permanent pension reduction per month they retired in advance of their NRA. In Section II, we show that UI entries at the ERA decrease as these penalties phase in. Cohorts born after January 1946 could no longer retire at an ERA of 60, even with a penalty. They saw their ERA increase in steps by birth month from 60 to 63, ending with cohorts born in December 1948. The ERA remained at age 63 for cohorts born between 1949 and 1951 until the entire UI pathway was eliminated for cohorts born on or after January 1st, 1952.

During our time period, firms and work councils also played a role in workers' early exits from the labor force. Facing rising unemployment paired with employment protection laws and powerful unions, firms sought to downsize employment through voluntary means by negotiating collective labor agreements (CLAs) and 'social plans' with their workforce. These plans were often shaped with the unemployment system in mind and often took the form of offering severance packages to older workers reaching a specific age to voluntarily quit the firm. Writing about a time when workers were eligible for a year of unemployment benefits prior to retiring at 60, [Trampusch \(2005\)](#) states, "employees agreed to voluntary redundancy (that is they agreed to become unemployed at age 59) and began to draw unemployment pension after the lapse of unemployment benefits [...] work councils were more than happy to facilitate the exit of older workers under the generous terms offered by the social security system. In fact, they often found themselves under considerable pressure from older workers who wanted to retire under the existing provisions." Appendix D.3 contains additional details on the role of firms, works councils, and CLAs.

I.C Data

We use German Social Security data – the Integrated Employment Biographies (IEB) – from the Institute for Employment Research. This data provides detailed information about employment start and end dates, earnings, unemployment insurance spells, and various demographic characteristics for all jobs covered by the social security system for the years 1975 to 2017.

Sample Selection Because retirement rules vary by birth date, we study individuals' labor market dynamics close to retirement age at the birth-year cohort level. While we ultimately use data from all birth cohorts from 1924–1963, for presentation purposes, we often focus on six cohorts that (a) represent periods of different UI generosity at older ages and (b) for which workers close to the bridge-to-retirement age faced stable UI policies: 1924, 1929, 1935, 1945, 1950, and 1952. Later, we will fit our model to three of these (1929, 1935, and 1950) and use the remaining three cohorts to visualize how our model performs out-of-sample in different regimes.¹⁵ The specific institutional features affecting these six cohorts are summarized in Table 1 Panel A and discussed further in the next section.

We restrict our focus to West German males with a stable employment history at age 50. We focus on men because for most of our sample men and women faced different retirement rules due to the presence of

¹⁵This helps keep exhibits focused and digestible. Our eventual policy counterfactuals will be based on model simulations for all cohorts between 1924 and 1963 and we also show model fit for each of these cohorts in the Appendix.

the women’s pension.¹⁶ Additionally, we study West Germany as we do not observe employment histories for East Germany before reunification. Because eligibility for the maximum PBD and for various retirement pathways depends on having made sufficient social security contributions in prior years, we select individuals who are employed on their 50th birthday and have worked continuously over the previous three years without any UI claims. This employment history restriction increases the likelihood that these individuals are eligible for the maximum possible UI PBD, which can require up to six years worked out of the previous seven years. It also means our focus is on older workers with high labor force attachment. Lastly, we also exclude individuals who are employed (initially, at age 50) in industries known for having special retirement policies or CLAs linked to age 55. Namely, we exclude mining and steel. For cohorts born in or after 1937, when CLAs expanded, we also exclude several additional industries with likely CLAs linked to early retirement at age 55. Appendix A describes this sample selection process in more detail.

Primary Analysis Sample In order to study UI inflows and non-employment durations, we generate a monthly balanced panel of each birth cohort that tracks an individual’s labor market status since age 50. We also generate an analogous quarterly panel that comprises the moments that are used in our structural estimation. To do so, we center the data around the cohort- and individual-specific bridge-to-retirement age, so that the first month after the bridge-to-retirement age starts with the exact date an individual faces a bridge to retirement. For all months, we assign individuals to one of five exclusive labor market states. Individuals can be employed (E), which includes all social security reliable employment, or in registered unemployment (UI), which consists of all periods of UI receipt. In addition, individuals can be outside of these observed employment and unemployment states.¹⁷ Here we distinguish between non-observed unemployment (Nu), which entails up to 3-month interruptions between E and UI , and temporary withdrawal from the labor force (Nt), which includes temporary employment interruption as well as interruptions between E and UI lasting longer than three months. Finally, individuals can withdraw permanently from the labor force (Np), denoted by an exit from E or UI that is not followed by any other E or UI spell in the data. This allows us to follow individuals over time and construct all possible transitions between states.

For our reduced-form evidence and the structural model we focus in particular on two key moments for each birth cohort: i) inflows into unemployment insurance (UI inflows) at each monthly (or quarterly) age.¹⁸ ii) non-employment durations conditional on entering UI at a given monthly (or quarterly) age, which we construct as the number of months out of work before the next employment spell up to age 63. We winsorize this variable at 36 months. Appendix A contains additional details.

¹⁶Women could retire earlier, often with lower penalties, and without a preceding UI spell (our [Online Companion Appendix](#) reports results for women).

¹⁷We count only employment covered by social security, excluding marginal employment and other nonstandard employment relationships. The resulting outside category therefore includes marginal employment and second-tier unemployment assistance, as well as genuinely unobserved states such as retirement.

¹⁸In practice we will define UI inflows as inflows into UI or Nu because if workers are sanctioned at the beginning of UI entry, they appear as Nu in the data and the relevant transition from work to unemployment occurs at the E to Nu transition.

Regression Discontinuity Sample We also construct a separate UI inflow sample that we use to obtain regression discontinuity estimates of intensive-margin responses to PBD extensions. In our primary analysis sample our goal is to keep employment restrictions constant across cohorts while making it very likely that workers qualify for maximum PBD at each age. Hence, we restrict to workers with three years of continuous employment at age 50, as this restriction can be constantly observed and applied to all cohorts, even the earliest ones. Our RD sample differs slightly in that it imposes typically more restrictive work requirements to ensure the relevant discontinuity binds and, for power reasons, it pools all UI entries across calendar time under a stable PBD rule, so any given RD window spans multiple birth cohorts. Specifically, the RD sample imposes the following work history requirements (see also Schmieder et al. (2012)): we require that, at the time of separation, individuals worked in a social security reliable job for at least 12 months of the previous 3 years, worked for at least 52 months within the last 7 years, and had no intermittent UI spell in the previous 48 months. Appendix C contains additional details on the RD sample.

II Reduced-Form Evidence

This section documents how older workers respond to changes in UI PBD and retirement policy. In Section II.A, we show that UI inflows spike at the bridge-to-retirement age and that they respond to both UI and retirement rule changes. In Section II.B we present RD estimates of the effects of PBD extensions for older workers.

II.A Evidence of Extensive-Margin (UI Entry) Responses Under Different UI and Retirement Rules

Using over three decades of data, we document UI inflow responses at the bridge-to-retirement age (a kink in the lifetime budget constraint) across different birth cohorts facing different UI and retirement regimes.¹⁹ We focus on birth cohort-level data so as to keep retirement rules constant within samples. Since UI rules changed over time and were often tied to age at UI entry, UI entrants at different ages in the same cohort can have different maximum PBDs (see Table I.1).

Excess UI Entries at the Bridge-to-Retirement Age We first present evidence of sizable extensive-margin UI responses. Figure 2 shows the number of individuals entering UI by age for six selected cohorts, each chosen to represent a different institutional regime (see Table 1). We focus first on the 1924 cohort in panel (a). We see a large spike in UI inflows at the bridge-to-retirement age, the age at which individuals can

¹⁹Intuitively, this kink at the bridge-to-retirement age occurs because individuals retiring before the bridge age are forced to spend time relying on other income sources, such as a spouse or unemployment assistance (UA/UIB II) before their pension, whereas individuals who leave at or after $ERA - P$ can transition directly into retirement from UI. This reduces the value of an extra year of work after the kink, decreasing the slope of the budget constraint. In general, the size of the kink is exacerbated by the generosity of the UI system, the replacement rate gap between UI and UA/UIB II, and how generously time on UI is counted towards pension contributions. In practice, unemployment counts as an 80% contribution year calculated on pre-unemployment wages. Figure H.2 plots the evolution of stylized lifetime budget constraints for select cohorts experiencing different UI and pension regimes. Appendix D.2 describes how these budget sets are constructed.

transition into retirement immediately following UI expiration. Specifically, Figure 2 (a) shows a spike in UI inflows at age 59 when the ERA was 60 and maximum PBD was 12 months (1924 cohort). This ‘bridge-to-retirement’ pathway is indicated by the red and blue shaded areas under the figures, with red indicating the period over which someone exiting at the bridge-to-retirement age would be on UI and blue when they would be receiving a pension.

To quantify the extent of these extensive-margin responses, we calculate and report measures of the excess bunching mass using standard bunching techniques (Kleven, 2016).²⁰ The estimated excess bunching mass is indicated by the shaded gray area in Figure 2. We also report the standard measure of excess mass (i.e. relative to the mean estimated counterfactual) as well as the excess mass relative to the entire sample size (i.e. the cohort share). The estimated excess mass of UI inflows for the 1924 cohort implies that 5.7% of the 1924 cohort sample entered UI at age 59 as a response to the incentives created by the bridge-to-retirement kink.

Responses to PBD Changes, Holding Retirement Rules Fixed We next show that these UI inflows react to changes in maximum PBDs (and in turn a shift in the bridge-to-retirement age).

All cohorts born before 1937 faced the same pension rules. These cohorts could retire as early as age 60 with no actuarial penalties following a year of unemployment insurance. However, individuals in these cohorts experienced different maximum PBDs due to UI reforms. Panels (a), (b) and (c) show UI inflows for three such cohorts, facing different maximum PBDs when near bridge age: the 1924, 1929 and 1935 cohorts.

The UI inflow mass moves when PBD changes. Figure 2 shows how the spike in UI inflows at age 59 when the ERA was 60 and maximum PBD was 12 months (1924 cohort) moved to age 58 when maximum PBD was extended to 24 months (1929 cohort), and further to age 57 and 4 months when maximum PBD was extended to 32 months (1935 cohort).²¹ The excess UI inflows are 5.7%, 6.5% and 11.8% of the 1924, 1929, and 1935 cohort sizes, respectively. In other words, over 10% of the men meeting our sample criterion who were born in 1935 entered UI at the bridge age as a response to the kink in the lifetime budget set at this point. And, because UI durations tend to be long at these ages among all UI entrants, almost 25% of this cohort was receiving UI at around age 59 (see Figure H.3 (c)). As another point of comparison, Seibold

²⁰We calculate the excess mass as the cumulative difference between the observed UI entries and the counterfactual UI entries from the bridge-to-retirement age to the earliest possible pension claiming age. The counterfactual is calculated by fitting a polynomial within a (usually) 5-year age window on each side of the cutoff, leaving out the bridge-to-retirement region. Normalized bunching is the excess mass divided by the number of individuals in the cohort. Standard errors are calculated using bootstrap with 1000 replications. See Appendix B for more details on the estimation.

²¹While there is no comparably large bunching elsewhere, it is worth noting a few other features of these UI entry distributions. First, note that the UI retirement pathway requires being unemployed for at least 12 months, implying that there is still a small kink at age 59 in the budget set. Indeed, we note some spikes at 59 in panels (b) and (c), which are counted as part of the excess mass. Second, this figure clearly shows bunching in UI entries at other non-kink points, particularly at ages 55 and 57 for cohorts 1929 and 1935. These likely represent collective bargaining agreements to release or buy out workers once they turn 55 or 57. This type of bunching is almost entirely absent in the years leading up to and including 1982, consistent with the timing of the first major CLAs specifying retirement ages (see Trampusch et al., 2010). Since these spikes occur at points in a lifetime budget set where there is no kink, our model will not be able to capture these spikes.

(2021) estimates the excess mass of entering retirement (pension claiming) at the statutory retirement ages in Germany (relative to the counterfactual). He finds large normalized excess bunching mass ranging between 14 to 32 with a mean of 20. In other words, workers are on average 20 times more likely to enter pension receipt at the statutory retirement age than they would at the same age without a statutory retirement in place. Our normalized bunching masses for UI inflows at bridge-to-retirement ages are similarly large, with workers entering UI at the bridge to retirement age 21 to 33 times more often than they would without the bridge to retirement in place. These large behavioral responses reflect how common the UI pathway to retirement was among these earlier cohorts.

A notable feature of the bunching mass is its asymmetry around the bridge-to-retirement age, with a sharper drop-off on the left (younger) side and a more gradual tail on the right (older) side. To the right of the bridge age, shifting UI entry slightly earlier simply lengthens the period spent receiving UI before retirement. By contrast, to the left of the bridge age, entering UI earlier is additionally costly: for workers who remain non-employed until retirement, UI benefits will expire before pension eligibility, leaving them to spend additional time without UI income (or on much lower unemployment assistance).

Figures H.4 (a), (b) and (c) show that average non-employment durations (until age 63) are very close to the maximum possible non-employment duration, indicating that almost all of these entrants end up using UI as a bridge to retirement. For example, in Figure H.4 (a) the average non-employment duration is very close to 48 months at age 59, the number you would obtain if everyone were non-employed between age 59 and the cap of 63. Altogether, there is clear evidence of quantitatively large, extensive-margin responses to UI policy during this period when the ERA was 60.

Responses to Pension Rule Changes, Holding UI Rules Fixed Cohorts born after 1937 faced changing retirement rules. Cohorts born between 1937 and 1941 were still able to retire and claim pensions as early as age 60 (the ERA via UI was still 60), but faced increasing financial penalties from doing so due to a steadily rising NRA. Beginning in birth month January 1937, and increasing with each passing birth month until December 1941, the no-penalty retirement age via UI was gradually increased from 60 to 65. While one could still retire and claim pension at age 60, a permanent 3.6% reduction in pension payments was assessed for each year someone retired in advance of the no-penalty age. By the end of this phase-in period (i.e. for those born in December 1941), retiring at age 60 (instead of 65) would have meant accepting an 18% lower pension.²² These rules remained stable for birth cohorts 1942-1945, after which the earliest possible retirement age for retiring via UI increased and this pathway eventually closed down entirely.

Figure 3 plots excess UI inflows relative to cohort size for birth cohorts before, during, and after this penalty phase-in period. Specifically, in panel (a), we plot bunching masses *for each* birth cohort between 1933 and 1945 – i.e. for four years before the penalty phase-in, during the penalty phase-in, and four years after the penalty phase-in. Maximum PBD remained stable at 32 months for those close to the bridge age across all these cohorts. UI inflows were stable and large prior to penalty phase-in, then dropped gradually

²²Individuals who qualified for retirement via the long-term insured pathway could continue to retire at age 63, but also faced a 7.2% penalty for doing so if born in 1939 or later.

as the penalty was phased in, until they finally reached a new, lower, and relatively stable level between 1940 and 1945. Figure 2 (d) plots UI entries for the 1945 cohort, which exemplifies the new, lower rate of inflows (3% of the cohort or an excess mass of 13 times the counterfactual). In Figure 3 panel (b), we also report findings from a simpler approach that does not require fitting a bunching model and follows in the spirit of a difference-in-differences design.²³ Specifically, we simply add up the UI inflows between ages 57.33 and 60 for each cohort and plot these relative to cohort size in the blue series. Then, we add up the UI inflows between ages 52.33 and 55 for the cohort born five years later (keeping actual calendar time fixed), which serves as a possible control group for macroeconomic conditions (in the green series). The gap between the two series thus reflects differential changes in UI inflows between older and younger ages in the same calendar years. Panel (b) reinforces the findings from panel (a): UI inflows of older workers initially differed massively from those of younger workers under the high maximum PBD and relatively generous retirement regime, but this gap narrowed considerably following the introduction of the retirement penalty.

Finally, Figures 2 (e) and (f) show UI inflows for the 1950 and 1952 cohorts. Both cohorts were eligible for 2 years of maximum PBD over the relevant ages, but faced very different pension rules from earlier cohorts. The 1950 cohort was still eligible to retire at age 63 via the UI pathway (or the long-term insured pathway, which requires at least 35 years of pension contributions), but with a two-year penalty applied. And there is indeed evidence of bunching at age 61. We also see some bunching (not shaded) at age 63, two years before the no-penalty retirement age of 65. However, bunching masses are now much lower (on the order of 1% of the cohort size) than in earlier years. The 1952 cohort saw the UI pathway abolished entirely, meaning retiring at 63 via the long-term insured pathway (if qualified) was the earliest available option. Here, too, we see a spike and some excess mass on the order of 1% at age 61. For both cohorts, the combination of the early retirement penalty, not actually being able to retire much earlier than the likely alternative for our sample despite bridging via UI, and the declining macroeconomic unemployment rates, has led to still-present but far fewer excess UI entries among older workers than in earlier decades.

Overall, we observe clear and sizable bunching into UI at the bridge-to-retirement age. The bunching mass responds to both UI PBD changes and to changes in retirement rules. Under the relatively generous PBD and retirement policies of the 90s, excess UI inflows of older workers were particularly large, and they only declined substantially following many years of penalty and eventually retirement age increases, as well as declining maximum PBD, among other changes.

II.B Evidence of Intensive-Margin Responses from Regression Discontinuity Estimates

The preceding cohort-by-cohort UI inflow and non-employment duration moments are used to identify our structural model. However, in addition to changing UI inflows, PBD extensions also affect non-employment durations *conditional* on entering UI and these ‘intensive margin’ effects could vary with age and proximity

²³We do this because the underlying UI inflow graphs in each year can be somewhat noisy, likely due to the fact that policy changes occurring at the birth-month level rather than birth-year level, and to potential adjustment frictions. Furthermore, fitting these bunching masses requires making some subjective choices. See Appendix B and Figure H.5 for details on the cohort-specific inflow graphs and construction of the counterfactual.

to retirement. Before turning to the model, we briefly discuss and, to the extent that we have valid RD-induced variation, estimate these effects for older workers.

Prior work on UI extensions has predominantly estimated these intensive margin effects for prime-age workers.²⁴ While whether or not these effects vary with age is interesting in its own right, it is also crucial for our model estimation. Since our model starts with workers aged 50 and above and has a search component, it is helpful to be able to target well-identified moments for how the non-employment duration among older workers, conditional on claiming UI, differs when PBD is extended. Hence, we extend [Schmieder et al. \(2012\)](#)’s RD approach and estimate intensive-margin non-employment effects of UI at older age discontinuities for workers in their 50s, which will ultimately result in an age-52 RD moment that we have confidence in and use (in addition to the UI inflow and non-employment duration moments) to discipline search behavior in our model.²⁵

Beginning in 1987, there are 12 age cutoffs across four distinct periods at which we can potentially estimate the non-employment effects of UI extensions using RDs (see Table I.1), seven of which are for workers aged 50 and above. These estimates require the standard RD assumptions, including no sorting into UI around the cutoffs. Unfortunately, and consistent with the UI inflow responses seen above, this assumption is not always satisfied at older ages. Figure 4 (a) shows RD estimates of the jump in the density at the age threshold, revealing strong sorting at the 55+ cutoffs. Since the model and data condition on being employed at age 50, we would like to be able to target estimates at ages above 50. Fortunately, there are two candidate RD estimates past age 50, at ages 52 and 54, that are plausibly identified, with the age 52-RD in 1999-2006 ultimately being the one we have the most confidence in and decide to target.²⁶

For completeness and to see if we can detect any hints of patterns across ages, at each age cutoff below age 55, we estimate the following RD specification:

$$y_{ia} = \delta \mathbf{1}(a_i \geq A) \Delta PBD + f(a) + X_i \beta + \varepsilon_{ia} \quad (1)$$

where y_{ia} is the non-employment duration (capped at 36) for individual i of age a , a_i is the age at the time of UI entry (measured on the daily level), and $\mathbf{1}(a_i \geq A)$ is a dummy variable indicating that an individual is above the age threshold A , where benefits are extended discontinuously by ΔPBD months. In this specification, δ measures the effect of a one-month increase in PBD. We specify $f(a)$ as a linear function, allowing different slopes on each side of the cutoff. X_i is a vector of additional controls. We use a rectangular kernel and cluster standard errors at the day level. We use a bandwidth of two years but restrict it to one year on the right side of the 49 and 54 age cutoffs during the 1987-1999 period due to the presence

²⁴In our context, [Schmieder et al. \(2012\)](#) analyze age discontinuities up until age 49 and [Schmieder and Trenkle \(2020\)](#) at age 50.

²⁵Note that our need to restrict to non-violated cutoffs also raises econometric challenges. If, for example, the degree of density violation is correlated with the intensive margin response, extrapolating estimates to close-but-violated cutoffs might be misleading. We abstain from such comparisons. In the model section, we target the estimate of age-cutoff 52 to match behavior at that age, and let the model simulate both intensive-margin and extensive margin responses at other ages.

²⁶This exhibits a somewhat smoother density close to the cutoff than the 54 cutoff (Figure H.6). In addition, unlike at the age 54 cutoff where we see hints of imbalance on pre-unemployment wages, all predetermined variables are balanced at the age-52 cutoff and the standard errors of the estimates are slightly lower.

of other discontinuities at ages 50 and 55. We also exclude two months on each side of the cutoff – the donut hole – in all our regressions to minimize any potential bias due to sorting around the cutoffs. Self-contained Appendix C includes additional details as well as validity and robustness checks.

Figure 4 (b) plots the eight RD estimates for different age cutoffs with and without controls. These estimates are also reported in Table I.3. Each dot in the figures corresponds to a marginal effect of one additional month of potential UI duration estimated at an age cutoff. The estimates average 0.09, suggesting that for each month of additional UI, affected workers spend around three more days in non-employment. Estimates are relatively insensitive to controls. We do not have sufficient power to detect any clear variation by age, though we obtain the largest point estimates at the older ages.

We target the RD moment at age 52 (the larger dot in Figure 4 (a) and (b)) in our structural model, for which balance and density checks all pass and standard errors are relatively precise. Figure 4 (c) shows the RD figure for this target estimate, plotting mean non-employment duration (capped at 36 months) by age around the age-52 cutoff for the 1999-2006 period. The point estimate at this cutoff, and hence our target moment, is 0.128.²⁷

III Dynamic Labor Supply Model

The reduced-form evidence shows that UI inflows respond to both UI and retirement reforms. To quantify the aggregate and fiscal consequences, we develop a dynamic life-cycle model of employment, unemployment, job search, and retirement and fit it to the preceding UI-inflow, non-employment-duration, and RD moments.

III.A Model Set Up

States and Value Functions Workers can be employed (E), unemployed (U), or out of the labor force (O), an absorbing state. We refer to U and O jointly as non-employment (N). Starting at age 50, the state variables are productivity p_t and cumulative unemployment duration d^U . Workers begin eligible for the maximum PBD and do not reaccumulate eligibility after reemployment. Thus, d^U determines both remaining UI eligibility and pension benefits, allowing the value functions to be written in terms of p_t and d^U .²⁸

Workers have a utility function $u(\cdot)$, are paid $w_t(\cdot)$, and experience disutility from working (η). The value of employment is:

$$V_t^E(p_t, d^U) = u(w_t(p_t)) - \eta + \beta E_{p_{t+1}} [\max \{V_{t+1}^E(p_{t+1}, d^U), V_{t+1}^N(d^U)\}] \quad (2)$$

For tractability, we assume workers have all the bargaining power and firms make zero profits so that $w_t = p_t$

²⁷The age-54 moment has a fair amount of bunching inside the donut hole. While excluding the donut hole likely yields plausible results and almost the same point estimate as the age-52 moment, the age-52 moment is arguably identified more cleanly.

²⁸Tracking eligibility across repeated spells would require additional state variables. Because repeated unemployment spells separated by long employment spells are rare, this simplification substantially reduces the state space at little empirical cost.

in all periods.²⁹ We also assume that workers have correct beliefs about future UI and retirement policies, which seems plausible given the slow, cohort-based phase-in of retirement reforms and that for our key cohorts the bridge age does not occur close to UI reforms. Workers will separate from their jobs whenever the expected value of future non-employment exceeds that of employment. This could occur for several reasons: workers could receive a low productivity draw (p_t) such that the employment relationship is no longer better than the worker's outside option. Alternatively, outside options could improve, e.g., due to an increase in UI or retirement benefits, which can push up $V_t^N(d^U)$ for workers close to the retirement age and increase the rate of job separations. We also allow for exogenous job destruction at the rate Λ_t . To operationalize this, workers face a (large) negative productivity shock with probability Λ_t .³⁰ Otherwise, they draw a productivity level p_t from a lognormal distribution.

We model unemployment as a fully dynamic process. This approach enables us to capture the duration of UI benefits and labor supply responses to changes in the structure of UI in a natural way.³¹ When workers enter unemployment they engage in costly job search and receive payments $B(d^U)$. Prior to UI benefits exhaustion ($d^U < P$), a worker receives benefits ($B(d^U) = b$). Afterwards, the individual receives y^u ($B(d^U) = y^u$), which can be interpreted as unemployment assistance. An unemployed individual searches for a job and chooses an optimal level of search effort s which is normalized to the probability of finding a job. Generating search effort comes at a cost $\psi(s, d^U)$, which is increasing and convex in search effort. Finally, whether or not an individual receives a job offer, he can decide to retire at the end of the period. If he remains unemployed d^U increases by one period. The value of unemployment is thus:

$$V_t^U(d^U) = u(B(d^U)) + \max_s \{ \beta s E_{p_{t+1}} \max [V_{t+1}^E(p_{t+1}, d^U + 1), V_{t+1}^N(d^U + 1)] + \beta(1 - s)V_{t+1}^N(d^U + 1) - \psi(s, d^U) \} \quad (3)$$

Let $\psi_s(s, d^U) \equiv \frac{\partial \psi(s, d^U)}{\partial s}$ denote the marginal cost of search. At an interior solution, optimal search effort is given by:

$$s^* = [\psi_s(\cdot, d^U)]^{-1} (\beta E \max [V_{t+1}^E(p_{t+1}, d^U + 1), V_{t+1}^N(d^U + 1)] - \beta V_{t+1}^N(d^U + 1))$$

At any point, a worker can choose to transition out of the labor force O , which is an absorbing state. The value of O depends primarily on the value of one's pension y_t^p as determined by prevailing retirement institutions. This value depends on the work history (d^U) and the age at which the worker retires. Specifically,

²⁹ Alternatively one could assume Nash bargaining over the surplus, but in that case, there is no closed form solution for the expected value of employment and solving the model becomes computationally challenging. Since we are not trying to match wages, this simplification strikes us as a worthwhile trade-off.

³⁰ We also assume that workers have foresight with respect to the future exogenous job destruction probability Λ_t . This seems like a reasonable trade-off for tractability since it mostly determines UI inflows for younger workers. As a robustness check, we will also show results where Λ_t is constant over time.

³¹ Other structural life-cycle papers (e.g. Haan and Prowse, 2010; García-Pérez and Sánchez-Martín, 2015; Michelacci and Ruffo, 2015) typically assume workers receive UI forever or model UI as a Markov process with a fixed transition probability to exhaustion. Our approach has the added benefit that our parameter estimates for the job search part can be compared with previous estimates of job search models (e.g., Paserman, 2008; DellaVigna et al., 2017, 2022).

for a worker who lives until T^{Last} and is eligible to receive a pension at T^{ERA} , the value function of being out of the labor force is:

$$V_t^O(d^U) = \begin{cases} \sum_{k=t}^{T^{ERA}-1} \beta^{k-t} u(y^o) + \sum_{k=T^{ERA}}^{T^{Last}} \beta^{k-t} u(y_t^p) & \text{if } t \leq T^{ERA} \\ \sum_{k=t}^{T^{Last}} \beta^{k-t} u(y_t^p) & \text{if } t > T^{ERA} \end{cases} \quad (4)$$

The value of the pension depends on the relevant, cohort-specific retirement institutions in addition to the individual's work history (d^U). Appendix F.5 details how V_t^O is calculated for each cohort.

Finally, the value of non-employment is defined as:

$$V_t^N(d^U) = \max(V_t^U(d^U), V_t^O(d^U)) \quad (5)$$

Heterogeneity in the disutility of work We let η vary across workers according to a cohort-specific normal distribution with mean $\bar{\eta}_{cohort}$ and a common standard deviation η_{sd} . Cohort-level transitions and non-employment durations integrate over this distribution; Appendix E provides implementation details.

How does the model generate bunching? Workers enter UI when $V_t^N > V_t^E$. As retirement approaches, the UI bridge raises the value of non-employment relative to employment, causing many η types to exit at the bridge age. Productivity shocks spread these exits across nearby ages and generate earlier separations.

III.B Assumptions and Parameters

We now briefly discuss the key functional forms and distributional assumptions used in our baseline model and lay out the parameters we estimate and those we fix based on institutional features. Self-contained Appendices E-F provide additional details.

Productivity p_t is drawn from a mixture distribution in which workers have Λ_t probability of facing a large negative productivity shock such that $\ln(p_t) = -L$, which destroys the job with certainty. Meanwhile, with probability $1 - \Lambda_t$, workers draw a productivity level p_t from a lognormal distribution. The CDF of this mixture variable is $F_t(\ln(p_t)) = \Lambda_t \mathbf{1}\{\ln(p_t) \geq -L\} + (1 - \Lambda_t) F_{p, \sigma_p}^N(\ln(p_t))$, where F_{p, σ_p}^N is the normal CDF with mean p and standard deviation σ_p . This allows for exogenous job destruction at the rate Λ_t . In practice, we allow this exogenous job destruction rate to vary with the national male unemployment rate (UR). Specifically, $\Lambda_t = \frac{1}{1 + e^{-(\lambda_1 + \lambda_2 \bar{U}_{Rt} + \lambda_3 \Delta \bar{U}_{Rt})}}$, with parameters λ_2 and λ_3 allowing Λ_t to vary with the level and year-on-year change in the national male UR.

We assume workers have log utility $u(\cdot) = \ln(\cdot)$. Firms pay the worker $w_t = p_t$ in all periods. Workers draw disutility η from a normal distribution ($\eta \sim N(\eta_{mean, cohort}, \eta_{sd})$). The search cost function is based on DellaVigna et al. (2022) with some added flexibility. Specifically, we assume:

$$\psi(s, d^U) = k_0 + k_1 \mathbf{1}(d^U = 0) + e^{k_2 \times d^U} \times k_3 \frac{s^{1+\gamma}}{1+\gamma}, \quad (6)$$

where k_0 is a fixed cost of being in unemployment, k_1 a fixed cost of entering unemployment for the first time, k_2 allows search to become more costly over the unemployment spell, and k_3 and γ govern the slope and curvature of the job search function. With these functional form assumptions, the model leads to closed-form solutions (which can be found in Appendix E) for all transition probabilities and can be solved via backwards induction from the last period.

We fix the remaining inputs using institutional rules and administrative data. Table 1 Panel A reports the cohort-specific institutions and calibrated values. We set mean net monthly earnings to EUR 1,950, UI benefits to EUR 1,170 for later cohorts and EUR 1,230 for earlier cohorts, unemployment assistance to EUR 500, and pre-pension out-of-the-labor-force income to EUR 50. Contribution years and pension replacement rates come from the German Pension Register data. The model applies each cohort's ERA, NRA, pension penalties, contribution rules, and available retirement pathways and allows workers to choose the highest-value feasible pathway. Appendix F.4 documents these choices and the pension-value calculations.

We estimate the following 13 parameters: the standard deviation of the productivity distribution (σ_p); three parameters that allow the exogenous job destruction rate Λ_t to vary with the level and year-on-year change in the national male unemployment rate ($\lambda_1 - \lambda_3$); five parameters in the search cost function ($k_0 - k_3$, and γ); and four parameters governing the η distribution: $\bar{\eta}_{1929}$, $\bar{\eta}_{1935}$, $\bar{\eta}_{1950}$, and η_{sd} (which does not vary by cohort).

III.C Identification and Estimation

We implement a two-step estimation procedure. In the first step, we jointly estimate 13 parameters using 307 moments from the 1929, 1935, and 1950 cohorts, which faced different UI and retirement regimes. These include 10 parameters that are common across cohorts and the cohort-specific mean disutility of work for each of the three primary cohorts. In the second step, holding the common parameters fixed, we estimate only the cohort-specific mean disutility of work, $\eta_{mean,cohort}$, for each of the remaining cohorts using their UI inflow and non-employment-duration profiles.

Identification In the first step, the age profiles of quarterly E to U inflows, across cohorts and labor-market conditions, are most informative about the dispersion of productivity shocks, the fixed costs of unemployment and UI entry, exogenous job destruction, and the distribution of disutility of work. Age-specific non-employment durations primarily discipline the marginal search-cost parameters, while the age-52 RD estimate of $\frac{\partial Nonemp}{\partial P}$ provides additional discipline on γ , which determines the responsiveness of search effort to the returns to search. These mappings are not one-to-one. Appendix G reports parameter-to-moment influence measures (Figure H.15 and Table I.10) and moment-to-parameter sensitivity measures (Figure H.16 and Table I.11), following Andrews et al. (2017). In the second step, each additional cohort's inflow and duration profiles discipline the only additional free parameter, that cohort's mean disutility of work, $\eta_{mean,cohort}$.

Estimation We implement both estimation steps using a minimum distance estimator to match our key empirical reduced-form moments. Denoting the parameters of the structural model as ξ , the vector of moments predicted by the model as $m(\xi)$, and the vector of observed moments as $\hat{m}$, the estimator chooses parameters $\hat{\xi}$ that minimize the distance

$$(m(\xi) - \hat{m})' W (m(\xi) - \hat{m}),$$

where W is a weighting matrix.

For the first step, we focus on three cohorts: 1929, 1935, and 1950, for whom we match the quarterly E to U flows and expected non-employment durations (until age 63) of workers entering unemployment in each quarter. Furthermore, we use the RD estimate for $\frac{\partial Nonemp}{\partial P}$ for men at the age-52 cutoff of 0.128 (Table I.3) to inform the intensive-margin effect of UI for the 1950 cohort.³²

We weight the moments using their empirical sampling variation: cohort-specific covariance matrices for the E -to- U transitions and diagonal variance matrices for non-employment durations and the RD moment. We upweight the well-identified age-52 RD moment by 35 so that the fitted model preserves the intensive-margin response to PBD.³³

After fitting the model to these primary cohorts, as a second step, we refit our model to all other cohorts between 1924 and 1963. When doing so, we only re-estimate a single parameter per cohort, which is the mean of that cohort's η distribution ($\eta_{mean, cohort}$). In this estimation exercise, our target moments are transitions from E to U and non-employment durations. Refitting allows different cohorts to have different outside options or labor force attachment in a way that is not otherwise captured by retirement and UI institutions or other parameters. See Appendix F for details about the estimation.

III.D Discussion of Modeling Choices

Our baseline focuses on labor-market transitions and job search and abstracts from savings, wage heterogeneity, household decisions, and retirement norms.

First, workers consume current income. Evidence of limited consumption smoothing around UI entry and exhaustion, and from search models with present bias, suggests that this can be a useful approximation (Ganong and Noel, 2019; Gerard and Naritomi, 2021; DellaVigna et al., 2017). Evidence from retirement reforms also generally points to modest savings responses (Lachowska and Myck, 2018; Etgeton et al., 2023; García-Miralles and Leganza, 2024; Haan and Prowse, 2014). As a partial check, column (7) of Table 3 raises consumption while out of the labor force; the main policy conclusions remain similar.

Second, the baseline targets transitions and non-employment durations rather than wages, so η_i captures unobserved heterogeneity in workers' propensity to enter UI. Column (9) replaces this heterogeneity with severance pay, motivated by evidence that firms offered heterogeneous severance packages to older workers

³²The moments and the parameters they primarily discipline are summarized in Table I.6.

³³This follows the spirit of Armstrong and Kolesár (2021) and DellaVigna et al. (2022). Results are quantitatively similar with a diagonal weighting matrix or without upweighting. Appendix F provides implementation details.

exiting through the UI bridge (Grund, 2006). Auxiliary regressions link severance pay to pre-unemployment wages and firm size, and we additionally target the corresponding wage and firm-size moments measured at age 50. This variant uses observed wage and firm-size patterns to discipline selection into UI but does not use wages to identify the full productivity distribution. It produces broadly similar policy conclusions.

Finally, the baseline abstracts from household decisions and retirement norms. Focusing on men likely captures an important margin because men in these cohorts were typically older and higher earning than their spouses. Column (8) of Table 3 adds a utility shift at the statutory retirement ages emphasized by Seibold (2021); its estimate is near zero and the policy predictions are essentially unchanged.

IV Estimation Results and Model Validation

In this section, we summarize the estimation results and gauge the ability of the model to fit the in-sample target moments as well as the out-of-sample moments of non-targeted cohorts. We also show how the model can be used to construct unemployment rates for counterfactual policy scenarios.

IV.A Estimation Results and In-Sample Fit

Table 1 Panel B reports the estimated parameters with bootstrapped standard errors. Standard errors are generally small, suggesting local identification of the main parameters. We estimate a substantial standard deviation of the productivity shock σ , which is necessary to generate the observed diffuse bunching around the UI bridge age. We also find a sizable fixed cost of UI entry k_1 , consistent with norms or stigma around UI take-up or the bureaucratic cost of applying; without this cost, the model predicts a large inflow just before retirement.³⁴ Search costs display meaningful duration dependence through k_2 , while the slope k_3 and curvature $\gamma = 0.82$ imply relatively elastic search effort with respect to the returns to search, in line with the prior literature.³⁵ The λ parameters allow job destruction to track the business cycle, letting the model fit changes in unemployment inflows over time without overstating search responses. Finally, we estimate substantial heterogeneity in worker types, with the expected cross-cohort pattern: higher η_{1935} and η_{1950} coincide with weaker labor markets and generally higher inflows, consistent with greater average disutility of work (or lower job values) in those cohorts.³⁶ That η_{1935} is similar to η_{1950} also suggests that the

³⁴Evaluated at the marginal utility of consumption for the average worker ($1/w$), $k_1 = 5.66$ corresponds to approximately EUR 11,000, consistent with UI entry being unattractive for short spells but worthwhile for longer bridge-to-retirement spells. Since our model assumes hand-to-mouth consumers, k_1 may also partly capture unmodeled wealth effects: workers with liquid savings can self-insure through short unemployment spells without entering UI, behavior that our model can only rationalize through a large entry cost. This is particularly relevant for older workers, who are the most likely to have accumulated liquid wealth.

³⁵Our $\gamma = 0.82$ closely matches the single-type δ -discounting reference-dependent estimate in DellaVigna et al. (2017) (0.81) and is comparable to the composite curvature reported in DellaVigna et al. (2022) using German data. The estimated slope of the search cost function ($k_3 = 51$) is also similar in magnitude to the medium-cost searcher in the standard three-type model of DellaVigna et al. (2017).

³⁶The flow utility of employment can be written as $\ln(w \cdot e^{-\eta})$, so η is equivalent to an effective wage reduction: workers behave as if they earned $w \cdot e^{-\eta}$ with no disutility. At average net wages of EUR 1,950, the 1929 cohort's $\bar{\eta} = 0.42$ implies an effective wage of EUR 1,281/month.

model explains lower bunching in the later cohorts through institutions rather than through a cohort-specific parameter.

Figure 5 gauges our estimated model’s in-sample fit by comparing simulated E to U transitions and simulated non-employment durations to their empirical counterparts for the three cohorts matched in the estimation (1929, 1935, and 1950). Overall, our model captures the key empirical patterns of interest: it predicts UI inflow bunching at the bridge-to-retirement age and generally gets the size of the bunching mass right, captures overall E to U transition trends, and matches older workers’ actual mean non-employment duration. It also matches finer features of the data, such as the dip in non-employment duration for the 1950 cohort between ages 56 and 58 when maximum PBD decreased. The model does not, however, fit every feature of the data. It systematically underfits UI inflow spikes at ages prior to the bridge-to-retirement age (e.g., 55 and 57 for the 1929 cohort), likely because these spikes reflect collective labor agreements linked to specific ages rather than the budget set kinks our model captures. It also has some difficulty with non-employment durations at younger ages, overpredicting for the 1935 cohort and under-predicting for the 1950 cohort.

Our model also targets our RD estimate of $\frac{\partial Nonemp}{\partial P}$ at age 52 for the 1950 cohort (0.128), calculated as the simulated change in non-employment duration among new UI entrants from an extra month of maximum PBD, holding the worker type distribution constant among new entrants so as to mimic a pure intensive-margin effect. The model fits this moment very well (0.127). Table I.7 reports simulated $\frac{\partial Nonemp}{\partial P}$ at other ages. For a given cohort it starts between 0.11 and 0.14 at age 50 and increases with age, a pattern also seen in the data (Table I.3), albeit noisily. At the oldest ages it declines, even reaching 0 for the pre-1950 cohorts, since nearly all UI entrants at these ages stay non-employed until age 63.

IV.B Out-of-Sample Performance

To simulate our model for out-of-sample cohorts, we require estimates of $\bar{\eta}_{cohort}$, the cohort-specific average disutility of work, which we obtain by refitting the model to match each cohort’s non-employment durations and transitions, holding all other parameters constant. Figure H.8 plots estimated $\bar{\eta}$ across all cohorts, revealing a relatively continuous pattern, with cohorts born prior to 1935 having lower disutility of work. The trends in $\bar{\eta}$ roughly mirror changes in the national unemployment rate between the mid-80s and early 2000s, approximately when these workers turned 60, consistent with $\bar{\eta}$ capturing changes in workers’ outside options not otherwise reflected in the model’s productivity distribution or job destruction rate.

The blue lines in Figure 6 show how the model performs out-of-sample for the remaining three focal cohorts in Figure 2: the 1924, 1945, and 1952 cohorts, which faced different UI and retirement institutions.³⁷ Despite using parameters estimated from other cohorts (aside from $\bar{\eta}_{cohort}$), our model performs well, broadly matching overall UI inflows, the spike in UI inflows at the bridge-to-retirement age, and non-employment durations. Overall, the model fit is similar both in- and out-of-sample. We discuss the red lines, which show a counterfactual PBD extension, in Section V.

³⁷ Figures H.9–H.12 show model fit for all cohorts from 1924 to 1963.

IV.C Constructing Model-Based Unemployment Rates

We now use model simulations for all birth cohorts between 1924 and 1963 to construct model-based, age-specific unemployment rates in calendar time (e.g., unemployment rates for ages 56-59 in 1994). These provide an additional out-of-sample validation exercise, since we can construct them both from the model and from the social security data and compare the two directly. We can also compare each series to the OECD unemployment rates in Figure 1. They also allow us to directly explore how different counterfactual scenarios affect older workers' unemployment trends. Specifically, we use E to U transitions and non-employment durations to predict the share of workers who are unemployed, or more precisely, non-employed following first UI entry, in each cohort $\times$ year cell, and then aggregate these predicted shares for any age range, doing so separately for the model's simulated moments and their empirical counterparts.³⁸

We plot these empirical and simulated unemployment rates separately for those aged 52-55 and 56-59 in Figure 7 (a). The empirical rates we construct roughly follow the OECD rates in Figure 1, and our model, fit to 40 cohorts holding all but one parameter constant across them, matches them remarkably well, which also indicates that we jointly fit non-employment durations and transitions well across all cohorts. The fit is nearly exact at younger ages and generally good at older ages, where we occasionally underfit the empirical rate, in earlier years because we under-predict the bunching mass at the bridge age and in later years because we under-predict non-employment durations. The model captures the striking 10pp rise in the unemployment rate of workers aged 56-59 between 1983 and 1994, its contrast with the much smaller rise for younger workers over the same period, and the equally striking decline for older workers between 1994 and the mid-2000s, when the rate for younger workers barely changed.

V Effects of UI and Retirement Policies on Aggregate Unemployment Rates

We now use our model to quantify how Germany's historical UI and retirement policies affected the aggregate unemployment rate. We begin with a simple exercise that extends PBD by one year, which illustrates the model's key margins of adjustment. We then show that Germany's PBD extensions and later retirement rule changes account for a large share of the rapid rise and subsequent fall in the old-age unemployment rate. We conclude this section with a discussion of the policy costs, specifically the fiscal externalities, of UI extensions for older workers.

V.A A First Look at the Effects of a PBD Extension

To provide an initial sense of how the model works, we simulate a one-year increase in maximum PBD for all cohorts. Figure 6 shows the resulting change (in red) for three cohorts. For younger workers, who mostly enter UI as a consequence of exogenous job loss, UI inflows do not change but non-employment durations

³⁸To simplify this procedure and to make it directly comparable to something we can derive from the administrative data, we do this calculation assuming a constant hazard of exiting unemployment, rather than allowing for the full duration dependence. By applying this approach uniformly to both the model simulations and empirical moments, we ensure that the two unemployment rates are directly comparable.

shift upwards, reflecting the standard intensive margin $\frac{\partial Nonemp}{\partial P}$ effect estimated in the RDs.³⁹ For workers close to retirement, the extension instead moves the bridge-to-retirement age left by one year (from 57 and 4 months to 56 and 4 months for the 1945 cohort, for example), causing some to enter UI up to a year earlier; since these individuals typically remain non-employed until retirement, non-employment durations at the new bridge age rise sharply. Aggregating across all cohorts, Figure 7 (b) shows that this extension raises simulated unemployment rates for those aged 52-55 by a modest 0.6pp in 1994, but raises the rate for those aged 56-59 by a comparatively large 2.5pp (Table 2). This provides a first glimpse of how the same UI extension can have substantially different impacts under differing circumstances.

V.B The Effects of Germany's UI and Retirement Policy Changes on Aggregate Older Workers' Unemployment

Rise between 1983 and 1994 We begin by asking what the rise in unemployment among older workers would have been had maximum PBD remained at the 1984 level of 12 months, rather than increasing to 32 months. Figure 8 (a) and (b) illustrate what this change does for the 1935 cohort (red dash-dotted line). The spike in UI inflows would have remained at age 59 instead of at 57 and 4 months, and the non-employment durations of workers in their late 50s would have been substantially lower. Figure 7 (c) shows how keeping PBD fixed at 12 months affects the overall unemployment rate of younger and older workers across all cohorts. We find that, at the peak in 1994, unemployment rates of workers aged 56-59 would have been 5.7pp lower (see Table 2). In other words, PBD extensions explain 5.7pp of the 9.9pp increase (or 58%) in the unemployment rate from 1983 to 1994.

In contrast to the large 5.7pp decline for workers aged 56-59, the effects on relatively younger and older workers are more muted. Table 2 shows that, had PBD remained at 12 months, the unemployment rate of workers aged 52-55 would have been 0.4pp lower in 1994 and that of workers aged 60-62 would have been 0.9pp lower. For younger workers this primarily reflects the intensive margin, as PBD changes do not meaningfully affect their inflows. For those in their 60s, inflows are the relevant margin, since most remain non-employed until retirement once on UI. Inflows do increase somewhat at these ages under the counterfactual (as can be seen for the 1935 cohort in Figure 8 (a)), but the changes are modest because entering UI very close to retirement is relatively unappealing.

Altogether, Germany's PBD increases account for a large share of the *overall* unemployment rate change for those aged 52-62: Table 2 column (4) shows that they explain half of the increase for this group (2.4 pp/4.6 pp). While other factors clearly mattered, Germany's PBD extensions played a primary role in increasing the unemployment rate of workers in their late 50s in the early 1990s by shifting the bridge-to-retirement age earlier.

³⁹In practice, $\frac{\partial Nonemp}{\partial P}$ is estimated both in the model and in the data on non-employment durations that are capped at 36 months, while this figure plots uncapped non-employment durations until age 63. The effect of changing PBD on these uncapped durations is larger.

Decline between 1994 and 2014 If changes in maximum PBD (together with the existence of the UI pathway) help explain much of the increase in the unemployment rate of older workers, what explains its more recent decline? In 1994, both UI and retirement institutions were near their most generous: workers aged 58 that year (the 1936 cohort) had a maximum PBD of 32 months and could retire via UI at age 60 without penalty. Thereafter, PBDs decreased, pension penalties for retiring at 60 began to kick in, and the earliest possible age for retirement via UI increased. Figure 7 (d) simulates a world in which none of these changes occurred, with pension and UI policies remaining stable at their 1994 levels. The dashed blue line shows that the unemployment rate of workers aged 56-59 would then have declined by only 3.0pp between 1994 and 2014, due to factors like changing economic conditions or other policy changes not considered, instead of the 11.8pp decline we observe (the solid, dark blue line).⁴⁰ Thus, the retirement and UI policy changes explain 8.8pp, or 75%, of the observed decline (see also Table 2).

We can also probe which of these specific policy changes, PBD decreases, the introduction of the retirement penalty, and the increase in the ERA, mattered most by changing one policy at a time. Relative to holding all institutions fixed at their 1994 levels, the reductions in PBD would have reduced the unemployment rate of older workers by 2.9pp. Holding PBD and early retirement ages at their 1994 levels but implementing the penalty for retiring via UI at 60 (due to the increasing NRA) would have reduced it by 5.4pp. Finally, simply increasing the earliest possible age for retirement via UI (which affected birth cohorts after 1945) would eventually have had a large 7.8pp effect, as the gradual ERA increase moves UI inflow bunching toward age (63-PBD) and far fewer workers enter UI between ages 56 and 59. As such, while PBD changes alone would have mattered, the retirement institution changes appear particularly relevant in explaining the decline of older workers' unemployment rates since the mid-1990s.

Altogether, these counterfactuals provide new insight into what drove the striking historical trends in Germany's older workers' unemployment rates (Figure 1): PBD extensions explain over half of the rise between 1983 and 1994, while retirement reforms and PBD cuts account for much of the subsequent decline.

Robustness In Table 3, we probe the robustness of these policy takeaways to alternative modeling choices. For each alternative, we re-estimate the model and repeat our policy simulations. Column (1) reproduces the baseline results. In Columns (2) and (3), we consider alternative weighting schemes. Using a diagonal weighting matrix (2) mechanically reduces the SSE but attributes a similar role to PBD increases in increasing unemployment and attributes more of the decline in the unemployment rate from 1994 to 2014 to changes in the institutional environment. Not upweighting the $\frac{\partial D}{\partial P}$ moment leads to a similar model fit with a predicted $\frac{\partial D}{\partial P}$ of 0.111, slightly below our RD estimate of 0.128. Shutting off the dependence of the exogenous shock on the unemployment rate in Column (4) (setting $\lambda_2, \lambda_3 = 0$) worsens overall fit substantially (SSE rises from 18 to 32 thousand) because the model can no longer track cyclical variation in

⁴⁰For intuition, Figures 8 (c) and (d) show how keeping UI and retirement institutions fixed at their 1994 levels affects the 1952 cohort. The dashed green lines show that had PBD remained at 32 months, and the ERA and NRA for retiring via UI at 60, the spike in UI inflows would have remained at age 57 and 4 months and non-employment durations would have been substantially higher.

UI entries for younger workers, but the fit around the bridge-to-retirement ages and the counterfactual policy predictions are nearly unchanged. Replacing the cohort-specific disutility levels η_{cohort} with a constant η_{mean} in Column (5) (which means the entire model is only ever fit to our three focal cohorts, and there is no refitting across all the other cohorts) unsurprisingly raises the SSE. Yet PBD changes still explain most of the 1983–1994 rise, while later retirement and PBD changes, although accounting for somewhat less than in the baseline, still explain 45% of the subsequent decline.

Columns (6)–(8) vary specific parameters. A higher UA replacement rate (Column (6), raising UA benefits from 500 to 750) slightly improves fit and, by making the UI pathway more attractive, attributes more of the unemployment variation to policy. Higher consumption out of the labor force (Column (7), raising home production when out of the labor force from 50 to 500), a partial stand-in for unmodeled savings, modestly dampens the impact of retirement policies and worsens fit, but the main message holds: policy changes still account for more than half of both the increase to 1994 and the decrease to 2014. Finally, in Column (8), we probe the retirement norms emphasized by [Seibold \(2021\)](#) by adding a utility boost for exiting the labor force at an exact retirement age (any ERA or NRA). Because this nests the baseline, the SSE falls, but negligibly; the estimated utility boost is slightly negative and insignificant, and the policy predictions are essentially identical. This is consistent with retirement via the UI pathway at the ERA not functioning as a norm in the way the statutory ages studied by [Seibold \(2021\)](#) did. The ERA via UI pathway was not, for instance, actively promoted by the pension system. Since we do not target retirements directly from employment (without UI), this is also not an ideal setting to identify norms at the NRA.

Finally, column (9) considers an alternative specification that uses observed wage and firm-size patterns to provide additional discipline on selection heterogeneity. The baseline model fixes wages at their mean and uses heterogeneous disutility of work to generate differences in workers’ propensity to enter UI. The alternative model instead introduces heterogeneous severance pay, motivated by evidence that firms—particularly larger firms—often offered severance packages to older workers exiting employment via the UI bridge. Workers with more generous severance packages have a higher outside option and are therefore more likely to leave employment. We relate this heterogeneity to workers’ pre-UI wages and firm size, measured at age 50, through auxiliary regressions, and additionally require the model to match the observed wage- and firm-size-at-age-50 patterns of UI entrants. Figure [H.13](#) shows that workers entering UI closer to the bridge age tend to have had higher wages and to work at larger firms, patterns that the alternative model reproduces. Despite imposing this additional discipline on who selects into UI, the model yields broadly similar policy takeaways.

V.C The Effects of UI Extensions Depend on Retirement Institutions

Several results so far suggest that the effects of the same UI extension might differ with the prevailing retirement policies, such as the earliest retirement age and the penalty for claiming early. We now probe this point more directly with two additional exercises.

First, to show that Germany’s PBD extensions would have mattered far less under less generous retire-

ment rules, we simulate a world in which the extensions are left in place but the early retirement pathway via UI is closed, so that age 63 rather than age 60 is the earliest age at which workers can claim a pension. Bridging into retirement via UI remains possible under this counterfactual, but now only at age 63 minus maximum PBD, for example, at 60 and 4 months for the 1935 cohort, when maximum PBD was 32 months, so it primarily affects UI inflows of workers aged 60 and up. The green dashed line in Figure 8 (a) shows the resulting behavior of the 1935 cohort: bunching now occurs at 60 and 4 months, and UI inflows and non-employment durations fall sharply for workers aged 60 and below. Combining simulations across all birth cohorts, Figure H.14 shows that closing the retirement via UI pathway would have had a dramatic effect on the unemployment rates of workers aged 56-59 prior to 2006. Had it not been available to pre-1946 cohorts, older workers' unemployment rates would have been far closer to those of younger workers (6.9% instead of 16% in 1994), *despite Germany's large PBD extensions*.⁴¹ Retirement institutions clearly shape the effects of UI extensions on workers at various ages.

As a final way to reinforce the dependence of UI effects on retirement policy, Table 4 considers how the effect of the same UI extension in 2014, holding all else equal, would differ under different retirement policies. Under the actual 2014 institutions (column (1)), extending PBD by 12 months increases the unemployment rate of workers in their early 50s by 0.34pp and of workers in their late 50s by 0.90pp, implying elasticities with respect to PBD of 0.28 and 0.45, respectively. The intensive-margin effect $\frac{\partial Nonemp}{\partial P}$ is 0.13 at age 52 and 0.17 at age 57. Column (2) reintroduces the UI retirement pathway, allowing retirement at age 60 with the early retirement penalty, keeping all else constant. The effect of the same UI extension on the unemployment rate of older workers now almost doubles (from 0.90 to 1.62pp) and the elasticity rises from 0.45 to 0.61. Column (3) shows that these patterns are even more pronounced if we reintroduce the UI pathway *and* repeal the penalty for retiring early (thus returning to the pre-2000s retirement institutions), with the same UI extension now increasing the unemployment rate by a full 3.01pp.⁴²

V.D Fiscal Externalities and the Marginal Value of Public Funds for UI Extensions

The preceding findings suggest that a welfare evaluation of UI extensions for older workers might look substantially different from one for younger workers, at least under certain retirement regimes. To formalize this, we turn to the Marginal Value of Public Funds (MVPF) framework of Hendren and Sprung-Keyser (2020) and calculate the fiscal externality (FE) and the MVPF associated with a UI extension for workers aged 50 and above.

The MVPF of a policy that spends an extra dollar on UI by extending PBDs is the willingness to pay of the recipients divided by the net cost: $MVPF_{PBD} = \frac{\sum_i WTP_{PBD}^i}{NetCost}$ (Hendren and Sprung-Keyser, 2020). Using the framework in Le Barbanchon et al. (2024), the willingness to pay for an extra dollar of PBD

⁴¹The convergence in the unemployment rate after 2006 stems from the fact that the ERA via the UI pathway increased from age 60 to 63 between the 1946 (aged 60 in 2006) and 1948 birth cohorts, and was formally closed starting with the 1952 cohort.

⁴²Interestingly, $\frac{\partial Nonemp}{\partial P}$ at age 57, which captures an intensive-margin effect, actually declines relative to column (1), since in this policy environment many of the workers displaced at age 57 would not have returned to the labor market even prior to the PBD extension. Despite this, the overall elasticity of the unemployment rate with respect to PBD increases relative to column (1) due to the importance of extensive margin, inflow effects.

spending, transferred to UI exhaustees (such that $t > P$), is the marginal rate of substitution between the relevant unemployed and employed states: $WTP^{PBD} = \frac{u'(c_{u,t>P})}{v'(c_e)} = 1 + \frac{u'(c_{u,t>P}) - v'(c_e)}{v'(c_e)}$, where $u(\cdot)$ is the utility of consumption while unemployed and $v(\cdot)$ while employed. Following [Le Barbanchon et al. \(2024\)](#), we set $WTP^{PBD} = 1.36$ as our numerator throughout. This comes from using [Kolsrud et al. \(2018\)](#)'s estimate of a 9.1% average consumption drop for the long-term unemployed together with a CRRA utility function with $\gamma = 4$.⁴³ Our focus is on estimating the denominator for the older workers in our setting. Since behavioral responses generate a negative fiscal externality, the denominator is $NetCost = 1 + FE$. Prior work has estimated the net costs of PBD extensions for prime-age workers across several different European settings, including Germany, under the assumption that there are no extensive margin responses ([Schmieder and von Wachter, 2016](#), also reported in [Le Barbanchon et al. 2024](#)). This work estimated fiscal externalities for German prime-age workers between 0.38–0.42, or a net cost of about 1.41; the mean net cost across all European studies is 2.31. Dividing the WTP by these net costs yields an average MVPF of 0.967 for Germany and 0.691 across all European studies (Table I.8).

How does this picture change for older workers once we allow PBDs to affect UI inflows? For each birth cohort, we compare the baseline simulation under actual PBD rules with a counterfactual that increases PBD by one period (3 months). We calculate the mechanical cost by applying the extended benefit schedule to baseline UI inflows and baseline non-employment survival. Thus, only spells that would continue beyond the original exhaustion date generate mechanical payments. We then re-solve the model under the longer PBD, allowing both UI inflows and non-employment durations to adjust. Let dB denote the resulting change in UI benefit payments, dR the associated loss in labor-tax revenue, and $MC = dB|_{mechanical}$.⁴⁴ The behavioral cost is $BC = dB + dR - MC$, so $FE = \frac{BC}{MC} = \frac{dB + dR - dB|_{mechanical}}{dB|_{mechanical}}$. Table I.9 reports these cohort-level components and provides further details on the aggregation and fiscal calibration.

We estimate large fiscal externalities for PBD extensions among older workers. We focus first on the 1935 cohort, which faced the most generous retirement and UI schemes we study as well as relatively high and rising unemployment. We obtain a fiscal externality of 4.62, for a net cost of 5.62, almost four times the net cost for prime-age workers from prior work in the same context, implying an MVPF for older workers of 0.242, compared to 0.967 for younger workers.⁴⁵ Figure H.17 places these numbers alongside other social programs ([Hendren and Sprung-Keyser, 2020](#)) and US and EU averages for UI (documented in Table I.8). UI extensions for older workers in our context stand out for their high net cost and low MVPF.

To highlight the role of inflow responses, we next calculate the fiscal externality when shutting them down. Taking the 1935 cohort under actual PBD policies, we hold UI inflows at each age fixed at baseline

⁴³We use this as the baseline to facilitate comparisons with prior estimates. Using the same 9.1% consumption drop under log utility (as in our structural model) would imply a WTP of approximately 1.10, and hence somewhat lower MVPFs.

⁴⁴An important input for these calculations is the effective tax rate on the employed, which determines the revenue loss from non-employment. As [Lawson \(2017\)](#) pointed out, using UI taxes alone severely understates this loss. We therefore follow [Schmieder et al. \(2016\)](#) and [Hendren and Sprung-Keyser \(2020\)](#) and use the average tax wedge on labor.

⁴⁵While the WTP for older workers might differ, evidence from the U.S. suggests consumption losses at unemployment are smaller for older workers ([Michelacci and Ruffo, 2015](#)), and older workers entering UI at the bridge age in our setting have slightly higher pre-UI wages (Figure H.13) and may receive severance pay; if anything, these would suggest using a lower WTP for older workers and hence an even lower MVPF.

and increase non-employment durations at each entry age by the amount implied by our preferred RD estimate of $\frac{\partial D}{\partial P} = 0.128$ at age 52, applied to the change in PBD at each age. This isolates the intensive-margin duration response, yielding FE components comparable to the calculations for prime-age workers in [Schmieder and von Wachter \(2016\)](#). Under this exercise, the behavioral cost is driven entirely by longer non-employment spells among *existing* UI entrants (through lost tax revenue and extended benefit receipt), while the mechanical cost reflects the additional benefit payments to those who would have exhausted their benefits even without behavioral changes. We obtain a far lower net cost of 1.90 (Table [I.9](#)), much closer to the 1.41 for prime-age workers.⁴⁶ Hence, much of the fiscal externality arises from changing UI inflows, implying that counterfactuals that only alter survival functions, like [Johnston and Mas \(2018\)](#), would not be very accurate for older workers in this context.

Because behavioral costs vary with retirement institutions and contextual factors like the macroeconomic unemployment rate, fiscal externalities also differ across cohorts. Figure [H.18](#) reports FE and the associated net costs for all cohorts. Net costs are always substantially above those of younger workers in the same context and those obtained when shutting down inflow effects. But they do vary: net costs started out lower for the 1929 cohort, when PBD was still short at most ages and exhaustion rates were high, peaked above 5 for cohorts in the mid 1930s facing large PBD, generous retirement, and high unemployment, and fell to around 3 as retirement rules made UI entry in one’s 50s less attractive and unemployment fell. Altogether, our evidence suggests that the non-employment effects of UI extensions can be large and fiscally expensive for older workers, especially under the right institutional and macroeconomic conditions.

VI Conclusion

We specify a dynamic life-cycle labor supply model that explicitly accounts for transitions between employment, unemployment, and retirement and how they are affected by the structure of UI benefits and parameters of the old-age pension system. We estimate this model using empirical moments of the German labor market for forty birth cohorts under widely varied policy regimes. The model shows that unemployment insurance extensions played a large role in explaining Germany’s remarkable historical rise in the unemployment rate of older workers. The model also suggests that changes to both the UI and retirement systems played an important and underappreciated role in the German “labor market miracle” after 2005 by reducing older workers’ unemployment rates. These large policy effects arise because retirement and UI institutions interact — the retirement policies that prevailed for older workers in the 1990s encouraged UI inflows among older workers and heightened the behavioral impact of additional UI extensions. In contrast, later retirement rules made retirement via UI less attractive. Altogether, our work suggests that changes in retirement institutions can meaningfully alter the effects of UI extensions and that the fiscal externalities associated with UI extensions can, under the right conditions, be substantially larger than typically understood.

⁴⁶The remaining gap is driven by the fact that older workers in this cohort have much larger baseline PBDs than younger workers with similar non-employment durations, which lowers the UI exhaustion rate and therefore the mechanical cost of a policy change.

An advantage of our setting is that we can study how changes in UI and retirement rules alter the effects of UI within the same broader context, but this also raises questions about what to expect in other contexts. We study a case with relatively generous UI rules and where firm- and sectoral-level labor agreements play important roles. The extent to which retirement institutions affect UI responses likely depends on the incentives of firms and other such contextual factors. For example, in a setting like the United States UI system which uses experience rating and generally does not allow for UI following quits, we might expect dampened bridge-to-retirement effects. Nevertheless, we suspect our general message that the design of retirement institutions can meaningfully alter the effects of UI to be relevant in other countries. Evidence consistent with UI inflow bunching into retirement (our key behavioral mechanism) has been documented in other European settings (e.g. [Kyyrä and Wilke, 2007](#); [Tuit and van Ours, 2010](#); [Baguelin and Remillon, 2014](#); [Kyyrä and Pesola, 2020](#)). While the fiscal externalities we estimate for older workers are substantially larger than what prior work has found for prime-age workers, they are comparable in size to those estimated by [Inderbitzin et al. \(2016\)](#), who study a PBD extension for those aged 50-57 in Austria. These findings, which reflect similar underlying economic behavior, strengthen our confidence that our broader conclusions are not confined to the German setting.

We conclude with several avenues for future work. We model separations as being efficient in that the joint surplus would turn negative if the employment relationship were to continue. [Jäger et al. \(2023\)](#) raise the possibility that this need not be the case. Exploring the nature of separations and the role that layoff protections, CLAs and works councils play would help to make progress on the normative implications of our findings. Further, we study only one of many inputs into optimal UI design, and our partial equilibrium approach leaves open several questions about possible general equilibrium effects. As one example, it is possible that, even though Germany's maximum PBD extensions in the 80s increased the non-employment duration of older workers, they might also have paved the way for younger workers to retain or get jobs while smoothing older workers' transitions out of employment. Last, other institutional changes besides retirement policies, such as changing UI benefit levels or post-UI welfare benefits, could also significantly alter the effects of UI extensions for all ages. Quantifying and modeling how much these other institutional changes matter could help policymakers better predict the effects of UI policy changes in new institutional environments.

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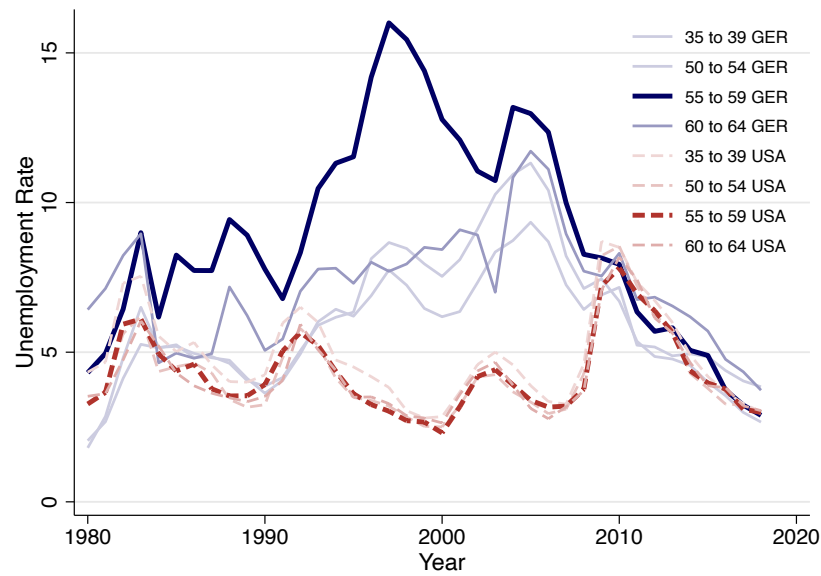
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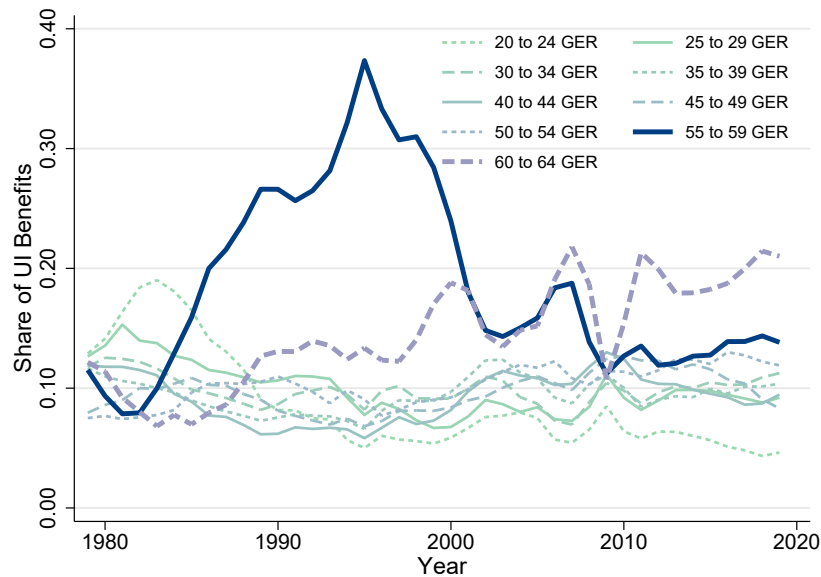
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Figures

Figure 1: Unemployment Rates and Share of Total UI Benefits by Age Group



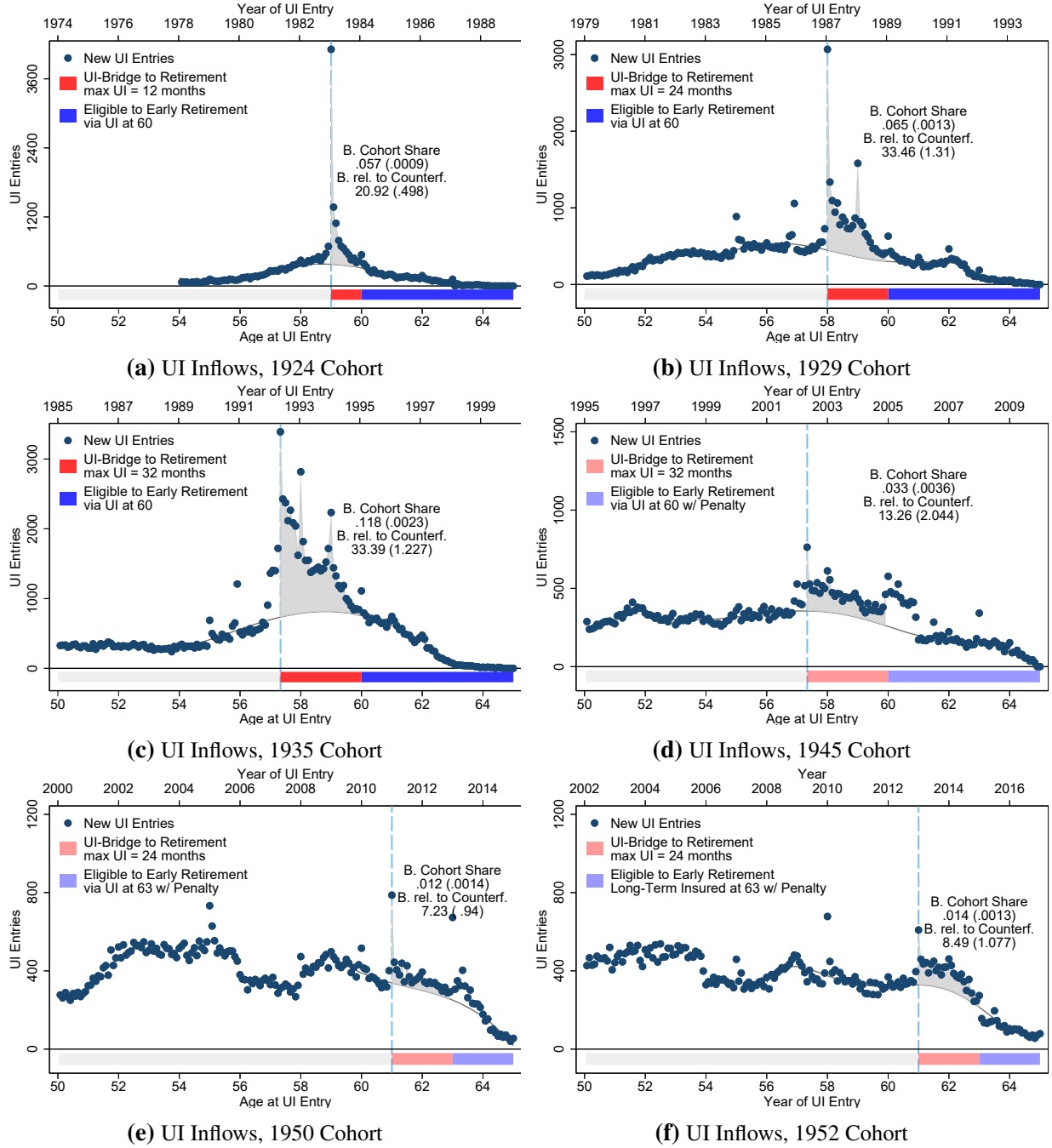
(a) Male Unemployment Rates by Age Group: West Germany and U.S.A



(b) Share of Total Benefits by Age Group in West Germany

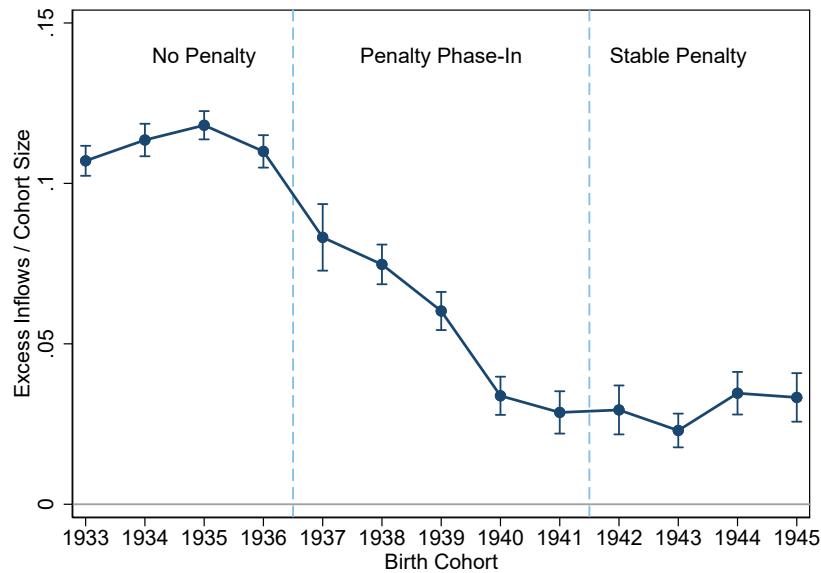
Notes: Panel (a) shows the male unemployment rate for select age groups in West Germany and the USA from 1980 until 2018, using data from the OECD. Panel (b) shows how UI payments are distributed across age and over time among West German men. Each line plots the share of UI payments in a given year that are paid out to UI recipients in the stated age group. In each year, the shares across all age groups add up to one. Results are based on own calculations using a 2 percent random sample of the Integrated Employment Biographies (IEB).

Figure 2: UI Inflows by Age for Different Cohorts in Germany, Men

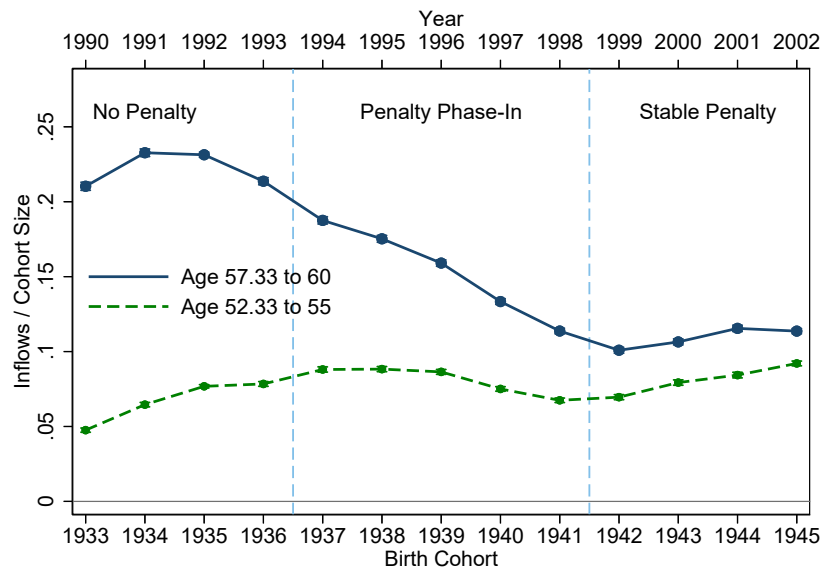


Notes: This figure plots the number of UI inflows per month (transitions from employment (E) to unemployment (UI or Nu)) by age at entry for different cohorts of West German men in our sample. The red shaded bar under each subfigure denotes time spent on UI if one starts receiving UI at the bridge-to-retirement age, indicated by the vertical dashed blue line. The blue bar denotes time spent receiving a pension if one starts receiving their pension as early as possible (via the UI pathway in panels (a)-(e), and via the long-term insured pathway in panel (f)). Lighter colors in panels (d)-(f) indicate periods where early pension receipt is penalized. The excess bunching mass (shaded in gray) is calculated as cumulative difference of the observed UI entries minus the counterfactual UI entries between the bridge-to-retirement age and the earliest possible pension age. For further details on the estimation of the counterfactual, see Appendix B. We report two different normalizations for this excess mass, the first relative to the number of individuals in the cohort-sample, and the second relative to the mean counterfactual. Standard errors (in parenthesis) are obtained via bootstrap, by resampling from the actual distribution with replacement 1000 times.

Figure 3: Evolution of UI Entries Around Phase-In of Retirement Penalty



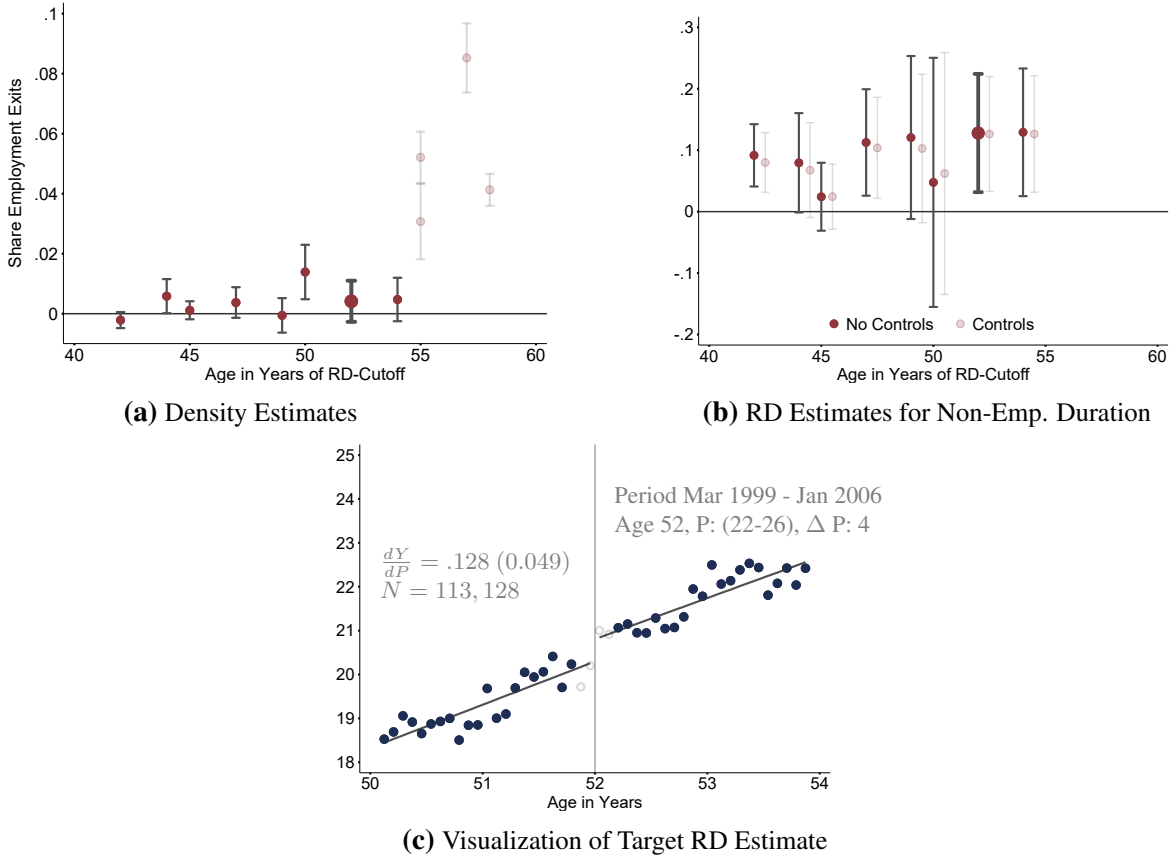
(a) Excess Bunching Mass Relative to Cohort Size



(b) UI Inflows Relative to Cohort Size

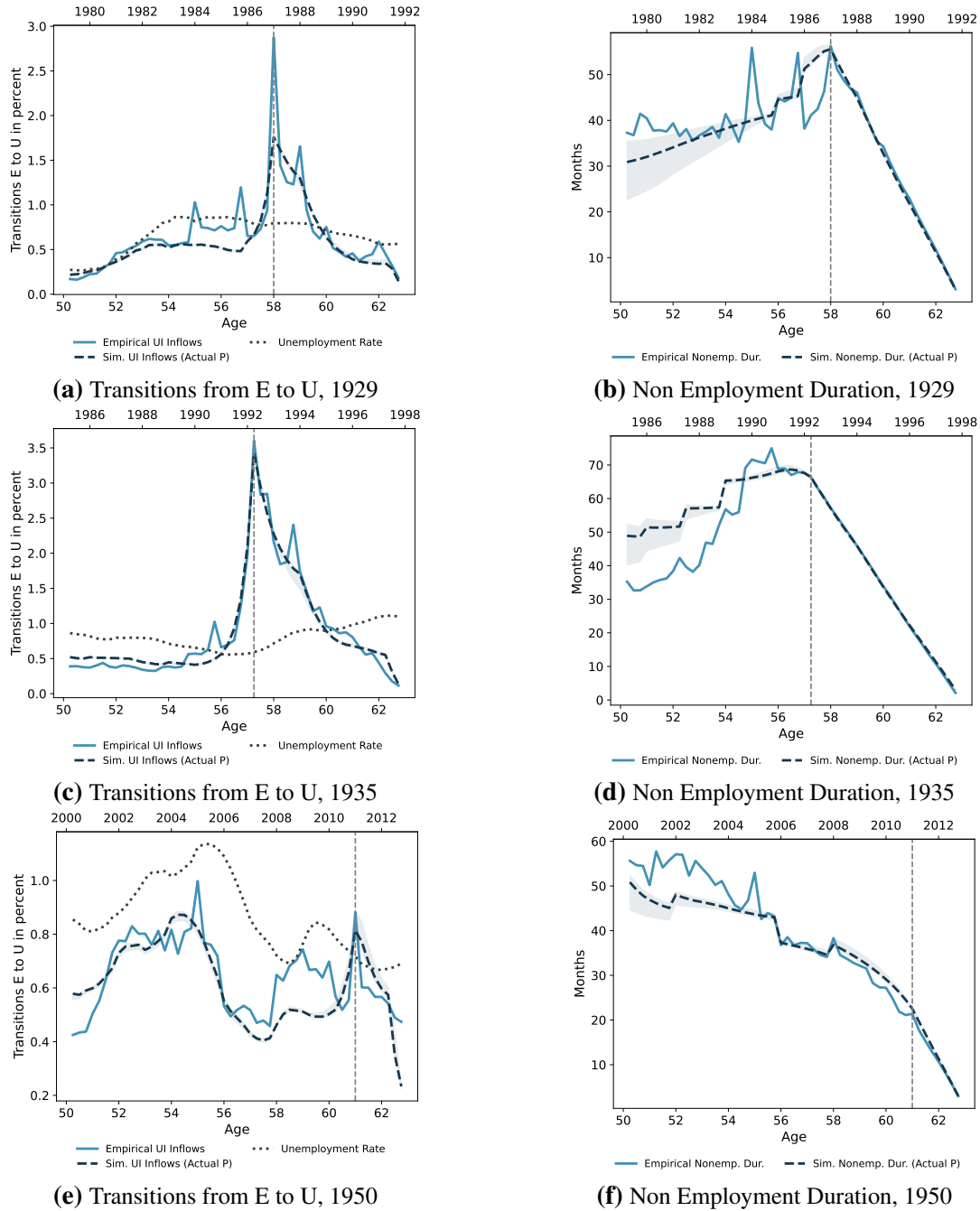
Notes: This figure shows the evolution of UI entries before, during and after the introduction of the penalty for early retirement via unemployment insurance. Panel (a) shows bunching estimates of the period-specific excess number of UI inflows relative to the number of individuals in the respective cohort. The excess mass is estimated using standard bunching techniques, with details laid out in Appendix B. 95% confidence intervals constructed via bootstrap with 500 replications are depicted. As an alternative to relying on bunching estimates, Panel (b) directly plots UI inflow shares (UI inflows divided by cohort size). The blue series plots the share of UI entries between age 57.33 and 60 for the stated birth cohort. The green series shows the share of UI entries between age 52.33 and age 55 for a 5-year younger cohort, so as to keep calendar time fixed. As such the birth cohort axis applies only to the blue series. 95% confidence intervals are depicted for each series in panel b), but are extremely small given these are based on raw counts in large data.

Figure 4: RD Estimates of the Effect of PBD Extensions on Non-Emp. Duration, Men



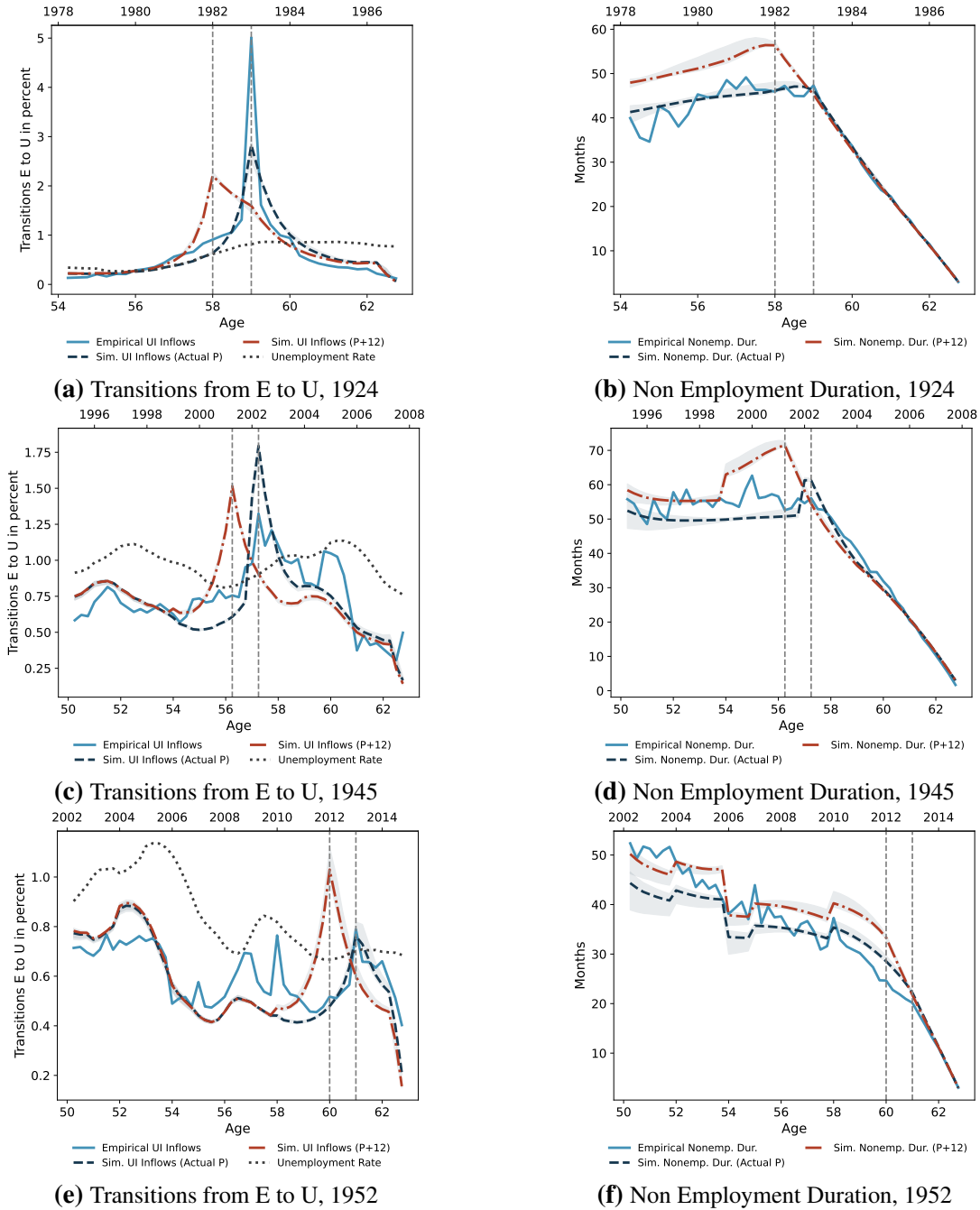
Notes: This figure plots RD estimates and corresponding density tests of the effect of a one-month PBD extension at each possible age cutoff. Panel (a) shows estimates of density discontinuities at each cutoff. The grayed out cutoffs at age 55 and above are cutoffs we exclude from our analysis due to the observed density discontinuities. The age 52 cutoff is highlighted in bold. Panel (b) plots RD estimates of an extra month of PBD on months spent non-employed (capped at 36 months), with grayed out coefficients corresponding to estimates with controls. Both panels show 95% CIs. Panel (c) shows how mean non-employment duration (capped at 36 months) varies around the age 52 cutoff, between Mar 1999-Jan 2006, at which PBD is discontinuously extended by 4 months (from 22 to 26). The solid line shows the best linear fit on each side of the cutoff, omitting the closest 2 months on each side. The jump at the cutoff corresponds to our RD estimate. The RD estimate (scaled ΔPBD) is .128 with a standard error (clustered on the age-day level) of 0.049 based on a sample size of 113,128 observations. This RD estimate constitutes our target parameter in the structural estimation. The point estimates are slightly smaller than those in [Schmieder et al. \(2012\)](#), primarily because our sample includes men only. See Table I.3 for RD-estimates at all cutoffs.

Figure 5: In-Sample Fit of Life-Cycle Model for Transitions from Employment to UI and Non-Employment Durations (capped at age 63)



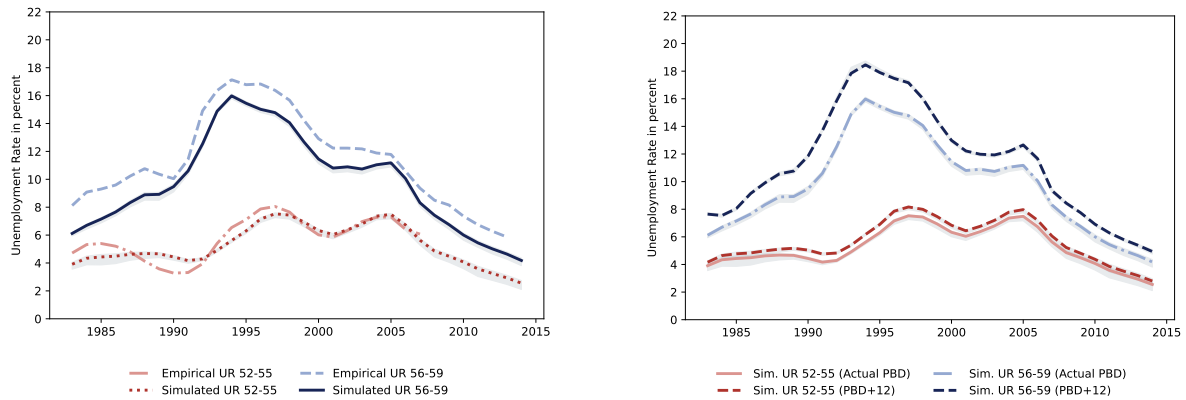
Notes: This figure compares our quarterly model-generated moments to their corresponding empirical moments for in-sample cohorts (1929, 1935, 1950). Panel (a) compares the transitions from employment to unemployment for the 1929 cohort whereas panel (b) compares non-employment durations for the 1929 cohort. Panels (c) and (d) show the same comparisons for the 1935 cohort, and panels (e) and (f) for the 1950 cohort. Non-employment duration is measured as time non-employed until age 63. Gray bands around simulated moments are 95% bootstrap confidence intervals, constructed by drawing target moments from a multivariate normal distribution with mean equal to the empirical moments and covariance equal to their estimated covariance matrix.

Figure 6: Out-of-Sample Fit of Life-Cycle Model for Transitions from Employment to UI and Non-Employment Durations, Baseline Model and Counterfactual 1: $P + 12$



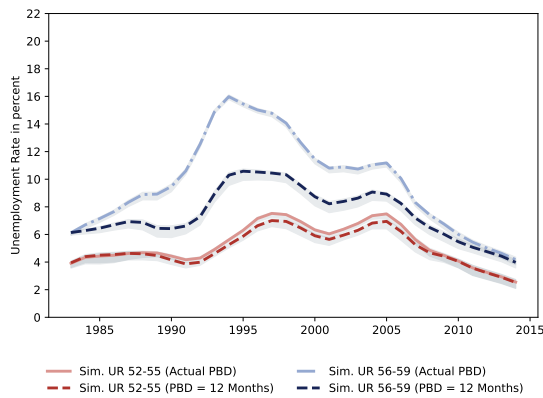
Notes: This figure compares our quarterly model-generated moments to their corresponding empirical moments for select out-of-sample cohorts (1924, 1945, 1952). Model-generated moments include the baseline specification (dashed blue line) and a counterfactual model where we increase potential benefit duration of UI by 12 months at all ages (dash-dotted red line). Panel (a) shows transitions from employment to unemployment for the 1924 cohort whereas panel (b) shows non-employment durations (until age 63) for the 1924 cohort. Panels (c) and (d) show the same comparisons for the 1945 cohort, and panels (e) and (f) for the 1952 cohort. Gray bands around simulated moments are 95% bootstrap confidence intervals.

Figure 7: Empirical and Simulated Unemployment Rates by Age Group

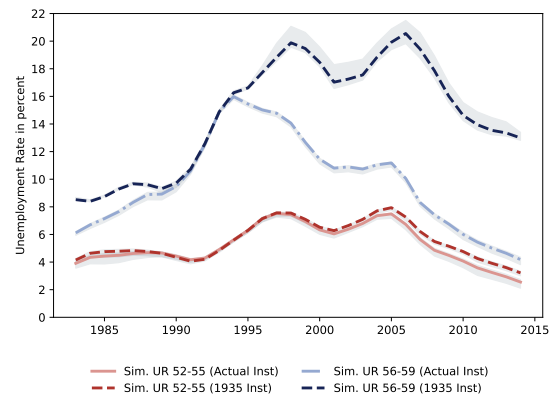


(a) Empirical and Model-Based Unemployment Rates

(b) Simulating a constant UI PBD increase of 12 months



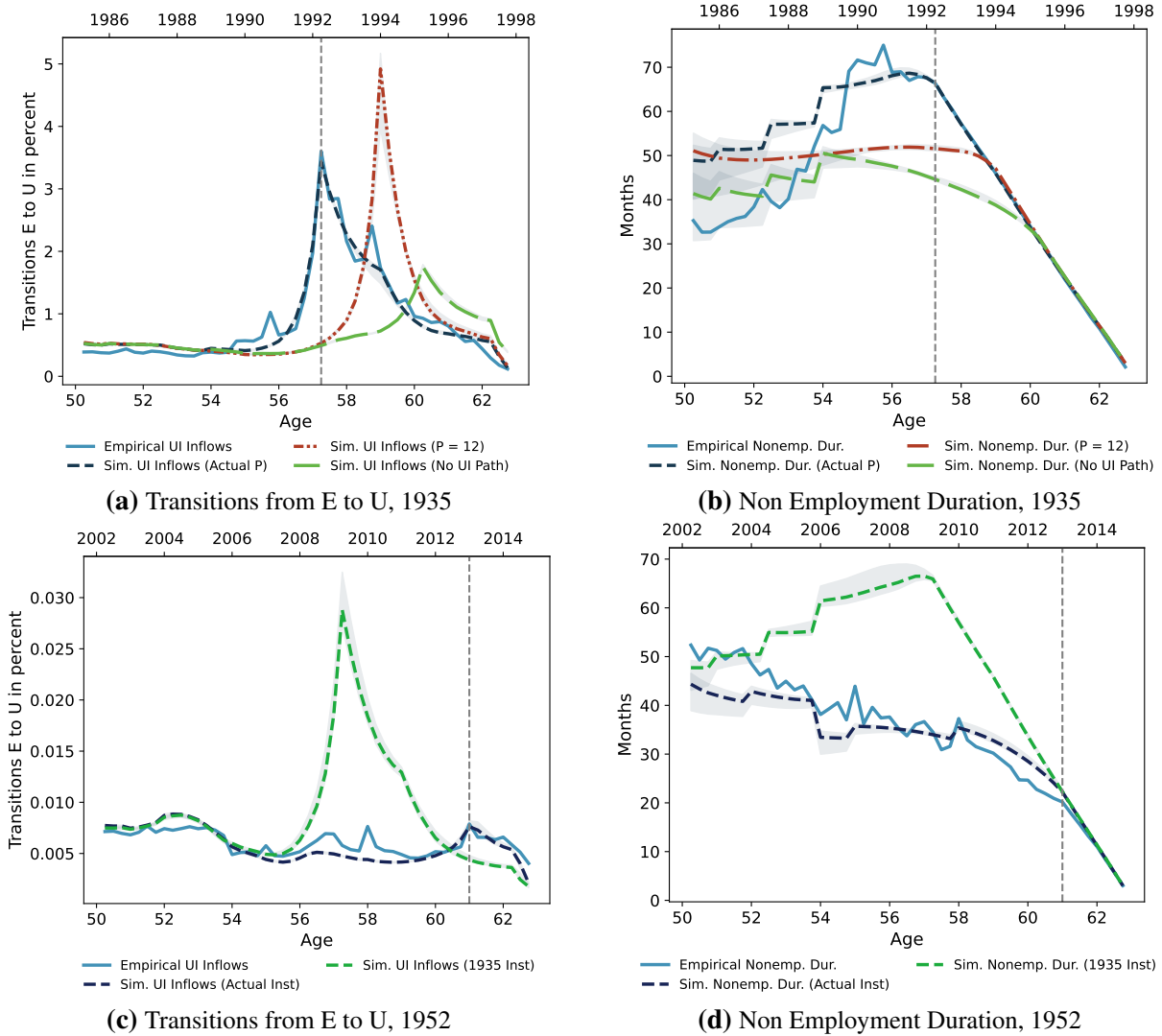
**(c) The Influence of Germany's PBD extensions:
Had PBD remained fixed at 12 months**



**(d) The Influence of Policy Changes after 1994:
Had UI and retirement policies remained fixed in 1994**

Notes: Panel (a) shows the empirical and simulated unemployment rate from the model for two age groups: 52-55 years old and 56-59 years old. Panel (b) shows the simulated unemployment rate under the actual institutions and the simulated unemployment rate when maximum potential benefit durations are increased by 12 months at all ages. Panel (c) shows the model-simulated unemployment rate at baseline (solid lines) and for a counterfactual in which PBDs never increased and instead remained fixed at 12 months (dashed lines). Panel (d) plots simulated unemployment rates for the baseline specification (solid lines) and a counterfactual model where we keep retirement rules (penalty, ERA, and NRA) and PBD fixed at 1994 levels (dashed lines). Gray bands around simulated moments are 95% bootstrap confidence intervals.

Figure 8: Model Simulations for Counterfactual Policies



Notes: To show how the policy counterfactuals we consider are working in our model, this figure compares our baseline model-generated moments to simulated moments under various counterfactuals for two illustrative cohorts (1935, 1952). In panels (a) and (b), for the 1935 cohort, model-generated moments include the baseline specification (dashed blue line), a counterfactual model where we keep PBD fixed at 12 at all ages (dash-dotted red line), and a counterfactual model where we leave PBD to evolve as it did in actuality but instead imagine that the UI pathway into retirement never existed (long-dashed green line). Actual empirical moments from the data are depicted in solid light blue. In panels (c) and (d), for the 1952 cohort, we compare the baseline model output (dashed dark blue line) to a counterfactual model (dashed green line) where we keep retirement rules (penalty, ERA, and NRA) and PBD fixed at their 1994 levels. Actual empirical moments from the data are depicted in solid light blue. Gray bands around simulated moments are 95% bootstrap confidence intervals.

Tables

Table 1: Institutional and Estimated Model Parameters

Panel A: Institutional Parameters (Men)						
	1924	1929	1935	1945	1950	1952
Statutory retirement age	65	65	65	65	65+4/12	65+6/12
ERA (earliest possible) for long-term insured*	63	63	63	63	63	63
NRA (no penalty) for long-term insured	63	63	63	65	65+4/12	65+6/12
Penalty for retire at ERA for long-term insured	0	0	0	0.072	0.084	0.09
ERA (earliest possible) via UI	60	60	60	60	63	-
NRA (no penalty) via UI	60	60	60	65	65	-
UI Bridge Age	59	58	57+1/3	57+1/3	61	61**
PBD at ERA via UI bridge age	12m	24m	32m	32m	24m	24m**
UI replacement rates on net wages at UI bridge age	0.63	0.63	0.63	0.60	0.60	0.60
Conversion rate to UI replacement rate on gross wages	0.65	0.65	0.65	0.65	0.65	0.65
Pension replacement rates per year of contribution on gross wages	0.0104	0.0104	0.0100	0.0099	0.0095	0.0094
Pension contribution years at age 54 cond. on being emp. at 50	32.5	32.5	32.8	31.8	31.6	31.1
Penalty for retiring at the ERA via UI	0	0	0	0.18	0.072	-

Panel B: Estimated Model Parameters												
σ	Search Costs					Job Destruction			Disutility of Work			
	k_0	k_1	k_2	k_3	γ	λ_1	λ_2	λ_3	$\bar{\eta}_{1929}$	$\bar{\eta}_{1935}$	$\bar{\eta}_{1950}$	η_{SD}
2.18	0.61	5.66	1.12	51.27	0.82	-6.61	0.17	0.15	0.42	0.73	0.75	0.51
(0.10)	(0.06)	(0.12)	(0.26)	(5.65)	(0.02)	(0.02)	(0.00)	(0.01)	(0.04)	(0.01)	(0.04)	(0.05)

SSE = 18,302

Notes: Panel A outlines the institutional parameters used in our structural model for our 6 focal birth-year cohorts based on the Sozialgesetzbuch (SGB) Sechstes Buch (VI) (see Appendices D and F for more details). *Individuals were eligible for the long-term insured pathway after 35 years of retirement contributions. **The old-age pension for unemployment pathway is abolished for cohorts born in 1952 and after; therefore, the bridge age via UI here refers to the age at which individuals can take the full UI and then transition directly into receiving a pension for the long-term insured. Panel B shows the parameter estimates for the baseline model. σ is the standard deviation of the productivity shock. k_0 is the fixed cost of job search; k_1 the cost of UI entry; k_2 the exponential time trend in search cost; k_3 the slope parameter of search cost; γ the elasticity of search cost. The $\bar{\eta}$ s are the cohort-specific mean of the disutility of work distribution and η_{SD} is the common standard deviation of the disutility of work distribution. Standard errors are reported in parentheses and are calculated by bootstrapping by drawing sample moments from a multivariate normal distribution with a mean equal to the sample moments and a covariance matrix equal to the sample covariance matrix.

Table 2: Quantifying the Effects of Germany's UI and Retirement Policy Changes on Unemployment Rates

	(1)	(2)	(3)	(4)
	Age 52-55	Age 56-59	Age 60-62	Age 52-62
Unemployment Rate				
1983, Actual Inst.	3.9%	6.1%	10.3%	6.7%
1994, Actual Inst.	5.6%	16.0%	12.7%	11.3%
1994, Actual PBD+12	6.2%	18.5%	12.9%	12.5%
1994, PBD=12	5.3%	10.3%	11.8%	8.9%
1994, No UI Path	5.3%	6.9%	6.9%	6.3%
2014, Actual Inst.	2.6%	4.2%	5.5%	4.0%
Change in UR from 1983 to 1994				
Overall change	1.7 pp	9.9 pp	2.4 pp	4.6 pp
Change due to PBD change	0.4 pp	5.7 pp	0.9 pp	2.4 pp
Change due to other reasons	1.3 pp	4.2 pp	1.5 pp	2.2 pp
Change in UR from 1994 to 2014				
Overall change	-3.1 pp	-11.8 pp	-7.2 pp	-7.4 pp
Change due to PBD and Retirement Policies	-0.7 pp	-8.8 pp	-9.3 pp	-6.0 pp
Change due to other reasons	-2.4 pp	-3.0 pp	2.1 pp	-1.4 pp
Change due to PBD change	-0.2 pp	-2.9 pp	0.0 pp	-1.1 pp
Change due to UI ERA change	-0.4 pp	-7.8 pp	-5.1 pp	-4.4 pp
Change due to penalty	-0.2 pp	-5.4 pp	-7.1 pp	-4.0 pp

Notes: The table shows model-generated output (levels and changes in unemployment rates) for the different counterfactual policy simulations discussed in Section V. Results are presented for four age groups: age 52-55, age 56-59, age 60-62, and age 52-62 in columns (1)-(4), respectively. pp stands for percentage point changes.

Table 3: Robustness to Alternative Model Specifications

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Baseline Men	Diagonal Weighting	No Upweighting	No Cyclical Shocks $\lambda_2, \lambda_3 = 0$	Constant Disutility η	Higher UA	Higher Home Production	Retirement Norms	Severance Pay
Model Fit									
SSE	18,302	15,195	18,240	32,263	27,122	17,955	20,930	18,279	43,084
dD/dP age 52	0.127	0.129	0.111	0.127	0.128	0.127	0.131	0.125	0.180
Number of Parameters	13	13	13	11	11	13	13	14	14
Unemployment Rate (Age 56-59)									
1983, Actual Inst.	6.1%	6.2%	6.2%	6.4%	5.5%	6.2%	6.0%	6.1%	5.1%
1994, Actual Inst.	16.0%	16.2%	16.2%	15.8%	13.1%	16.3 %	15.9%	16.0%	12.9%
1994, Actual PBD+12	18.5%	19.0%	18.7%	18.2%	20.1%	19.0%	17.8%	18.4%	18.3%
1994, PBD=12	10.3%	10.2%	10.3%	10.1%	6.8%	10.3%	10.5%	10.3%	6.2%
1994, No UI Path	6.9%	6.7%	7.0%	6.4%	6.5%	7.0%	7.4%	6.8%	7.6%
2014, Actual Inst.	4.2%	4.3%	4.2%	4.3%	4.4%	4.4%	4.3%	4.2%	3.7%
Change in UR (Age 56-59) from 1983 to 1994									
Overall change	9.9pp	10.0pp	10.0pp	9.3pp	7.6pp	10.1pp	9.8pp	9.9pp	7.8pp
Change due to PBD change	5.7pp	6.0pp	5.9pp	5.7pp	6.3pp	6.0pp	5.4pp	5.7pp	6.7pp
Change due to other reasons	4.2pp	4.0pp	4.1pp	3.6pp	1.3pp	4.1pp	4.5pp	4.2pp	1.1pp
Change in UR (Age 56-59) from 1994 to 2014									
Overall change	-11.8pp	-11.9pp	-11.9pp	-11.4pp	-8.7pp	-11.9pp	-11.5pp	-11.8pp	-9.2pp
Change due to PBD and Ret. Policies	-8.8pp	-11.1pp	-8.2pp	-8.8pp	-3.9pp	-10.5pp	-6.2pp	-8.6pp	-3.6pp
Change due to other reasons	-3.0pp	-0.8pp	-3.7pp	-2.6pp	-4.9pp	-1.5pp	-5.3pp	-3.1pp	-5.7pp
Change due to PBD change	-2.9pp	-3.9pp	-2.8pp	-2.8pp	-2.6pp	-3.6pp	-2.1pp	-2.9pp	-2.2pp
Change due to UI ERA change	-7.8pp	-9.9pp	-7.3pp	-7.8pp	-3.2pp	-9.0pp	-5.3pp	-7.7pp	-2.5pp
Change due to penalty	-5.4pp	-7.2pp	-5.1pp	-5.2pp	-3.3pp	-8.4pp	-5.2pp	-5.3pp	-2.6pp

Notes: The table shows key simulation results for alternative samples and models. Column (1) shows the results for the baseline specification for men for ease of comparison. Column (2) estimates the model using a diagonal weighting matrix. Column (3) presents results for a specification that does not up-weight the intensive-margin RD-moments at age 52 compared to other moments. Column (4) sets the parameters λ_2 and λ_3 that are meant to capture cyclical shocks to zero. Column (5) provides results where the disutility of work parameter is set constant across all cohorts. Column (6) increases the value of second-tier unemployment benefits (UA) to a higher value from 500 to 750 Euro per month. Column (7) increases the value of home production from 50 to 500 per month. Column (8) includes a retirement norm parameter that adds additional utility for exiting the labor force at an exact retirement age (ERA or NRA). Column (9) shows results from a model where workers are offered a severance pay at the time of leaving a job in the form of an annuity that pays while either being unemployed or OLF. This model replaces heterogeneity in the disutility η with heterogeneity in severance pay and estimates a standard deviation of severance pay across individuals and cohort specific mean severance pay. This model also targets wage heterogeneity in the form of wage-at-age-50 moments.

Table 4: Illustrating Interactions between UI and Retirement Policies

	(1)	(2)	(3)
	Baseline Institutions in 2014	UI Pathway ERA at age 60 w/ Penalty	UI Pathway ERA at age 60 No Penalty
Age 52-55			
UR, Actual PBD.	2.79%	3.01%	3.30%
UR, PBD + 12	3.13%	3.42%	3.70%
Change in UR	0.34 pp [0.29, 0.38]	0.41 pp [0.36, 0.46]	0.40 pp [0.35, 0.45]
dD/dP at age 52	0.130 [0.127, 0.133]	0.150 [0.148, 0.152]	0.162 [0.160, 0.163]
Elasticity of UR w.r.t PBD	0.28	0.32	0.29
Age 56-59			
UR, Actual PBD.	4.61%	5.82%	11.04%
UR, PBD + 12	5.51%	7.44%	14.06%
Change in UR	0.90 pp [0.80, 1.01]	1.62 pp [1.45, 1.79]	3.01 pp [2.74, 3.28]
dD/dP at age 57	0.177 [0.174, 0.179]	0.180 [0.177, 0.183]	0.147 [0.138, 0.153]
Elasticity of UR w.r.t PBD	0.45	0.61	0.60

Notes: The table reports model-simulated effects of PBD extensions in the year 2014, first leaving institutions as they were in 2014 and then for counterfactual scenarios in which retirement rules are more generous (but all else is held constant). Numbers in brackets are 95% confidence intervals. Column (1) shows simulation results (UR under regular PBD and under PBD+12 months, dD/dP , and the elasticity of UR with respect to PBD) from our model for the actual institutional and other parameters in 2014. To simplify matters and for comparability purposes, we set $P=24$ for all ages considered at baseline, so that $P + 12$ corresponds to 36 months PBD for all ages. Column (2) shows the same results but alters institutions to allow for retirement via UI at age 60 with penalty (relative to the actual NRA). Column (3) is like (2) but also eliminates the penalty for retiring at 60.

Appendix For Online Publication

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A Data Appendix for Cohort Data

Data We use German Social Security data – the Integrated Employment Biographies (IEB) – from the Institute for Employment Research. This data provides detailed information about employment start and end dates, earnings, unemployment insurance spells, and various demographic characteristics for the years 1975 to 2017. We use IEB-Versions v14.00 and (especially for the later cohorts) v16.00.

Sample Selection We use the labor market history of selected birth-years to track individual labor market dynamics when approaching retirement age. Each birth year is called a cohort which we construct separately for men (who make up about 60% of individuals meeting all other sample restrictions) and women (who make up around 40%). We study all birth-year cohorts between 1924 and 1963. For illustration purposes, we highlight cohorts that i) represent periods of different UI generosity at older ages and ii) are not directly affected by a UI reform close to retirement. These focal cohorts are 1924, 1929, 1935, 1945, 1950, and 1952. The relevant institutional features faced by each cohort are summarized in Table 1 Panel A with full details on UI and retirement policies and reforms shown in Tables I.1 and I.2. For each of these cohorts we select all individuals with a stable employment history on their 50th birthday. Specifically, we select individuals that are in social security reliable employment on their 50th birthday and have at this point worked in regular social security reliable employment continuously over the previous three years without any UI receipt during this period which amounts to 60% of all individuals observed in any type of employment.⁴⁷ In addition, we exclude some industries that are known for having special early retirement practices. Namely we exclude mining and steel construction (about 5% of workers after applying the stable employment restriction). For cohorts 1937 and later we exclude additional industries that have excess exits from employment at age 55 in the 1941 cohort based on visual inspection (about 15% of workers). This approach should partially screen for industries with CLAs that specified an early retirement agreement at age 55. In particular we exclude the following three digit industry codes based on the 2008 industry classification: 291 (manufacturing of cars), 201 (production of base chemicals), 351 (electricity supply), 701 (business administration), 234 (production of other porcelain and ceramics), 642 (holdings), 212 (production of other pharmaceuticals), 204 (production of cleaning and toilet products), 192 (petroleum refinement) and 262 (production of data processing devices).

States and Transitions for a Monthly Balanced Panel We generate a monthly balanced sample of each birth cohort that tracks an individual’s labor market status since age 50.⁴⁸ We center the data around the cohort- and individual-specific bridge to retirement age, so that the first month after the bridge to retirement age starts with the exact date an individual faces a bridge to retirement.

For all months, we assign individuals to one of five exclusive labor market states. Individuals can be employed (E), which includes all social security reliable employment, or in registered unemployment (UI),

⁴⁷For the 1924-1927 cohorts we start later, at their 54th-51st birthdays, respectively, due to not having data prior to 1975 and requiring 3 years to establish stable employment.

⁴⁸We also generate a complementary quarterly panel that we use in the structural estimation.

which consists of all periods of UI receipt. In addition, individuals can be outside of the observed E and UI states.⁴⁹ We distinguish between non-observed unemployment (Nu), which entails up to 3-month interruptions between E and U , and temporary withdrawal from the labor force (Nt), which includes temporary employment interruption as well as interruptions between E and UI lasting longer than three months. Finally, individuals can withdraw permanently from the labor force (Np), denoted by an exit from E or UI that is not followed by any other E or UI spell in our data. If individuals are in multiple states in a given month – due to the transition date being in the middle of the month – we select one state with the rule that UI is preferred over Nu which is preferred over E , which is preferred over Nt and Np . If an individual has, for example, an employment spell (E) in the first half of a month and an Nu spell in the second half of the month, the individual is assigned Nu for the month. We construct all possible transitions between states where a transition is defined by comparing the current and previous state of an individual.

For simplicity, we later condense these five states into three: Employment (E), Unemployment (UI or Nu), and Non-Employment (Nt or Np). The main reason for combining Nu and UI , is that if workers are sanctioned at the beginning of an UI entry, they would appear as Nu in the data and the relevant transition from work to unemployment occurs at the E to Nu transition.

B Bunching Estimation Details

We estimate the excess bunching mass using standard bunching techniques (Kleven, 2016). We fit a polynomial to estimate a counterfactual distribution of monthly UI entries around each cohort’s UI -bridge age, leaving out the bridge period—the interval between the UI -bridge age (early retirement age minus potential benefit duration) and the earliest possible retirement age. When estimating the polynomial, we extend the leave-out region beyond the bridge period, i.e. below the UI -bridge age or above the earliest retirement age, using cohort-specific parameters. These extensions account for example for non-sharp bunching patterns in specific cohorts. We also exclude round ages with visible sharp bunching outside the regular bridge period. Table B.1 summarizes the period-specific parameters.

We calculate the excess mass as the cumulative difference between actual and counterfactual UI entries over the bridge period (shaded region in figures 2 and H.5). We report two normalized excess mass measures: First, the cohort-size adjusted excess bunching mass (CBM), calculated as excess mass divided by cohort size, estimates the share of each cohort entering UI in response to the UI -bridge. Second, the normalized bunching mass (NBM) divides the excess mass by the counterfactual distribution’s UI entries at the UI -bridge age. Standard errors are calculated via bootstrap resampling with replacement (number of draws are indicated in figure notes).

Our setting differs conceptually from other bunching applications because bunching may arise from two sources: individuals relocating their exit timing to the bunching region (creating missing mass elsewhere) or additional exits by individuals who would not have entered UI absent the UI -bridge but transition di-

⁴⁹This includes other states such as marginal employment or second-tier unemployment assistance that could sometimes be observed in the data as well as states that are genuinely never observed in the data, such as retirement.

rectly from employment to retirement. Given this possibility of additional UI-exits, we do not impose the restriction that the excess bunching mass equals a missing mass outside of the bunching region.

Table B.1: Bunching Parameters

Institutional Parameters																	
Cohort	'24	'29	'33	'34	'35	'36	'37	'38	'39	'40	'41	'42	'43	'44	'45	'50	'52
Ret. Age (yrs)	60	60	60	60	60	60	60	60	60	60	60	60	60	60	60	63	63
PBD (mo.)	12	24	32	32	32	32	32	32	32	32	32	32	32	32	32	24	24
UI-Br. Age (yrs)	59	58	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	57 $\frac{1}{3}$	61	61
Cohort-Specific Bunching Parameters																	
Poly. deg. (p)	5	5	5	5	5	3	5	5	5	3	5	5	5	5	5	3	5
BW below (mo.)	60	60	60	60	60	60	60	60	60	36	60	60	60	60	60	24	60
BW above (mo.)	60	60	60	60	60	36	60	60	60	60	60	60	60	60	60	48	48
L-O below (mo.)	3	3	0	0	3	18	30	0	3	30	3	3	3	3	3	0	3
L-O above (mo.)	3	3	0	3	3	9	3	3	3	12	3	12	3	3	12	9	3
Ages excl.	55,56,57	55,56,57	55,56,57	55,56,57	55,56,57	55,56	55,56	55,56	54,55,60	54,55,60	54,55,60	54,55,60	54,55,60	55	57	60,63	58,60

Notes: Cohort-specific bunching parameters for estimating counterfactual UI-entries and excess mass. Abbreviations: Ret.=Retirement, PBD=potential benefit durations, UI-Br.=UI-Bridge, Poly. deg.=degree of polynomial, BW=bandwidth, L-O=leave-out, Bunch.=bunching, Ages excl.=round ages that are excluded due to bunching outside of UI-bridge period.

C Additional Details and Results for the RD Specification

This section describes the sample used for the RD analysis, validity tests, the main findings, and associated robustness checks.

C.1 Data and Sample Construction

We construct an inflow sample into UI receipt based on the IEB, largely following [Schmieder et al. \(2012\)](#), with two main differences: First, we also include older individuals. Second, to be consistent with our cohort data, we also exclude individuals who were employed in mining or steel construction prior to job loss.

We select West German individuals that, based on their pre-UI history, are eligible for the maximum (age- and cohort-specific) potential benefit duration (PBD), as summarized in [Table I.1](#). In particular, we restrict to individuals who worked at least 12 months in a social security reliable job for the previous 3 years and also worked 52 months within the last 7 years with no intermittent UI spell in the previous 48 months. We further restrict to cases of UI take-up within 28 days after job separation. Our main sample restricts to male individuals, but we provide complementary evidence for females and for the pooled sample of men and women.

As [Table I.1](#) illustrates, the German UI system has had several historical periods with different age-specific PBDs. We select all age cutoffs below age 55 from the 1987 period onwards.⁵⁰ This leaves us with 8 age cutoffs from 3 periods. For the remaining cutoffs that seem to exhibit density violations — namely

⁵⁰We discard some earlier periods because of their short duration and some open questions regarding the implementation, especially unclear evidence for a first stage.

the age cutoffs 54 and 52 — we further exclude years from the end of the period where the violation is most severe. For the 54 age cutoff we exclude the last 5 years (07/1995 - 03/1999) of the period 07/1987-03/1999, for the 52 age cutoff, we exclude the last year (04/2005-01/2006) in period 03/1999-01/2006.

Differences from the Primary Analysis Sample The main differences between the RD sample and the primary analysis sample are two-fold: First, in the RD-sample we pool all cohorts that belong to one constant PBD regime in order to increase power instead of showing results for each cohort separately. Second, we apply somewhat different employment restrictions. In the RD-sample we set employment restrictions *at the time of separation* such that individuals would be eligible for full PBD at all of the age cutoffs.

Restricting to 12 months of social security contributions in the previous three years allows for more gaps in the recent history than the primary analysis sample, that restricts to three years of continuous working history. At the same time, the RD-sample has a larger requirement for total employment duration by requiring more than 4 years of working history in the previous seven years. An additional difference regarding the employment restriction is *when* it applies. In the RD-sample, the restriction applies at the time of job-separation, the time for determining UI eligibility. In the primary analysis sample in contrast the restriction applies at age 50 without reinforcing the restrictions later, which allows for a constant sample within cohort.

Outcomes Our main outcome is an individual's non-employment duration, measured as the duration in months between the start of UI receipt and the start of the next job. We topcode values above 36 to reduce the influence of outliers and to be consistent with prior work. In addition, we use several predetermined variables for balance checks and/or as control variables. In particular, we use the daily pre-UI wage, a dummy for foreign nationality, the years of education, years of firm-, industry-, and occupation- specific tenure as well as the time in months between job loss and UI claim.

Main Specifications For each cutoff, we estimate a separate RD specification. The main specification employs a two-year bandwidth on each side of the cutoff with the exception of the 49 and 54 age cutoffs, where it is only one year on the right due to other policy discontinuities above one year. Because of sorting, especially at some of the older cutoffs, we use a donut-hole approach and exclude 2 months just to the left and right of each cutoff. We control for a linear trend in the running variable which is allowed to differ on each side of the cutoff. We estimate the model via OLS, clustering standard errors at the age (in days) level. We also provide a range of robustness checks and alternative specifications discussed in the next section.

C.2 Description of Findings

Validity Checks Before turning to the main findings, we conduct balance and density checks. Figure H.6 explores the smoothness of the density around the cutoffs for men, plotting the number of UI entries by age separately around each cutoff. There is some evidence of sorting directly around the cutoffs, i.e. a missing mass directly left to the cutoff and an excess mass right to the cutoff. This sorting appears somewhat

stronger for older workers and females.⁵¹ Importantly, though, sorting is mostly restricted to the plus or minus 2 months on each side of the cutoff that are excluded in the main specification. There appears to be no or at most small evidence of a density shift for men.

To further quantify the presence (or absence) of a shift in a density, Column (1) of Table I.4 reports estimates of the marginal increase in the number of UI entries which is rescaled around the sample mean for each of the cutoffs to make the estimates more comparable between periods and cutoffs. For males, most estimates are precisely estimated and very close to zero. The strongest exception is the 50 cutoff in the most recent period, where the estimated increase in the density is about 1.5% relative to the mean. We also examine whether pre-determined variables are balanced across the cutoffs in columns (2) - (7) of Table I.4. In particular we check for balance in the daily pre-UI wage, a dummy for foreign nationality, years of education, years of firm-, industry-, and occupation- specific tenure as well as the time in months between job-loss and UI claim. Most estimates are insignificant and close to zero, with most estimates precise enough to rule out economically meaningful sorting along the dimensions considered. The one notable exception is a positive effect on pre-UI wages at the age 54 cutoff for both males and females and at the age 52 cutoff for females.

Main Findings Figure H.7 plots mean non-employment duration as a function of age so that our RD estimates can be inspected visually.⁵² The linear specification used on each side of the cutoff appears to be a reasonable approximation of the underlying conditional expectation.

Estimates of the effect of a one month increase in PBD on non-employment duration are reported in Table I.3. Column (1) shows the main results without controls and column (2) shows it with controls. Most estimates are in a similar ballpark as those in Schmieder et al. (2012), with estimated effect sizes for older workers tending to be slightly (though not statistically significantly) larger. For example, the baseline estimate at the age 42 cutoff in Period 07/1987-02/1999 implies an increase in non-employment duration of 0.092 months for an additional month of PBD (s.e.=0.026), whereas the estimated effect at age 54 is 0.129 months (s.e.=0.053). Adding controls barely moves the coefficients. If anything, the effect sizes tend to get a little smaller, though the differences are not statistically significant.

Additional Robustness We complement our findings with a number of robustness checks, reported in Table I.5. In particular, we examine the robustness to the inclusion of more granular controls including detailed industry and regional controls (Column (2)), extending the excluded area around the cutoff to 3 months (Column (3)), reducing the bandwidth to one year (Column (4)), and using a triangular kernel instead of a uniform one (Column (5)). Overall, our findings are relatively robust: most estimates are similar, or at least in the same ballpark, as the baseline estimates, though sometimes less precisely estimated.

⁵¹ Figures exploring the smoothness of the density for the women's sample and the pooled (men and women) sample are available upon request.

⁵² Figures plotting mean non-employment duration for the women's sample and the pooled (men and women) sample are available upon request.

D Additional Institutional Details

D.1 Pension Institutions and Pension Reforms in Germany

Over our sample period, several pension reforms altered the incentives to claim pension early and the various pathways into retirement. Table I.2 summarizes the reforms for all of the different pathways over our study period (elaborated upon below). There are six main pathways: Standard old-age pension, old-age pensions for the long-term insured, old-age pensions due to unemployment (and part-time work), old-age pensions for women, old-age pensions for disabled workers, old-age pension for especially long-term insured. The 5 non-standard pathways allow for early retirement under specific conditions. Each pathway has its own eligibility conditions, normal retirement age (NRA), or the age at which pension can be drawn without penalties, and early retirement age (ERA), the earliest possible age pension can be drawn.

Standard old-age pension: Workers can claim the standard old-age pension (SGB VI §235) at age 65 throughout our sample period. The eligibility condition is at least 5 years of contributions. The 2007 reform introduced a stepwise increase in the statutory retirement age from 65 to 67 for cohorts born between 1947 and 1964. These changes began in 2012 and will be complete in 2030 (See SGB VI §235(2)).

Old-age pension for long-term insured: The long-term insured pathway allows workers with at least 35 years of contributions to claim pension as early as age 63 (SGB VI §236). The NRA without penalty for early claims was 63 until the 1936 cohort. It was increased gradually, in monthly steps, from age 63 to 65 for cohorts 1937 to 1938 and remained at 65 until the 1948 cohort. The NRA was again increased to 65 and 3 months for the 1949 cohort and will increase at the same pace as the SRA for cohorts 1950 to 1964, reaching age 67 in 2030. The ERA, meanwhile, remained stable at age 63. Hence, workers eligible for this pathway could always claim as early as age 63, however they faced an actuarial adjustment in the form of a 0.3% permanent pension reduction per each month they retired in advance of the NRA.

Old-age pension due to unemployment or part-time work: Cohorts born before 1952 could claim pensions early via this pathway (SGB VI §237). The eligibility requirements for the UI pathway were: 1) at least 15 years of contributions, at least 8 of which must have occurred in the past 10 years, and 2) being unemployed for at least one year after the age of 58 and a half, or in old-age part-time work.⁵³ The ERA was 60 for cohorts younger than 1946 and then started to gradually increase, in monthly intervals, from 60 to 63 for cohorts 1946 to 1948. It then remained at age 63 until it was abolished for cohorts born in or after 1952 (SGB VI appendix 19). The NRA for claiming a pension without penalty was 60 until the 1936 cohort. It increased gradually from 60 to 65 between the 1937 and 1941 cohorts, and then remained at age 65 until this pathway was abolished.

Old-age pension for women: Women with at least 15 years of contributions, of which at least 10 must have occurred after age 40, were eligible for the women's pathway. The ERA remained at 60 throughout the sample period until this pathway was abolished for cohorts born in or after 1952. The NRA was 60 until

⁵³The part-time work component is granted by the partial retirement law (Altersteilzeitgesetz), which provided a maximum public subsidy for up to five years if older workers switch from full-time to part-time work. This program was enacted in the mid-1990s and was suspended in 2009.

the 1939 cohort, when it began to gradually increase, reaching 65 for the 1944 cohort (SGB VI appendix 20). The NRA then remained at age 65 until the pathway was abolished. Notice that these changes occurred later than those for the UI pathway, so that the women's pathway always offered early retirement on more generous terms.

Old-age pension for disabled workers: Workers who have lost their earnings capacity can claim the old-age pension for disabled workers. This pathway is also referred to as the invalidity pathway. Eligibility requires a loss of at least 50% of earnings capacity and at least 35 years of qualifying periods, which can include, for example, periods spent raising a child under age 10. It allows eligible, severely disabled persons to claim a pension before the statutory retirement age. The ERA for this pathway was 60 throughout the sample period and is scheduled to gradually increase to age 62 between the 1952 and 1963 cohorts. The NRA was 60 for workers born between 1920 and 1940. It was raised gradually by 1 month for each month of birth from 60 to 63 for cohorts 1941 to 1943, and remained at 63 until the 1951 cohort (SGB VI appendix 22). It is scheduled to gradually increase from age 63 to 65 for the 1952 to 1963 cohorts.

Old-age pension for especially long-term insured: The 2014 pension reform introduced the old-age pension for the especially long-term insured. Since July 2014, this pathway allowed workers with at least 45 contributory years to draw a pension without deductions as early as age 63. The first cohort that could use this pathway is the 1951 cohort. From birth cohort 1953 onwards, the NRA increases by two months for each birth cohort reaching 65 for persons born in 1964.

The last way for workers to leave the labor force and receive regular payments is via disability insurance. Disability insurance is available for workers with at least 5 years of contributions of which at least 3 need to be in the 5 years prior to claiming. Disability insurance can be claimed at any age. Workers who are officially recognized as having low earnings capacity, which entails permanently not being able to work more than 3 hours per day in any job, can claim disability insurance. For active DI recipients, benefits are converted into an old-age pension when they reach statutory retirement age. In Germany, the health assessment for disability insurance is relatively strict. About half of applications are rejected. Therefore, using disability pensions as a pathway to retire is difficult and typically not an attractive option.

D.2 Budget Set Calculations for Figure H.2

Here we detail how we calculate the lifetime budget constraints depicted in Figure H.2. Note that these are primarily used for illustrative purposes, though the structural model uses related components. We assume individuals earn a constant (after tax) wage w and at retirement receive total pension payments $y^R(E)$ and UI payments $y^{UI}(E)$, where E is age at exiting employment. Thus, the total years worked is $S = E - s$, where s indicates years of schooling.

This yields a budget constraint of the form

$$C = w(E - s) + y^{UI}(E) + y^R(E)$$

Let ρ be the replacement rate per year of pension contribution on net wage. In other words, each year of

work with wage of w will increase pension benefits $y^R(E)$ by ρw . Each year spent on UI increases pension benefits $y^R(E)$ by $0.8 \times \rho w$. We assume individuals take their full UI duration upon exit and then rely on UA until they retire at age T^R . For illustration purposes, we assume UA provides zero income. In the model, we will assume UA yields 500 per month (y^u) and workers spend $T^R - E - P$ on UA if there is a period without other income support before they can claim pensions.

The budget constraint is thus given by:

$$C = w(E - s) + \underbrace{bD + 0.8 \times \rho w D \times [T - \max\{T^R, E - s + T^u\}]}_{y^{UI}(E)} + \underbrace{\rho w(E - s) \times [T - \max\{T^R, E - s + T^u\}]}_{y^R(E)}$$

where D is UI duration, T^u is unemployment duration, and P is maximum potential UI duration, b is UI benefit level. By definition, $T^u = D \geq P$. The stylized budget sets in Figure H.2 assume that a worker always retire at the earliest possible retirement age ($T^R = ERA$).

Therefore,

$$C = Y = \begin{cases} w(E - s) + bP + \rho w \times (E - s + 0.8P) \times [T - T^R] & \text{if } E < T^R - P \\ w(E - s) + b(T^R - E) + \rho w \times (E - s + 0.8(T^R - E)) \times [T - T^R] & \text{if } E \geq T^R - P \end{cases}$$

$$\frac{dY}{dE} = \begin{cases} w + \rho w \times [T - T^R] & \text{if } E < T^R - P \\ w - b + \rho w(1 - 0.8) \times [T - T^R] & \text{if } E \geq T^R - P \end{cases}$$

If maximum potential UI duration P changes over the life cycle—for example, from P_1 to P_2 at age T^{RD} —the value of P just before the ERA determines the bridge age, $ERA - P_2$. The budget set is then:

$$Y = \begin{cases} w(E - s) + bP_1 + \rho w \times (E - s + 0.8P_1) \times [T - ERA] & \text{if } E < T^{RD} \\ w(E - s) + bP_2 + \rho w \times (E - s + 0.8P_2) \times [T - ERA] & \text{if } T^{RD} \leq E < ERA - P_2 \\ w(E - s) + b(ERA - E) + \rho w \times (E - s + 0.8(ERA - E)) \times [T - ERA] & \text{if } E \geq ERA - P_2 \end{cases}$$

When there exists a financial penalty to claim pension at ERA , we adjust the $y^R(E)$ by multiplying $(1 - (NRA - ERA) * 3.6\%)$.

Let's take the 1924 cohort as an example (where $P = 1$ and $T^R = 60$). Therefore, the budget set is

$$C = Y = \begin{cases} w(E - s) + bP + \rho w \times (E - s + 0.8P) \times [T - 60] & \text{if } E < 60 - P \\ w(E - s) + b(60 - E) + \rho w \times (E - s + 0.8 * (60 - E)) \times [T - 60] & \text{if } E \geq 60 - P \end{cases}$$

The baseline budget sets by cohort are constructed for the sample of married couples without dependent children. Given that in our sample, around 80% are married and around 15% have dependent children, the lifetime budget constraint for married couples without children is likely a reasonable approximation of reality. We use the following parameters: $s = 20$, $T = T_{last} = 78$ and $a = 0.8$. For the other parameters, we use the same institutional parameters as described in Appendix section F.4.

In Figure H.2 (a)-(c), representing the 1924, 1929, and 1935 cohorts respectively, the NRA and ERA

for retirement via unemployment were age 60, but maximum PBD varied. In panel (d), representing the 1945 cohort, the ERA remained at 60 but the un-penalized NRA was increased to around 64, with slight variation by month of birth. This amounted to a financial penalty for retiring at age 60 of approximately 18% of gross lifetime pension benefits. In panel (e), representing the 1950 cohort, the ERA was increased to 63 and the NRA was 65.18. The penalty for retiring at age 63 via unemployment was thus 7.2%. In panel (f), representing the 1952 cohort, the pathway into retirement via unemployment was abolished, leaving the earliest possible retirement age as 63 for long-term insured workers with over 35 years of qualified contributions. The penalty for retiring at age 63 via the long-term insured pathway was 9%.

D.3 UI as a Bridge to Retirement, the Role of Firms and Worker Councils, and Other Ways to Retire Early

Evolution of the UI bridge over time The use of UI as a bridge to retirement dates back to the Weimar Republic. The “59 rule” originated in the economic crisis of 1929-1930, allowing white-collar workers to retire at age 60 after receiving UI for one year. After WWII, the rule was extended to blue-collar workers in 1957 (Trampusch, 2005; Trampusch et al., 2010). The popularity of UI as a bridge to retirement increased in the early 1980s. After the 1982 recession, using UI as a bridge to retirement became a popular way to manage layoffs (Trampusch et al., 2010). The increase of PBD in several steps from 12 to 32 months in 1987 for workers above 54 (see Table I.1) increased the attractiveness of this pathway and shifted the earliest age where one could use the UI pathway from 59 down to 57 and 4 months. In addition, the so-called “58-rule” came into effect at the end of 1985, which allowed workers to stay on UI without any job search obligations (Bundesgesetzblatt, 1985). It provides additional incentives to use UI as a bridge to retirement (Schneider and Stuhler, 2007). Starting in 1997, the reduction in the generosity and phase-out of the early retirement system after UI made the UI pathway less attractive (see section D.1). In addition, the 2006 UI reform cut back PBD for workers 55 and older from (up to) 32 months to a maximum of 18 months. PBDs were increased back to 24 months in 2010 (see Table I.1). The “58-rule” was abolished for new UI entries from 2007 onwards (Schneider and Stuhler, 2007), further decreasing the attractiveness of UI as a bridge to retirement. Thereafter and as of today, UI can still be used as a bridge to retirement, though at later ages and to less generous terms.

Public perceptions The norm of using UI as a bridge to retirement changed over time. Describing the situation before the oil crisis of 1973, (Trampusch, 2005, p. 206) writes “*The operation of early retirement (...) made it popular with a wide and diverse constituency. (...) The policy was widely seen as a particularly humane solution to structural adjustment...*”. With the increased usage of the bridge, this changed over time. The news magazine “Der Spiegel” described the situation in 1995 (Der Spiegel, 1995), when UI receipt for the affected age group (55-59) was at its historical high: the article — titled “*Sliding into retirement*” (German: Gleitend in die Rente, own translation) — emphasizes that using the bridge to retirement puts high pressure on the social security system making the current practice unsustainable, while also

displaying some sympathy for retiring early. The labor minister is cited as warning representatives of the Employer Organizations and Unions of “*misusing the retirement system*” who were at that time still making heavy use of the early retirement options via UI. The leader of the metal union (IG-Metal) at that time is quoted in defense of the UI pathway.

By 2017, the tone of news articles shifted again. A 2017 article by the Spiegel — now titled “*double dipping*” (German: Doppelt Kassieren, own translation) — adopted a considerably stronger stance against the usage of the bridge (Fröhlingsdorf, 2017). The article describes and denounces the practice of using UI as a bridge to retirement at a large private bank and a leader of the service union (Verdi) is calling out this practice.

Usage in practice and the role of firms, unions, and worker councils Firms’ incentives play an important role in workers’ early exit from the labor force during our time period. In Germany, older workers with long tenure benefit from strong layoff protections in Germany (see EPL Database (2015) for more details). Consequently, laying off older workers prior to retirement age often occurs with the workers’ explicit consent to the terms and conditions of the separation (see Fröhlingsdorf (2017) for a concrete example). This can occur on an case-by-case basis at the individual-level, but commonly involves different pillars of Germany’s industrial relations system, including Works Councils and managers on the establishment level as well as Unions and Employer Organizations on the sectoral level (see Jäger et al. (2022) for a review of these institutions and Trampusch (2005); Trampusch et al. (2010) for their role in using the UI bridge as a separation policy).

After labor shortages in the 1960s and 1970s and extremely low unemployment rates, the German labor market worsened sharply after the 1973 oil crisis and even more so during the 1982 recession. Shrinking labor demand led to rapidly rising unemployment. Facing employment protection laws and powerful unions and works councils, firms and employer organizations sought to downsize employment through voluntary means by negotiating collective labor agreements (CLAs) and ‘social plans’ with their workforce. These agreements commonly offered severance packages to older workers to voluntarily quit the firm and were often tied to a specific age threshold. While severance payments are not observed in our data, Grund (2006) provides survey-level evidence consistent with this pattern.

Whether or not a worker would be willing to accept a severance package depends on the worker’s outside option. In a labor market with high unemployment rates, like that in the 1980s and 90s, exiting a job in one’s late 50s often meant accepting never to find work again, making the availability of unemployment benefits a crucial factor. Firms and labor unions that negotiated were aware of the institutional setting and would take the structure of UI benefits into account when negotiating workforce reductions and exit packages as part of CLAs. Indeed, Trampusch (2005) states that as early as the 1970s, “employees agreed to voluntary redundancy (that is they agreed to become unemployed at age 59) and began to draw unemployment pension after the lapse of unemployment benefits [...] Enterprises made this option attractive by topping up unemployment benefits with redundancy payments [...] Social plans providing for early exit spread quickly during the employment crisis of the 1970s and 1980s [...] work councils were more than happy to facilitate

the exit of older workers under the generous terms offered by the social security system. In fact, they often found themselves under considerable pressure from older workers who wanted to retire under the existing provisions.”

These practices gained steam in the 1980s and 90s as unemployment spiked and UI benefits were expanded to a maximum of 32 months, and CLAs with severance pay provisions proliferated. Over this time sector-level collective bargaining agreements (CBAs) that defined the conditions of early-retirement practices became prevalent (Trampusch et al., 2010). Social plans often accompanied these agreements — agreements between works councils and the establishment management on how to manage separations— further cementing the usage of these rules (Trampusch, 2005). The details of such agreements, including the earliest exit age and the corresponding severance package, varied across sectors and even individual firms (see Trampusch, 2005, 2009), but tended to take age discontinuities induced by the UI and public pension system into account. Trampusch (2005) writes, “a side effect of the [law allowing older workers to draw unemployment benefits for a maximum of 32 months] was effectively to turn the previous ‘59 rule’ into a ‘57 rule’, as early retirement became even more attractive to firms. Now firms could retire employees at age 57. Workers could receive unemployment benefits for a period of thirty-two months, and then take advantage of the pension due to unemployment at age 60.”

In cases where firms encourage workers to exit at those age thresholds with severance packages, one can view these agreements as a mechanism of how age discontinuities lead to extensive-margin responses. Of course, other factors could also influence the precise details of social plans/CLAs and associated age limits, potentially leading to bunching in UI inflows at age thresholds not directly related to retirement or UI institutions.

Fröhlich et al. (2013) describe the use of different pathways into early retirement in the early 2010s in six different industries, including a detailed portrait of one firm in each sector. In two of the six sectors – the chemical industry and private banking – the portrayed firm had recently used UI as a bridge to retirement. In both cases, the bridge-to-retirement arrangements involved an explicit or implicit agreement between management and the works council and generous severance payments to top up UI benefits. These policies guaranteed a fixed replacement rate of the previous net wage (between 70% and 90%) and coverage of all social security and tax contributions for the period between layoff and the earliest possible retirement age, under the assumption that workers took up and fully exhausted their UI benefits. In the case of the portrayed bank, the policy explicitly offered assistance to workers in claiming UI benefits. For the same sector, (Fröhlingsdorf, 2017) reports high demand for the UI bridge among workers at a large firm, and a take-up rate of 96% among workers who were offered the policy. In this firm, the management decided whom to offer the policy on a case-by-case basis. Separately, Knuth and Kalina (2002) document high usage of the bridge in the manufacturing sector, among high-income workers, and in large (≥ 500 employees) establishments.

Alternative CLAs and retirement pathways The government also supported CLAs on early retirement in other forms, such as subsidizing employers’ costs of buying-out older workers through the so-called

partial retirement law (Altersteilzeitgesetz). This partial retirement law (Altersteilzeitgesetz) was enacted in mid-1990s and was suspended in 2009. Most CLAs on early retirement based on this law were not renewed. It worked by halving older workers' working time (either via part-time work or early retirement). The employer paid 50% of the previous full-time income and the state government provided the remaining 50% to the employers, but only under the condition that the vacancy was replaced by an unemployed person or a freshly trained apprentice. In addition, the government supported this early retirement option by topping up the pension contribution of the workers who entered early retirement. This partial retirement law provided a maximum public subsidy for up to five years. Combined with the ERA being at age 60, this requirement meant that the CLA early retirement option applied most directly to employees age 55 and older (Trampusch, 2005). Age 55, and to a lesser extent, age 56, became a common cutoff used in CLAs (in addition, of course, to CLAs based around the bridge-to-retirement age).

E Model Details

This appendix sets up and solves our labor supply model.

E.1 Model Set Up

States Workers can be in one of three states: Employed (E), Unemployed (U), or out of the labor force (O). We assume that once a worker drops out of the labor force, he will not return, hence O is an absorbing state. We call a worker Non-Employed N if the worker is either unemployed or out of the labor force.

We assume that workers produce output p_t in each period, where p_t is i.i.d. according to some distribution $F(p)$. Another important state variable in our model is the total unemployment duration of a worker d^U . In practice we will estimate our model starting at age 50, so that d^U will be the duration in unemployment since then. To keep the state space manageable, we also assume that workers are initially eligible for the maximum benefit duration but do not reaccumulate benefit eligibility if they are reemployed after losing a job. Under this assumption, d^U is sufficient to calculate both remaining UI benefit duration and an individual's pension if the person retires. A full accounting of benefit eligibility in the presence of multiple unemployment spells would require separately keeping track of d^U , the remaining benefit duration in each unemployment spell, and employment duration in each employment spell. This quickly becomes computationally very challenging due to the curse of dimensionality. As long as repeated unemployment spells with long in-between employment spells are rare, which they are in practice, our approach is only a very minor simplification that vastly reduces the computational complexity. We can therefore write the value functions for the firm and worker as functions of p_t and d^U , where d^U is deterministic, while p_t is uncertain.

Value Function For Employment Workers have a utility function $u(\cdot)$, are paid $w_t(\cdot)$, and experience disutility from working (η), which will be drawn from a cohort specific distribution. The Value Function for

Employment is:

$$V_t^E(p_t, d^U) = u(w_t(p_t)) - \eta + \beta E_{p_{t+1}} [\max \{V_{t+1}^E(p_{t+1}, d^U), V_{t+1}^N(d^U)\}] \quad (\text{E.1})$$

Workers will separate from their job whenever the expected value of future non-employment exceeds that of employment. This could occur for several reasons: workers could receive a low productivity draw (p_t) such that the employment relationship is no longer better than the worker's outside option. Alternatively, outside options could improve. For example, an increase in retirement benefits will push up $V_t^N(d^U)$ for workers close to the retirement age and can increase the rate of jobs ending.

Value Function For Unemployment When workers enter unemployment they engage in costly job search and receive payments $B(d^U)$. If the individual still has Unemployment Insurance benefits remaining ($d^U < P$), he will receive UI benefits ($B(d^U) = b$). If not, the individual receives y^u , which can be interpreted as unemployment assistance. An unemployed individual searches for a job and chooses an optimal level of search effort s , which is normalized to the probability of finding a job. Generating search effort comes at a cost $\psi(s, d^U)$ which is increasing and convex in search effort. Finally, whether or not an individual receives a job offer, he can decide to retire at the end of the period. If he remains unemployed d^U increases by one period. The Value Function for Unemployment is thus:

$$\begin{aligned} V_t^U(d^U) = & u(B(d^U)) + \max_s \{ \beta s E_{p_{t+1}} \max [V_{t+1}^E(p_{t+1}, d^U + 1), V_{t+1}^N(d^U + 1)] \\ & + \beta(1 - s)V_{t+1}^N(d^U + 1) - \psi(s, d^U) \} \end{aligned} \quad (\text{E.2})$$

Let $\psi_s(s, d^U) \equiv \frac{\partial \psi(s, d^U)}{\partial s}$ denote the marginal cost of search. Individuals choose search effort so that the marginal return to search equals the marginal cost up to the constraint that $s \leq 1$. For an interior solution, the first order condition for the optimal level of search effort s^* is:

$$\psi_s(s^*, d^U) = \beta E \max [V_{t+1}^E(p_{t+1}, d^U + 1), V_{t+1}^N(d^U + 1)] - \beta V_{t+1}^N(d^U + 1)$$

Since we assume that $\psi(\cdot, d^U)$ is increasing and convex in search effort, optimal search effort at an interior solution is:

$$s^* = [\psi_s(\cdot, d^U)]^{-1} (\beta E \max [V_{t+1}^E(p_{t+1}, d^U + 1), V_{t+1}^N(d^U + 1)] - \beta V_{t+1}^N(d^U + 1)) \quad (\text{E.3})$$

Value Function For Out of the Labor Force At any point, a worker can choose to transition to being out of the labor force O , which is an absorbing state. The value of O depends primarily on the value of one's pension y_t^p as determined by prevailing retirement institutions. y_t^p will depend on work history (d^U) and the

age at which the worker retires. Specifically, for a worker who lives until T^{Last} and is eligible to receive a pension at T^{ERA} , the value function for being out of the labor force is:

$$V_t^O(d^U) = \begin{cases} \sum_{k=t}^{T^{ERA}-1} \beta^{k-t} u(y^o) + \sum_{k=T^{ERA}}^{T^{Last}} \beta^{k-t} u(y_t^p) & t \leq T^{ERA} \\ \sum_{k=t}^{T^{Last}} \beta^{k-t} u(y_t^p) & t > T^{ERA} \end{cases} \quad (\text{E.4})$$

The value of the pension depends on the relevant, cohort-specific retirement institutions in addition to the individual's work history (d^U). Individuals accrue pension benefits while working and while on UI benefits (at 80%), but not otherwise. Persons retiring at the earliest allowable retirement age (ERA) but before the normal retirement age (NRA) begin receiving a penalty starting with the 1937 cohort. We assume all individuals in our sample are eligible for the long-term insured retirement pathway and eligible for the retirement via UI pathway as long as they have one year of unemployment history (d^U).⁵⁴ We allow individuals to choose the best retirement option available. In Section F.5 below, we outline in detail how we calculate V_t^O for each cohort.

Value Function For Non-Employment Finally the value of non-employment is defined as $V_t^N(d^U) = \max(V_t^U(d^U), V_t^O(d^U))$.

E.2 Heterogeneity in the Disutility of Work

We introduce an additional layer of heterogeneity (beyond the productivity distribution $F(p)$), by integrating the preceding model over a distribution of disutility of work types (η -types).

Under our distributional and functional form assumptions (laid out in detail next), the preceding model generates closed form solutions for all transitions between states (e.g. E to U) and can be used to calculate expected non-employment durations for a given value of η . We will assume individual workers draw their η from a cohort-specific distribution, and integrate transitions and non-employment durations over the entire distribution. Specifically, we will assume that η is normally distributed with mean $\eta_{mean, cohort}$ and standard deviation η_{sd} (which is fixed across cohorts). We implement this in practice by simulating the model for 25 different values of η and use Gaussian integration to approximate the full integral over the η distribution whenever we calculate cohort-level transitions and non-employment durations.

E.3 Distributional and Functional Form Assumptions

Here, we lay out the functional forms and distributional assumptions underlying our baseline model.

⁵⁴We implement these eligibility restrictions numerically by setting the fixed cost parameter $k_6 = 100$ for unemployment from age 63 onward, reflecting the maintained assumption of universal access to the long-term-insured pathway, and by subtracting 10^{11} from the value of unemployment at or after the ERA once the pathway-specific eligibility condition is met. These devices serve only to exclude unemployment states that are inadmissible under the maintained eligibility assumptions; they are fixed, not estimated, parameters.

Productivity p_t is drawn from a mixture distribution in which workers have Λ_t probability of facing a large negative productivity shock such that $\ln(p_t) = -L$, which destroys the job with certainty. Meanwhile, with probability $1 - \Lambda_t$, workers draw a productivity level p_t from a lognormal distribution. The CDF of this mixture variable is

$$F_t(\ln(p_t)) = \Lambda_t \mathbf{1}\{\ln(p_t) \geq -L\} + (1 - \Lambda_t) F_{p, \sigma_p}^N(\ln(p_t)),$$

where F_{p, σ_p}^N is the normal CDF with mean p and standard deviation σ_p . This allows for exogenous job destruction at the rate Λ_t . In practice, we allow this exogenous job destruction rate to vary with the national male unemployment rate (UR). Specifically,

$$\Lambda_t = \frac{1}{1 + e^{-(\lambda_1 + \lambda_2 UR_t + \lambda_3 \Delta UR_t)}},$$

with parameters λ_2 and λ_3 allowing Λ_t to vary with the level and year-on-year change in the national male UR.

We assume workers have log utility $u(\cdot) = \ln(\cdot)$. Firms make zero profits and hence pay the worker $w_t = p_t$ in all periods. Workers draw disutility η from a normal distribution ($\eta \sim N(\eta_{mean, cohort}, \eta_{sd})$).

The search cost function is based on [DellaVigna et al. \(2022\)](#) with some added flexibility. Specifically we assume:

$$\psi(s, d^U) = k_0 + k_1 \mathbf{1}(d^U = 0) + e^{k_2 \times d^U} \times k_3 \frac{s^{1+\gamma}}{1+\gamma} \quad (\text{E.5})$$

where k_0 is a fixed cost of being in unemployment, k_1 a fixed cost of entering unemployment the first time, k_2 allows search to become more costly later on in unemployment spells, while k_3 and γ govern the slope and curvature of the job search function.

E.4 Closed Form Solutions For Each Value Function

Value Function For Employment Let ω_{t,d^U} be the ‘reservation productivity’ such that $V_t^E(\omega_{t,d^U}, d^U) = V_t^N(d^U)$. Further, let $\bar{\omega}_{t,d^U} \equiv \frac{\ln(\omega_{t,d^U}) - p}{\sigma_p}$.

Since $V_t^E(p_t, d^U) = \ln(p_t) - \eta + \beta E_{p_{t+1}} [\max \{V_{t+1}^E(p_{t+1}, d^U), V_{t+1}^N(d^U)\}]$, plugging in ω_{t,d^U} for p_t and rearranging $V_t^E(\omega_{t,d^U}, d^U) - V_t^N(d^U) = 0$ gives:

$$\ln(\omega_{t,d^U}) = \eta - \beta E_{p_{t+1}} [\max \{V_{t+1}^E(p_{t+1}, d^U), V_{t+1}^N(d^U)\}] + V_t^N(d^U) \quad (\text{E.6})$$

Given the distribution of p_t :

$$\begin{aligned} E_{p_{t+1}} [\max \{V_{t+1}^E(p_{t+1}, d^U), V_{t+1}^N(d^U)\}] &= [\Lambda_{t+1} + (1 - \Lambda_{t+1}) \Phi(\bar{\omega}_{t+1, d^U})] V_{t+1}^N(d^U) \\ &+ [1 - \Lambda_{t+1} - (1 - \Lambda_{t+1}) \Phi(\bar{\omega}_{t+1, d^U})] E \left[V_{t+1}^E(p_{t+1}, d^U) \mid \frac{\ln(p_{t+1}) - p}{\sigma_p} \geq \bar{\omega}_{t+1, d^U} \right] \end{aligned}$$

For the continuous component of the productivity distribution, the conditional expectation in the final term can be evaluated using the corresponding truncated-normal expectation. Using the fact that $E(X|Z < \bar{\omega}_{t+1}) = p - \sigma_p \frac{\phi(\bar{\omega}_{t+1})}{\Phi(\bar{\omega}_{t+1})}$ and $E(X|Z \geq \bar{\omega}_{t+1}) = p + \sigma_p \frac{\phi(\bar{\omega}_{t+1})}{1 - \Phi(\bar{\omega}_{t+1})}$ for a random variable $Z \sim N(0, 1)$ and for $X = \sigma Z + \mu \sim N(\mu, \sigma)$, we obtain:

$$E \left[V_{t+1}^E(p_{t+1}, d^U) \middle| \frac{\ln(p_{t+1}) - p}{\sigma_p} \geq \omega_{t+1, d^U} \right] = p - \eta \\ + \beta E_{p_{t+2}} \left[\max \{ V_{t+2}^E(p_{t+2}, d^U), V_{t+2}^N(d^U) \} \right] + \sigma_p \frac{\phi(\bar{\omega}_{t+1}(d^U))}{1 - \Phi(\bar{\omega}_{t+1}(d^U))}$$

And hence

$$E_{p_{t+1}} \left[\max \{ V_{t+1}^E(p_{t+1}, d^U), V_{t+1}^N(d^U) \} \right] = [\Lambda_{t+1} + (1 - \Lambda_{t+1})\Phi(\bar{\omega}_{t+1, d^U})] V_{t+1}^N(d^U) \\ + [1 - \Lambda_{t+1} - (1 - \Lambda_{t+1})\Phi(\bar{\omega}_{t+1, d^U})] \\ \times \left\{ p - \eta + \beta E_{p_{t+2}} \left[\max \{ V_{t+2}^E(p_{t+2}, d^U), V_{t+2}^N(d^U) \} \right] \right. \\ \left. + \sigma_p \frac{\phi(\bar{\omega}_{t+1}(d^U))}{1 - \Phi(\bar{\omega}_{t+1}(d^U))} \right\}$$

Similarly,

$$E_{p_{t+2}} \left[\max \{ V_{t+2}^E(p_{t+2}, d^U), V_{t+2}^N(d^U) \} \right] = [\Lambda_{t+2} + (1 - \Lambda_{t+2})\Phi(\bar{\omega}_{t+2, d^U})] V_{t+2}^N(d^U) \\ + [1 - \Lambda_{t+2} - (1 - \Lambda_{t+2})\Phi(\bar{\omega}_{t+2, d^U})] \\ \times \left\{ p - \eta + \beta E_{p_{t+3}} \left[\max \{ V_{t+3}^E(p_{t+3}, d^U), V_{t+3}^N(d^U) \} \right] \right. \\ \left. + \sigma_p \frac{\phi(\bar{\omega}_{t+2}(d^U))}{1 - \Phi(\bar{\omega}_{t+2}(d^U))} \right\}$$

And so forth, until the final period T^{Last}

$$E_{p_{T^{Last}}} \left[\max \{ V_{T^{Last}}^E(p_{T^{Last}}, d^U), V_{T^{Last}}^N(d^U) \} \right] = V_{T^{Last}}^N(d^U) = V_{T^{Last}}^O(d^U)$$

Hence, the value of employment in any given period can be determined using backward induction. For convenience, we define $\Omega_{t, d^U} \equiv E \max [V_t^E(p_t, d^U), V_t^N(d^U)]$. This allows us to express $V_t^E(p_t, d^U) = u(w_t(p_t, d^U)) - \eta + \beta \Omega_{t+1, d^U}$.

Altogether, these results and Equation E.6 imply: $\bar{\omega}_{t, d^U} \equiv \frac{\ln(\omega_{t, d^U}) - p}{\sigma_p} = \frac{\eta - \beta \Omega_{t+1, d^U} + V_t^N(d^U) - p}{\sigma_p}$.

Value Function For Unemployment Given the above, we can rewrite the value of unemployment as a function of

$$V_t^U(d^U) = u(B(d^U)) + \max_s \{ \beta V_{t+1}^N(d^U + 1) + \beta s (\Omega_{t+1,d^U+1} - V_{t+1}^N(d^U + 1)) - \psi(s, d^U) \}$$

and

$$s^* = [\psi_s(\cdot, d^U)]^{-1} (\beta \Omega_{t+1,d^U+1} - \beta V_{t+1}^N(d^U + 1))$$

Transitions Individuals can be in any of the following N_s states: employed with $d^U = 0$ to $d^U = T$, unemployed with $d^U = 0$ to $d^U = T$, or out of the labor force. Let $\mathbf{h}_t \equiv (h_{t,E,d^U=0}, \dots, h_{t,E,d^U=T}, h_{t,U,d^U=0}, \dots, h_{t,U,d^U=T}, h_{t,O})$ be the vector describing the number of individuals across states at each time period. Let the $m_{t,i,j}$ be the probability of an individual transitioning from state i at time t to state j at time $t + 1$. Let $\mathbf{M}_t$ be the transition matrix across states where $m_{t,i,j}$ is the element of the i^{th} row and j^{th} column.

The transition matrix describes the evolution of the number of individuals across states:

$$\mathbf{h}_{t+1} = \mathbf{h}_t \mathbf{M}_t$$

Define $\zeta_{d^U} \equiv \Lambda_{t+1} + (1 - \Lambda_{t+1})\Phi(\bar{\omega}_{t+1,d^U})$ and $\zeta_{d^U+1} \equiv \Lambda_{t+1} + (1 - \Lambda_{t+1})\Phi(\bar{\omega}_{t+1,d^U+1})$

The transition matrix $\mathbf{M}_t$ is given by:

	Employed d^U	Employed $d^U + 1$	Unemployed d^U	Unemployed $d^U + 1$	OLF
...					
Employed d^U	$1 - \zeta_{d^U}$	0	$\zeta_{d^U} \mathbf{1} \left(\begin{matrix} V_{t+1}^U(d^U) \\ \geq \\ V_{t+1}^O(d^U) \end{matrix} \right)$	0	$\zeta_{d^U} \mathbf{1} \left(\begin{matrix} V_{t+1}^O(d^U) \\ > \\ V_{t+1}^U(d^U) \end{matrix} \right)$
...					
Unemployed d^U	0	$s[1 - \zeta_{d^U+1}]$	0	$\left\{ (1-s) + s(\zeta_{d^U+1}) \right\} \times \mathbf{1} \left(\begin{matrix} V_{t+1}^U(d^U+1) \\ \geq \\ V_{t+1}^O(d^U+1) \end{matrix} \right)$	$\left\{ s\zeta_{d^U+1} + (1-s) \right\} \times \mathbf{1} \left(\begin{matrix} V_{t+1}^O(d^U+1) \\ > \\ V_{t+1}^U(d^U+1) \end{matrix} \right)$
...					
OLF	0	0	0	0	1

As an example, a transition from employed with $d^U \rightarrow$ unemployed with d^U occurs with $prob[V_{t+1}^E(p_{t+1}, d^U) < V_{t+1}^N(d^U)] \mathbf{1} (V_{t+1}^U(d^U) \geq V_{t+1}^O(d^U))$.

This can be simplified to:

$$= prob[V_{t+1}^E(p_{t+1}) < V_{t+1}^N(d^U) + \eta - \beta \Omega_{t+2,d^U}] \mathbf{1} (V_{t+1}^U(d^U) \geq V_{t+1}^O(d^U))$$

$$= F(V_{t+1}^N(d^U) + \eta - \beta \Omega_{t+2,d^U}) \mathbf{1} (V_{t+1}^U(d^U) \geq V_{t+1}^O(d^U))$$

Recall $\bar{\omega}_{t+1,d^U} = \frac{V_{t+1}^N(d^U) + \eta - \beta \Omega_{t+2,d^U} - p}{\sigma_p}$, hence:

$$= [\Lambda_{t+1} + (1 - \Lambda_{t+1})\Phi(\bar{\omega}_{t+1,d^U})] \mathbf{1} (V_{t+1}^U(d^U) \geq V_{t+1}^O(d^U))$$

$$= \zeta_{d^U} \mathbf{1} (V_{t+1}^U(d^U) \geq V_{t+1}^O(d^U))$$

Model Output: Aggregate Transition Probabilities and Non-Employment Durations We first simulate the model for 25 different realizations of the distribution of disutility of work. For each of them, we calculate simulated moments such as transitions between employment statuses and non-employment durations. For transitions, we sum across the elements of the transition matrix that correspond to each moment. For non-employment durations, we employ a backwards induction procedure that assumes that all workers are not employed by the last period, and then it considers the probability of entering non-employment recursively. This approach allows us to generate the expected value for non-employment duration for new entrants into UI ($d^U = 0$) for every period. After calculating these moments, we aggregate all realizations by integrating over the distribution of η using Gaussian integration.

F Estimation Details

F.1 Estimation Procedure

In-Sample Cohorts We estimate the model structurally, using a minimum distance estimator to match the empirical reduced form moments from Section II. Denote as ξ the parameters of the structural model. Furthermore, let $m(\xi)$ be the vector of moments predicted by the model as a function of the parameters ξ , and by $\hat{m}$ the vector of observed moments. We estimate the model using 3 cohorts: 1929, 1935, and 1950 on quarterly data. The moments $m(\xi)$ we use for matching are i) the quarterly transition probabilities of workers from E to U (i.e. UI or Nu in the data) between age 50 and 63, ii) the non-employment durations (calculated from job exit until age 63), and iii) $\frac{\partial Nonemp}{\partial P}$ at age 52 = 0.128 for the 1950 cohort (from Table I.3).⁵⁵

The estimator chooses the parameters $\hat{\xi}$ that minimize the distance:

$$(m(\xi) - \hat{m})' W (m(\xi) - \hat{m}) \quad (F.1)$$

where W is a weighting matrix. We simulate all transitions using the empirical data to construct the full covariance matrix for the transitions. We use diagonal covariance matrices based on the estimated standard errors for the non-employment durations and for $\frac{\partial Nonemp}{\partial P}$.

For the intensive-margin RD moments, we use a larger weight ($\times 35$) since this is a causal estimate that we have significant confidence in given the research in this paper and many other well identified estimates from the literature and we want to make sure our fitted model generates realistic predictions for intensive-margin responses. We omit the first and last quarter from the estimation.

Out-of-Sample Cohorts In the second step of the model we refit our model to all other cohorts by estimating a single parameter per cohort - the mean of that cohort's η distribution ($\eta_{\{mean, cohort\}}$). For this

⁵⁵While we observe UI receipt, we cannot distinguish unemployment from OLF after UI benefits are exhausted. For this reason we simply distinguish between non-employment and employment, which we can easily generate from the model predictions by pooling the unemployed and OLF states.

estimation exercise, our target moments are transitions from E to U and non-employment durations. Since this parameter was already estimated within our in-sample cohorts, refitting does not change the model parameters for our in-sample cohorts, but allows different cohorts to have different outside options / workforce attachment that are not otherwise captured by other features of the model and institutional parameters. We also employ a minimum distance estimator using the same specifications previously described.

F.2 Estimated Parameters

We estimate the following parameters: standard deviation of the distribution of productivity σ_p ; parameters of exogenous job loss shock $\lambda_1 - \lambda_3$; search cost function parameters $k_0 - k_3$ and γ ; and parameters for the cohort-specific distribution of disutility of work $\eta_{mean,1929}$, $\eta_{mean,1935}$, $\eta_{mean,1950}$, and η_{sd} .

F.3 Numerical Optimization

The model is simulated in Python. We carefully optimized our code using the Python package Numba to pre-compile the code, which greatly speeds up computation times. We then estimate the model by numerically minimizing the objective function (Equation F.1). For this, we rely on the optimization package **estimagic** (see Gabler (2022)), which provides an elegant way to search for global minima using a multi-start algorithm, that can be distributed over many computing cores and nodes and allows for easily switching between alternative local optimizers. For our problem, we found that two derivative-free least squares optimizers work well: Derivative-Free Optimizer for Least-Squares Minimization (DFO-LS) (Cartis et al., 2018) and POUNDERS (Wild, 2015). A noteworthy practical point is that these least-squares optimizers perform vastly better than a wide range of black box optimizers that we tried (such as Newtonian, quasi-Newtonian, trust-region, and genetic algorithms).

Our algorithm is the following: We use 18 compute nodes with 28 cores each. We then draw 280 random starting values on each node using latin hypercube sampling (to guarantee good coverage of the parameter space). On each node, we then pick the 28 best starting values (lowest SSE) and run a local minimizer (in half the cases DFO-LS in the other half POUNDERS) on them with a walltime of 10 hours. The total compute time is thus $18 \times 28 \times 10 = 5040$ hours. We can assess convergence by comparing the best solutions from each of the 18 nodes. They are fairly close to each other, both in terms of SSE and the parameter estimates, suggesting that we reliably find a global minimum or at least a point very close to the global minimum.

F.4 Institutional and Other Non-Estimated Parameters Used in the Model

We set $T^{Last} = 78$ and $\beta = 0.95$.

Average Wages/Productivity: Mean (net) wages are set at euro 1,950, so the mean of the p_t distribution is the logarithm of 1950. This implies an approximate gross wage of 3000, which is in line with average gross

wages for men aged 50-60 with a UI spell (3,282 across all 6 select cohorts). We use a constant conversion rate between gross and net wages of 0.65.⁵⁶

UI and UA replacement rates: UI reforms in the past decades also changed the UI replacement rates. The replacement rates on net wages stay at 63% for an individual without children and 68 % for an individual with children until the end of 1993. Starting January 1994, the replacement rates were reduced to 60% and 67%, respectively. Since most of our sample will no longer have eligible children, we use the 63% and 60% rates. We apply the UI replacement rates on net wages for each cohort based on when they reach the UI bridge age. In practice, this means we set $b = 1230$ for 1936 and earlier cohorts and $b = 1170$ for 1937 and later cohorts. We set $y^u = 500$, which is approximately half of what one would receive if on UA with no deductions. We halve the amount as evidence in [Schmieder et al. \(2012\)](#) suggests that, due to deductions, average UA benefits actually received are substantially below the 53% nominal replacement rate on net wages, and only 50% of UI exhaustees take up UA.

Pension replacement rate: ρ represents the pension replacement rate on gross wages per one additional year of employment. We calculate the values for an average earner born in the cohort based on the pension benefit formula in Germany. For each cohort, we take the value of ρ in the years when they are between 60 and 63 years old, which we calculate on a cohort-by-cohort basis as described below. Several pension reforms in the past decades have changed the pension benefit formula.

Before 1992, the pension benefit size was determined by four factors: the relative earnings of the insured, the aggregate annual pension value, the number of insurance years, and an adjustment factor, which was set at 1.5 for old-age pensions. For an average earner with 45 years of contribution, the gross annual pension benefit was the annual pension value $\times 45 \times 1.5$. Therefore, the pension replacement rate on gross wage is (annual pension value $\times 45 \times 1.5$)/average annual income. The pension replacement rate on gross wages per one additional year of employment is calculated from the monthly pension benefits net of health care and long-term care contribution (ssc) : (annual pension value $\times 1.5$)(1-ssc)/average annual income.

After 1992, the monthly pension benefit amount is obtained by multiplying the personal pension base by the monthly pension value (PV). The personal pension base is the sum of the earnings points (EPs) accumulated over the entire working history. For example, an average wage earner with 45 contribution years will accumulate 45 EPs. At the time of retirement, this personal pension base is scaled up by the pension value at the time of retirement, which is determined aggregately by factors such as the average wage of all insured, the contribution rate, and demographic changes. For example, one EP was equivalent to 29.21 euro per month in 2015. Therefore, the pension replacement rate on gross wage earnings was (45 $\times$ PV $\times$ 12)/average annual income. The pension replacement rate on gross wages of an additional year of contribution net of ssc is (PV $\times$ 12)(1-ssc)/average annual income.

We obtain the pension values, the average annual income of all insured, and health care and long-term

⁵⁶This conversion rate comes from the data. Specifically, for cohorts 1935 and later, we take all individuals in the cohort with a UI spell in the IEB-data aged 50-60 and compare their actual UI benefits to their gross income. For each cohort, we obtain an average gross replacement rate of 0.39, implying a constant conversion rate from net UI replacement rates to gross UI replacement rates of 0.65. We assume this conversion rate also applies to prior cohorts.

care contribution rates for the years 1980 to 2016 from the German pension statistics office and social code book VI. The pension values are from [Zahlen und Tabellen vom 1.1. bis 30.6.2020](#). The average annual income of all insured is from Appendices 1 and 2 of the social code book VI. The average social security contribution rates are from the [German pension insurance annual report 2019](#).

We set the income tax rate on pension benefit to zero for two reasons. First, for individuals who retire before 2005, pension income is tax-free.⁵⁷ Second, for individuals who retire after 2005, only 50 percent of their gross pension benefit is recognized as taxable income. However, there is an annual income threshold that is exempt from income tax, regardless of income sources. This threshold was 9000 euro per single individuals in 2018 and 7356 euro in 2005. For an average earner with 40 years of contributions who retires in 2005, the annual pension benefits are around EUR 12,500. Much of this amount is below the taxable income threshold, which is why it is reasonable to set the income tax for pension benefits to zero.

Using these data and assumptions we calculate a ρ for each cohort and we use these values in our model. The value of ρ is shown for select cohorts in Table 1 Panel A.

Years of contribution made before age 54. We obtain the average years of contribution at age 54 by using the scientific use file of the Insurance Account Sample (Versicherungskontenstichprobe, SUFVSKT) of the German Federal Pension Register. Each wave of SUFVSKT contains 5% random sample of individuals with an active public pension insurance account in Germany, who were between the ages of 30 and 67 at the time of data collection. Each wave also contains the earnings biographies from age 14 onwards, at a monthly frequency. For cohorts from 1935 to 1946, we calculate the average years of employment at age 54 for West Germans employed at age 50 using the wave SUFVSKT2002. We obtain the values for cohorts from 1947 to 1952 by using the waves SUFVSKT2010 and SUFVSKT2018. However, we cannot observe cohorts older than 1935 because the earliest publicly available SUFVSKT wave is 2002. Cohorts born before 1935 are older than 67 in 2002. To obtain reasonable values of employment years by age 54 for these older cohorts, we use the average values for cohorts from 1935 to 1940 as a proxy for the older cohorts' years of contribution made by age 54.

Discounted pension accrual rates while on UI and UA The time spent on unemployment insurance also increases pension benefits, because the UI agency contributes to the pension scheme on behalf of the unemployed. During the periods of claiming UI, contributions are paid on the basis of 80% of previous gross earnings ([SGBVI §166 Paragraph 1 No. 2](#)). Therefore, one additional year of time spent on UI increases the future pension benefits by $\rho \times 80\%$. During the periods of claiming unemployment insurance benefits 2 (UIB II), which is means-tested and paid at a lower rate, and unemployment assistance (UA), no financial contributions are counted towards the pension ([OECD: pension at a glance 2019](#)).

⁵⁷The proportion of the income subject to tax varies with the year of retirement at which the individual first started drawing the pension. Pensions starting before 2005 are tax-free. For pensions beginning in 2005, 50 percent of the gross pension benefit is recognized as taxable income. This portion remains fixed for the pensioners who retire in 2005 and subsequent years. Until 2020, the taxable part of the pension increases by 2 percentage points per year and from 2020 until 2040, it will increase by one percentage point per year. In 2015, 70% of the pension income is taxable. The statutory health and long-term care insurance contributions are exempt from the taxable income. For more details about the schedule, see [German statutory pension insurance website](#).

The model also takes as inputs the relevant earliest available retirement age (ERA), the age at which you can collect pension without penalties (NRA), and the accrual adjustment penalty for retiring at the ERA (simply a function of the difference between the NRA and ERA) for both the UI and long-term insured pathways.

F.5 Retirement Details: How We Calculate the Value of Out Of the Labor Force

To calculate the value of being out of the labor force (OLF), we first calculate the income from pension at any given point in time. This depends on the worker's contribution years (from employment, unemployment and welfare), working years, duration of unemployment d^U , reference income, pension replacement rates, potential UI duration, and the pension contribution discount while on UI. In the model, we take average contribution years from the data as described above at the starting age of the model and then allow individuals' contribution years to evolve based on individuals' simulated employment in subsequent years. Gross reference income is euro 3000 per month and pension replacement rates on gross income are listed in Table 1 Panel A. Contribution years on UI count for 0.8 and contribution years from UA count for 0. Pensions taken at the ERA but before the NRA are further penalized by 3.6% per year retiring in advance of the NRA. Once we know the value of the pension at each point in time, we generate an age-specific OLF income path, which comprises home production before retirement (y^o) and pension income (after early retirement penalties) after retirement.⁵⁸ This income stream will depend on cohort-specific institutional values such as early and normal retirement ages.

This whole procedure is done for each relevant pathway, namely, the UI pathway and the long-term insured pathway. That is, we calculate the present discounted value of OLF at each point in time for both pathways following Equation E.4. The worker then endogenously assigns the value of OLF to the pathway that provides higher value (if both are available and feasible, otherwise, as for later cohorts when the UI pathway was closed, this choice is determined for them).

For women, all is as above except we also allow women to take the women's pathway into retirement, which in practice will be as or more attractive than the UI pathway. Average contribution years prior to starting age in the model also differ for women.

G The Relationship between Moments and Model Parameters

This section documents the relationship between the 13 model parameters and the empirical moments used for their estimation. We proceed in two parts. First, we analyze the influence of marginal changes in model parameters on the moments (i.e., the partial derivatives $\frac{\partial m_i}{\partial \xi_j}$). Second, we conduct a sensitivity analysis, following Andrews et al. (2017), to quantify how the parameter estimates respond to changes in the empirical moments (i.e., $\frac{\partial \xi_j}{\partial m_i}$).

⁵⁸We set home production (y^o) to a low value, 50, so individuals in our model will typically remain employed or on UI/UA prior to the earliest age at which they could claim their pension, but model fit is relatively insensitive to the exact choice of y^o .

Influence of Model Parameters on Moments

We begin by examining how model parameters influence the simulated moments, a relationship captured by the Jacobian of the model function that generates simulated moments from the vector of model parameters $m(\xi)$. Each element of this Jacobian, $\frac{\partial m_i}{\partial \xi_j}$, represents the marginal effect of a parameter ξ_j on a simulated moment m_i . To facilitate comparison across parameters and moments of different magnitudes, our analysis focuses on the elasticities, defined as $\frac{\partial m_i}{\partial \xi_j} \times \frac{|\xi_j|}{m_i}$. For a tractable exposition, we focus on the 1935 cohort, as the insights are broadly applicable to other cohorts. The influence of the parameters on the 1935 inflow and duration moments are presented in Figure H.15 and Table I.10 for all cohorts and the dD/dP moment.

σ (Standard Deviation of Productivity Shocks) A larger σ increases job destruction for younger workers, as they are more likely to experience large negative shocks that result in job loss (Figure H.15(a)). As workers near retirement, the effect of σ on inflows to non-employment becomes more complex. Initially, a higher σ increases inflows before the bridge age, as some workers who would have otherwise waited until the bridge age leave their jobs earlier due to shocks. At the bridge age itself, however, the influence of σ becomes negative. The productivity shocks spread out exits around the bridge age: negative shocks precipitate earlier exits, while positive shocks delay them. This dilutes the sharp bunching pattern. At older ages, the influence of σ on inflows becomes positive again.

As shown in Figure H.15(b), σ has a relatively small effect on non-employment durations. For younger workers, higher σ slightly increases durations as they wait for more favorable productivity shocks before accepting a job. For older workers approaching retirement, higher σ alters the composition of the unemployed, leading to a higher share of workers who are effectively retiring, which in turn increases average durations.

k_0 (Fixed Cost of Job Search) A higher k_0 increases the cost of job search, making unemployment less attractive. Consequently, it reduces inflows to non-employment, particularly for younger workers and those approaching the bridge age. For workers just past the bridge age, a higher k_0 can lead to a small increase in inflows due to a substitution effect, as some workers delay their entry into unemployment.

k_1 (Fixed Cost of Entering UI) k_1 represents a one-time cost of entering UI. A higher k_1 discourages entry into non-employment, an effect that is most pronounced for workers near the bridge age who are on the margin between working and not working. As shown in Figure H.15(c), this parameter is crucial for preventing a large, counterfactual spike in UI entries just before retirement, when the returns to continuing work diminish. Since k_1 is a one-time cost at the beginning of a non-employment spell, it has a negligible effect on subsequent job search effort and thus on non-employment durations.

k_2 , k_3 and γ Parameters (Marginal Cost of Job Search) k_2 , k_3 and γ are the parameters governing the marginal cost of job search, since $\psi_s(s, d^U) = e^{k_2 * d^U} \times k_3 s^\gamma$. k_2 is the duration dependence of the cost of job search, where positive values of k_2 imply that the cost of job search increases the longer the job

search lasts. k_3 is a scalar multiplying the marginal cost of job search. For younger workers, increases in k_2 and k_3 increase the marginal cost of job search, and therefore increase unemployment durations. γ is the elasticity of the cost of job search. Since larger values of γ yield smaller marginal costs of job search (recall s is between 0 and 1), increasing γ reduces unemployment durations. Workers past the bridge age are mostly not searching anyways and thus there is not much of an effect on durations of any of the three parameters. The first order conditions of the model imply that $\frac{1}{\gamma}$ is the elasticity of search effort with respect to the returns to search. For this reason, γ is the key parameter determining the effect of UI extensions on unemployment durations, which can be clearly seen in Table I.10, where gamma has the largest impact on the dD/dP moment. Inflows are not much affected by any of the three parameters.

λ Parameters (Exogenous Job Destruction) The λ parameters govern the exogenous job destruction rate. λ_1 acts as a level shifter, increasing job destruction across all ages. In elasticity terms, its impact is largest for younger workers who have a lower baseline separation rate. λ_2 links job destruction to the unemployment rate, with a higher unemployment rate leading to more separations, an effect also concentrated among younger workers. Both λ_1 and λ_2 have a positive influence on non-employment inflows at all ages. The effect of λ_3 , which captures the role of the change in the unemployment rate, can be positive or negative depending on the direction of the change. The λ parameters primarily affect inflows and have only small, indirect effects on non-employment durations through compositional changes.

η Parameters (Preferences for Non-employment) The η parameters characterize the distribution of worker preferences for non-employment. A higher mean of this distribution, η_{mean} , implies that more workers are close to the margin of indifference between working and not working. Consequently, an increase in η_{mean} raises non-employment inflows at most ages. For the oldest workers (ages 61 and above), this effect is reversed due to a substitution effect: higher inflows at younger ages mean fewer workers are left to enter non-employment later. The standard deviation of the η distribution, η_{SD} , has a similar effect, as a wider distribution also increases the mass of workers near the indifference point. As the influence plots are shown for the 1935 cohort, only $\eta_{\text{mean},1935}$ and η_{SD} are influential, whereas the mean parameters for other cohorts have no effect. Regarding durations, a higher η_{mean} makes work less attractive, reducing search effort and thus leading to longer non-employment spells, particularly for younger workers.

Sensitivity of Model Parameters to Moments

We now turn to the sensitivity of parameter estimates to changes in the empirical moments. Following Andrews et al. (2017), sensitivity is calculated as $-(S'WS)^{-1}S'W$, where S is the Jacobian matrix of $m(\xi)$ and W the weighting matrix. We refer to individual elements of this sensitivity matrix as: $\frac{\partial \xi_j}{\partial \hat{m}_i}$. To facilitate comparisons of magnitudes, we follow Andrews et al. (2017)'s suggestion and scale each element by multiplying with the standard error of the respective moment and dividing by the parameter value: $\frac{\partial \xi_j}{\partial \hat{m}_i} \times SD(\hat{m}_i) \times \frac{1}{|\xi_j|}$, where $\hat{m}_i$ is an empirical moment. The interpretation of this scaled sensitivity is that it is

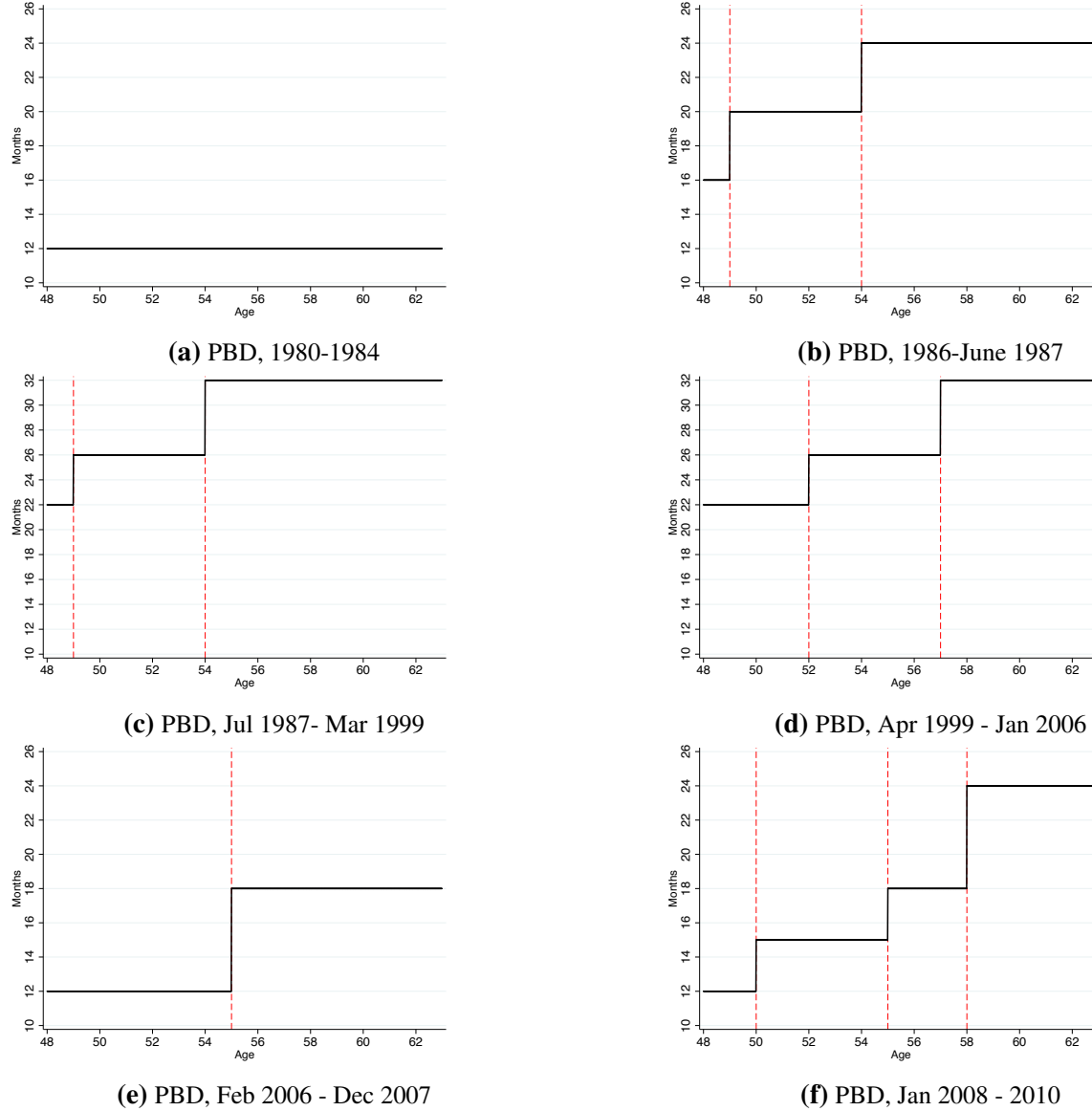
the percentage change in the parameter estimate for a 1 standard error change in the moment. The results are presented in Table I.11 and Figure H.16.

The interpretation of these sensitivity measures is complicated by the inter-dependencies among the parameters. Consider, for instance, the η parameters. Within the 1935 cohort, only $\eta_{\text{mean},1935}$ and η_{SD} directly affect inflows. However, a change in an inflow moment for this cohort will affect the estimated values of all η parameters. For example, if we were to increase the inflow moment at the bridge age for the 1935 cohort, the model could match this by increasing either $\eta_{\text{mean},1935}$ or η_{SD} . The positive sensitivity of $\eta_{\text{mean},1935}$ to the inflow moment at the bridge age (as shown by the positive sensitivity in Table I.11) suggests that an increase in inflows at the bridge age will lead to an increase in $\eta_{\text{mean},1935}$. However, increasing this parameter alone would also raise inflows at other ages in 1935. To counteract this, the model reduces η_{SD} . This reduction, in turn, lowers inflows in the other cohorts, which is then compensated by increases in $\eta_{\text{mean},1929}$ and $\eta_{\text{mean},1950}$.

Since these interactions are complex and we have many moments and parameters, we will refrain from going over all parameters in detail. However, overall, the sensitivity analysis broadly confirms the intuition from the influence analysis. For example, σ is identified primarily by inflow moments around the bridge age. Similarly, the search-related cost parameters k_0 and k_1 are more sensitive to inflow moments than to duration moments, while the parameters governing the marginal cost of search (k_2, k_3, γ) are sensitive to both. Finally, the η and λ parameters are, as expected, more sensitive to inflow moments than to duration moments.

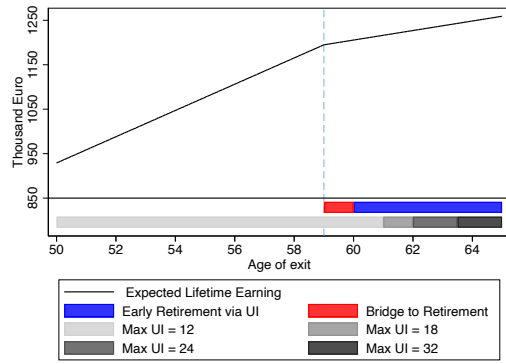
H Appendix Figures

Figure H.1: Maximum Potential UI Benefit Durations (PBDs) by Age for Different Time Periods in Germany

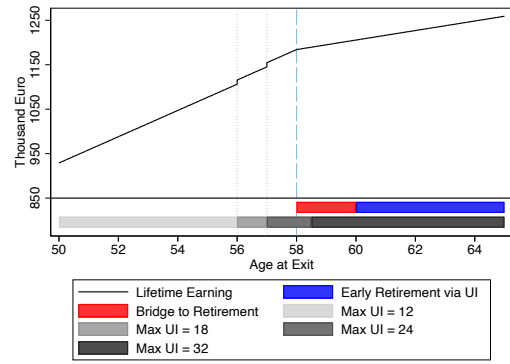


Notes: The figure shows how maximum potential unemployment insurance (UI) benefit durations vary with age and over time in Germany from 1980 to 2010. We drop the brief 1985 regime for presentation purposes. Each figure corresponds to a different UI regime. Appendix Table I.1 contains more detailed information on each institutional regime, including eligibility requirements and benefit levels. The vertical red dash-dotted lines mark the age cutoffs for increases in potential UI durations at different ages.

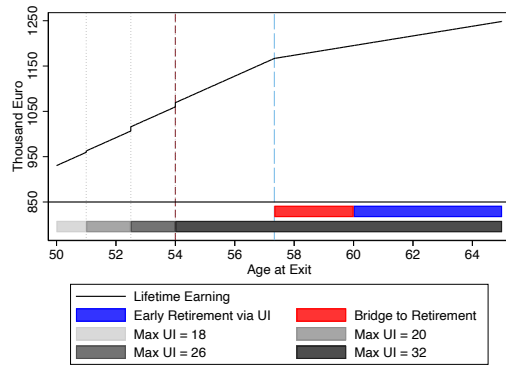
Figure H.2: Stylized Budget Sets for Different Cohorts in Germany, Men



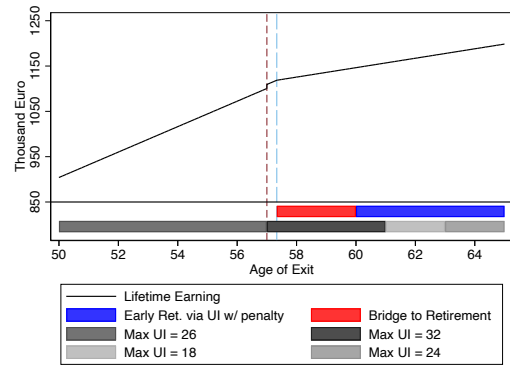
(a) Lifetime Income, 1924 Cohort



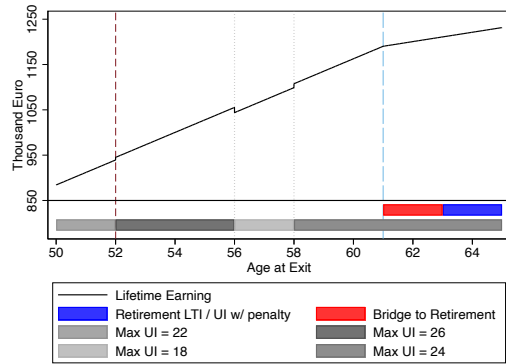
(b) Lifetime Income, 1929 Cohort



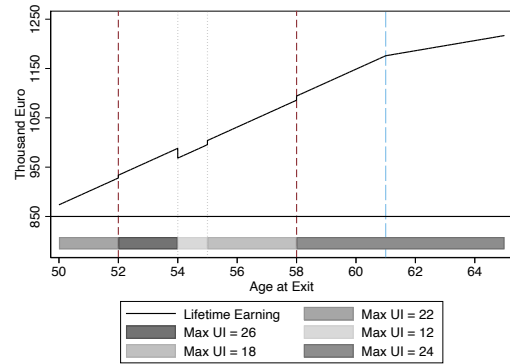
(c) Lifetime Income, 1935 Cohort



(d) Lifetime Income, 1945 Cohort



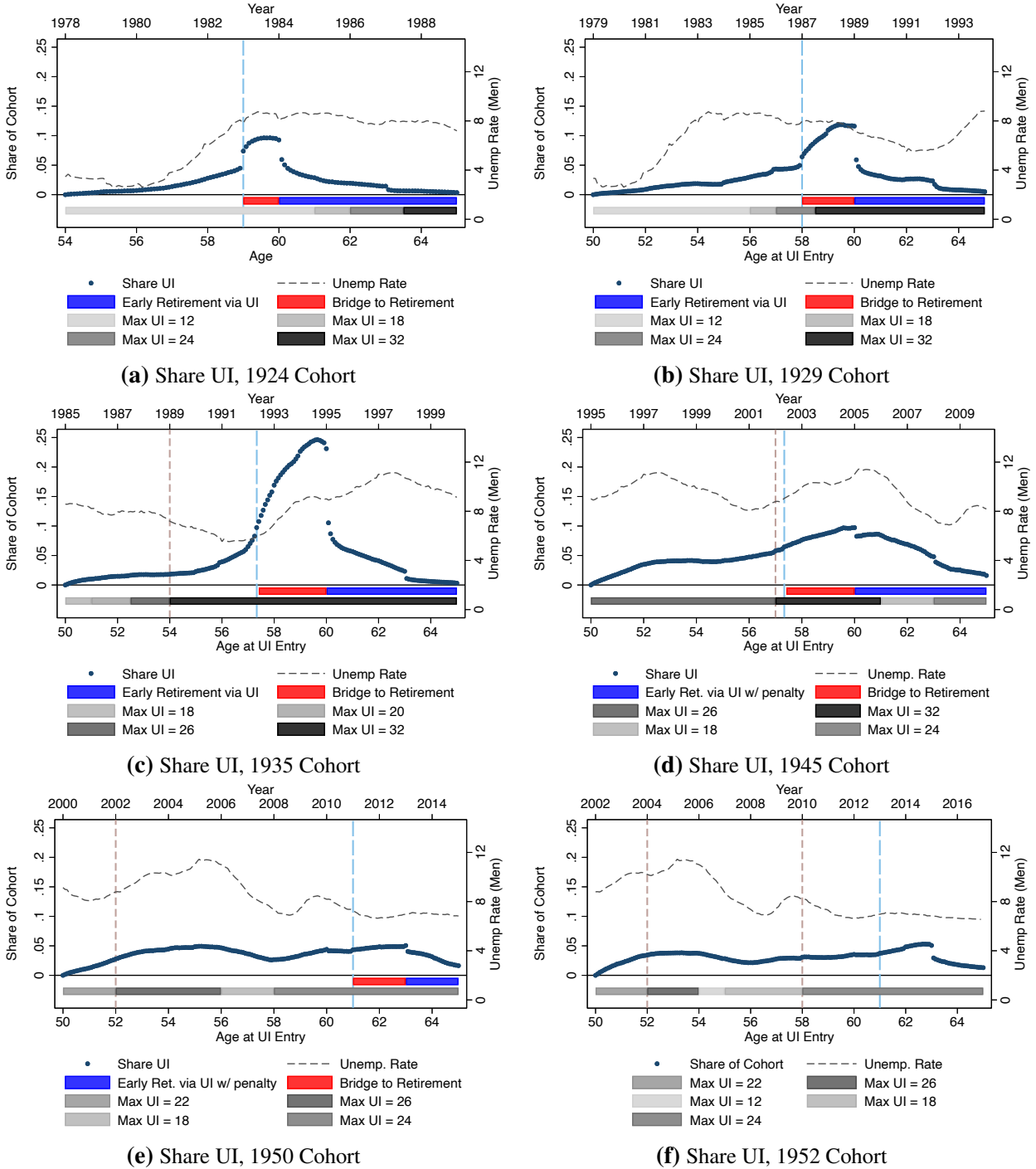
(e) Lifetime Income, 1950 Cohort



(f) Lifetime Income, 1952 Cohort

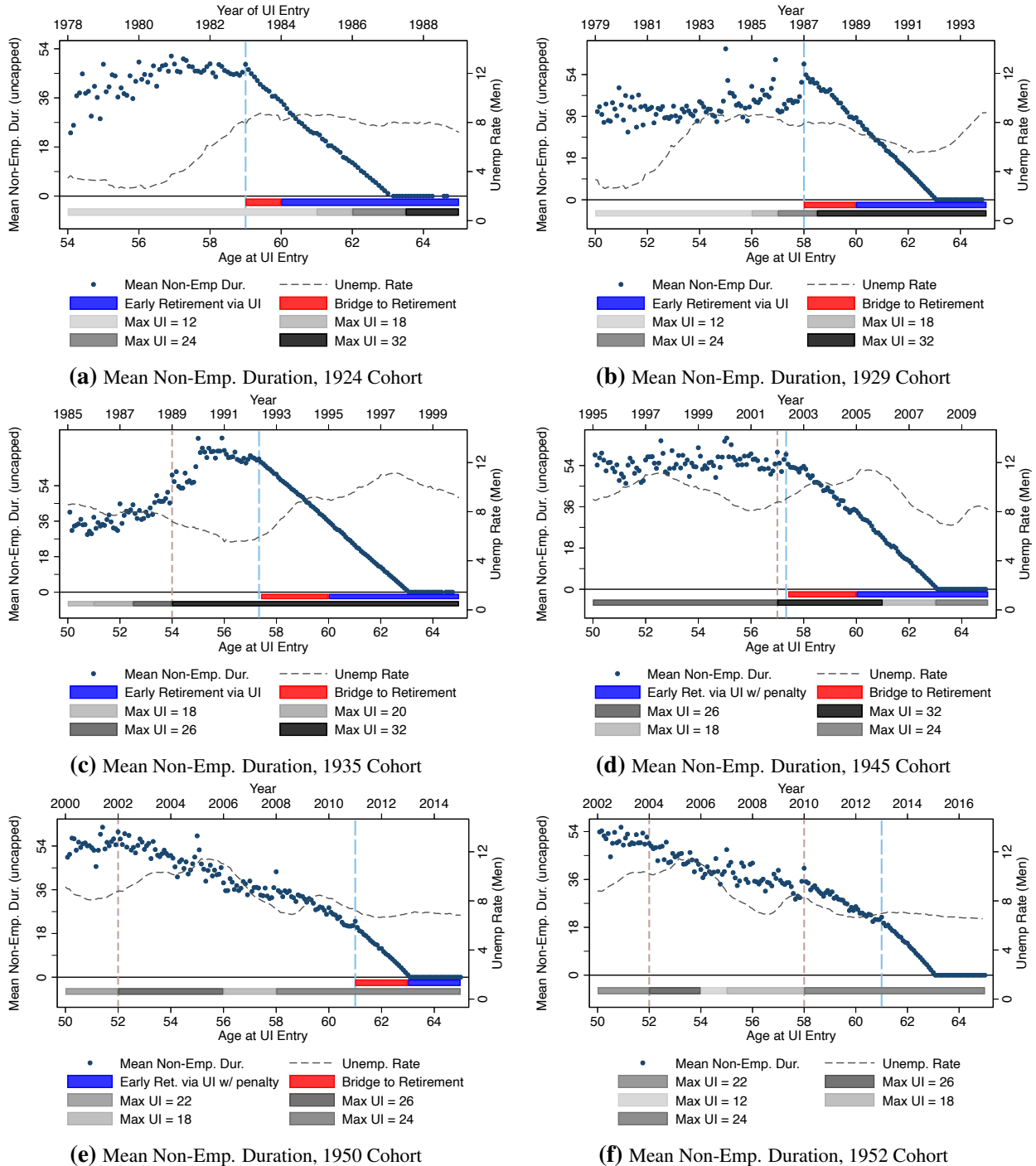
Notes: This figure plots stylized lifetime budget sets by age for different cohorts of West German men in our sample. The red bar under the figure indicates the period over which an individual could receive UI before drawing a pension if he entered UI at the bridge-to-retirement age (the blue dashed line). The blue bar indicates the period over which such an individual would receive their pension. The different shades of gray represent different maximum PBD eligibility for UI, which can change because of an existing age-cutoff (the red dashed line) or because of an overall UI policy change enacted in that year (the gray dotted line).

Figure H.3: Share UI by Age for Different Cohorts in Germany, Men



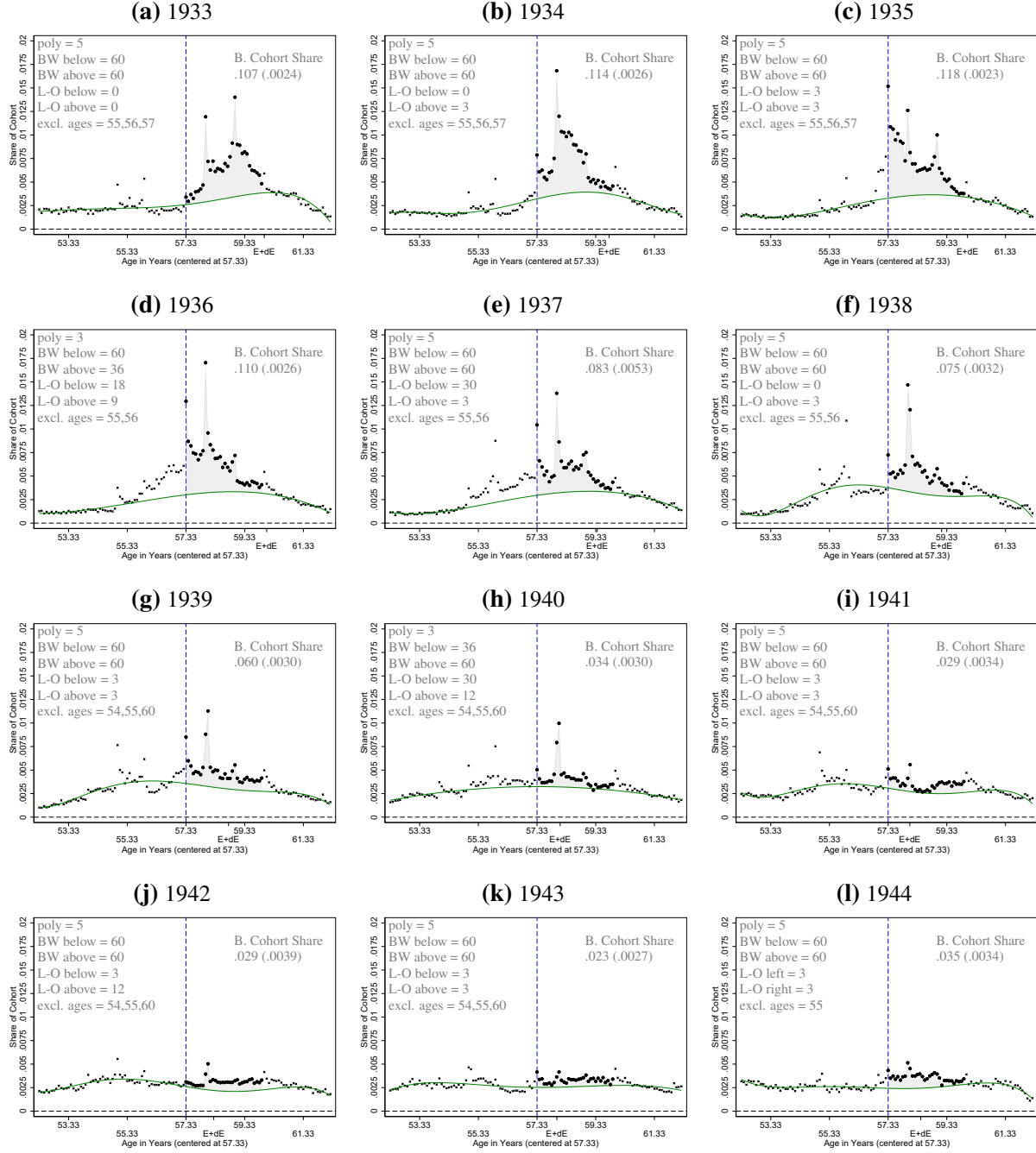
Notes: This figure plots the share of the cohort in UI by age for different cohorts of West German men in our sample (left axis). It also plots the male, seasonally adjusted unemployment rate as a dashed gray line (right axis). The red bar under the figure indicates the period over which an individual could receive UI before drawing pension if he entered UI at the bridge-to-retirement age (blue dashed line). The blue bar indicates the period over which such an individual would receive their pension. Different shades of gray represent different maximum PBD eligibility for UI, which can change because of an age-cutoff (red dashed line) or because of an overall UI policy change enacted in that year (gray dotted line).

Figure H.4: Mean Non-Emp. Duration by Age for Different Cohorts in Germany, Men



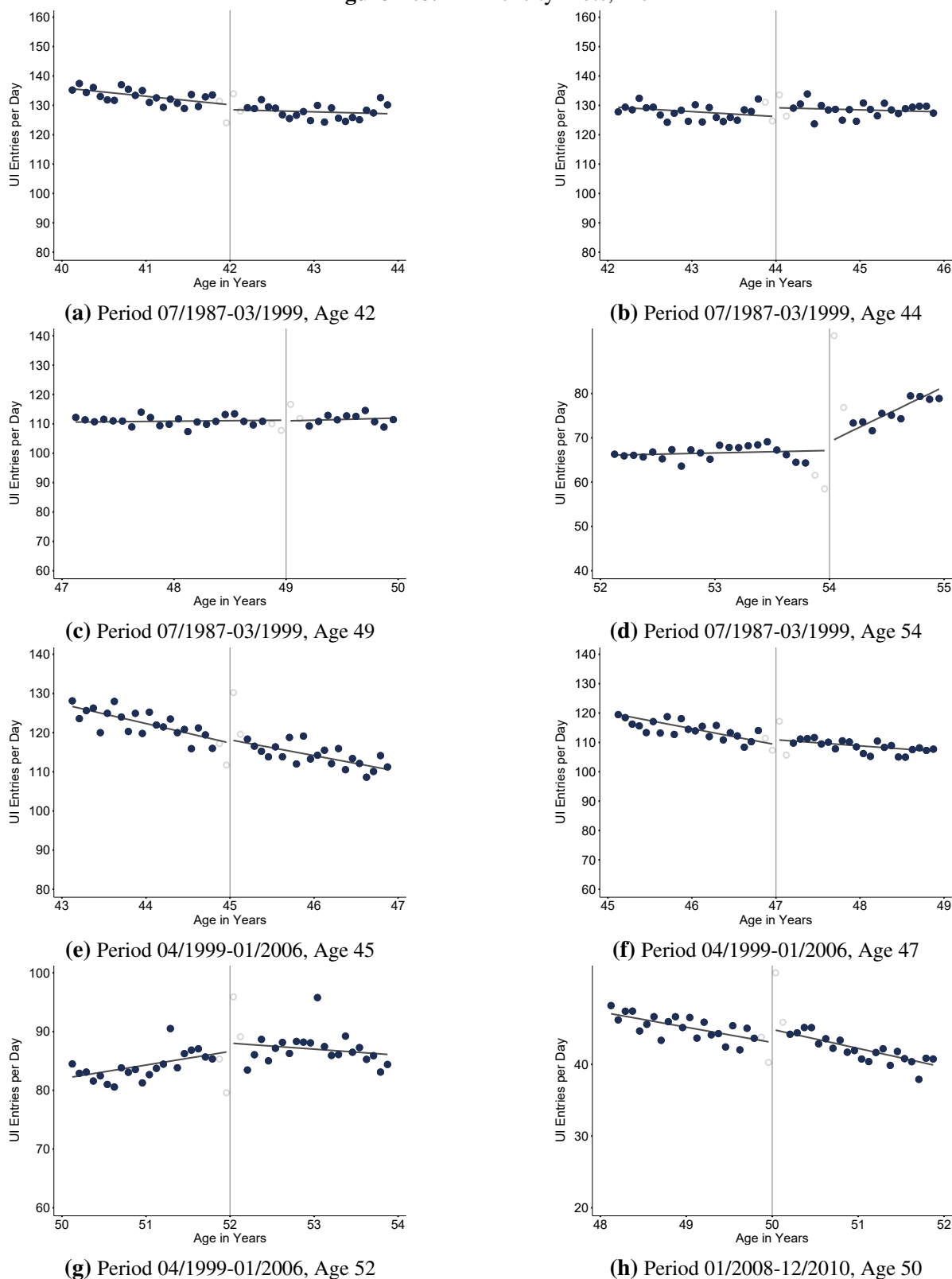
Notes: This figure plots mean non-employment duration (up to age 63) for different cohorts of West German men in our sample (left axis). It also plots the male, seasonally adjusted unemployment rate as a dashed gray line (right axis). The red bar under the figure indicates the period over which an individual could receive UI before drawing a pension if he entered UI at the bridge-to-retirement age (blue dashed line). The blue bar indicates the period over which such an individual would receive their pension. Different shades of gray represent different maximum PBD eligibility for UI, which can change because of an age-cutoff (red dashed line) or because of an UI policy change enacted in that year (gray dotted line).

Figure H.5: Bunching Figures for Cohorts Around the Retirement Penalty Phase-In



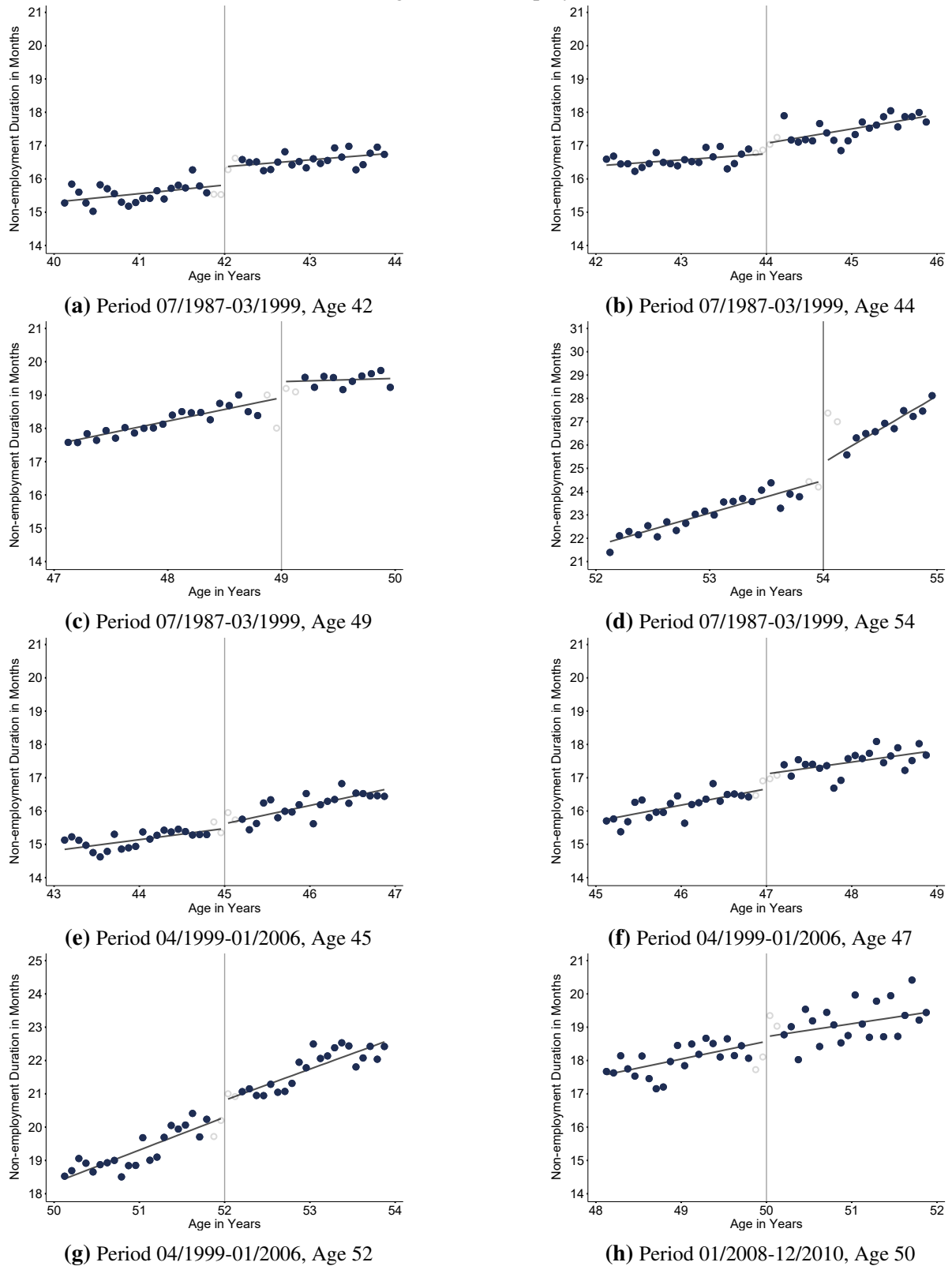
Notes: These figures shows cohort specific number of UI entries relative to the overall number of individuals in the cohort sample (black dots) the estimated counterfactual distribution (green solid line) and the resulting estimated excess bunching (shaded gray area) for cohorts displayed in figure 3 (cohort 1945 excluded due to redundancy to figure 2). The estimates of the excess bunching mass as a share of the cohort are depicted on the upper right of each sub-figure (standard errors are in parentheses). Cohort-specific parameters when estimating the counterfactual are depicted on the upper left. "Poly" refers to the degree of polynomial, "BW" the bandwidth, "L-O" to the leave-out region above and below the UI-bridge period. "excl. ages" refers to round ages outside the bunching region that are excluded due to the excess mass. For further details on the bunching estimation, see also Appendix B.

Figure H.6: RD Density Plots, Men



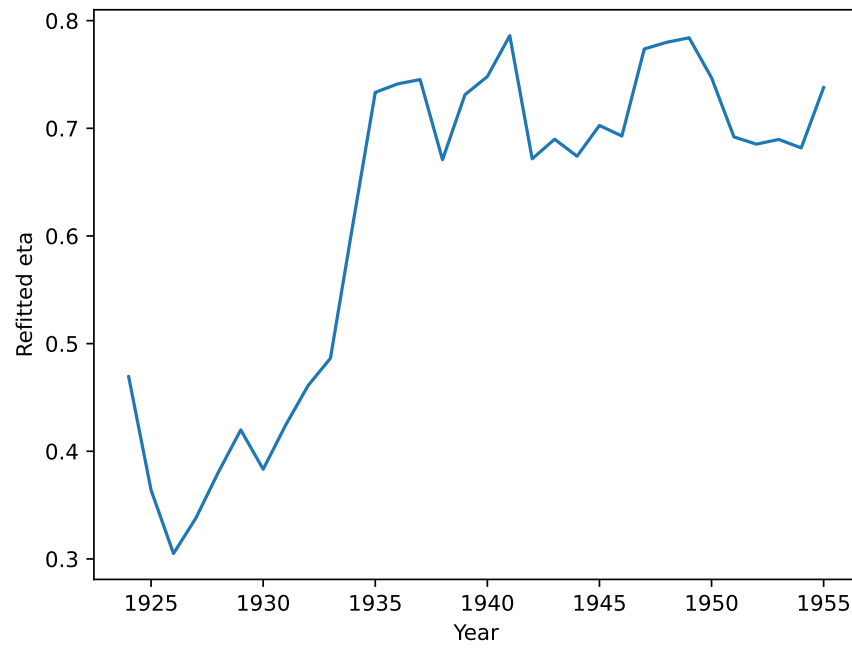
Notes: This figure shows the average number of UI entries by age. Each dot shows this mean over a one-month window. Transparent dots close to the cutoff mark the leave-out region and solid lines show the line of best fit in the running variable.

Figure H.7: RD Figures: Non-employment Duration, Men



Notes: This figure shows the average non-employment duration (capped at 36 months) around the different age cutoffs. Each dot shows this mean over a one-month window. Transparent dots close to the cutoff mark the leave-out region, and solid lines show the line of best fit in the running variable.

Figure H.8: Cohort-specific estimates of mean disutility of work ($\bar{\eta}$)



Notes: This figure plots our model's cohort-specific estimates of the mean disutility of work ($\bar{\eta}$). For the three in-sample cohorts, this is estimated directly along with all other parameters. For the out-of-sample cohorts, this single parameter is estimated (taking all other parameters as given) to fit E to U transitions and nonemployment durations.

Figure H.9: Empirical and Simulated UI Inflows for all Cohorts (Baseline Model)

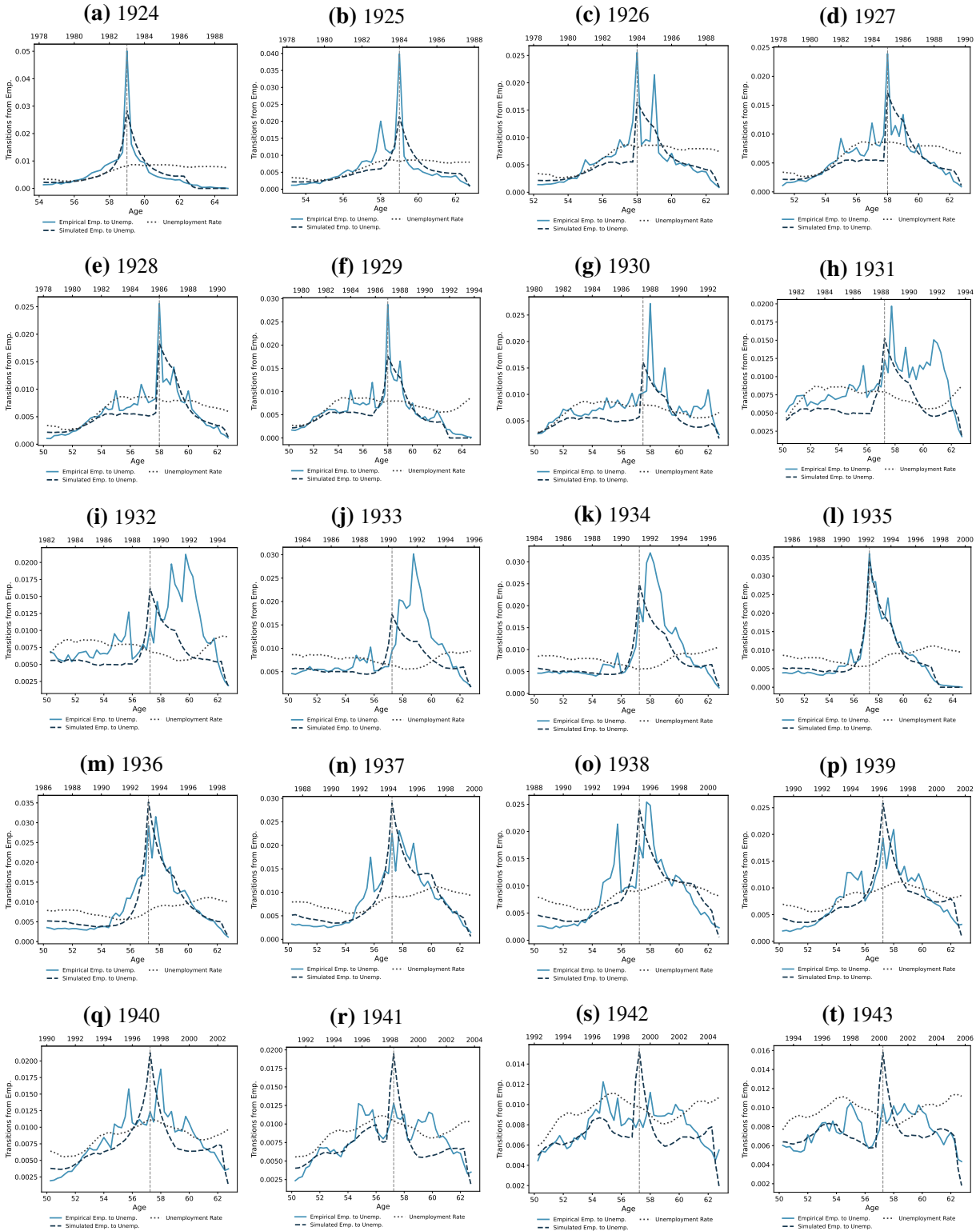
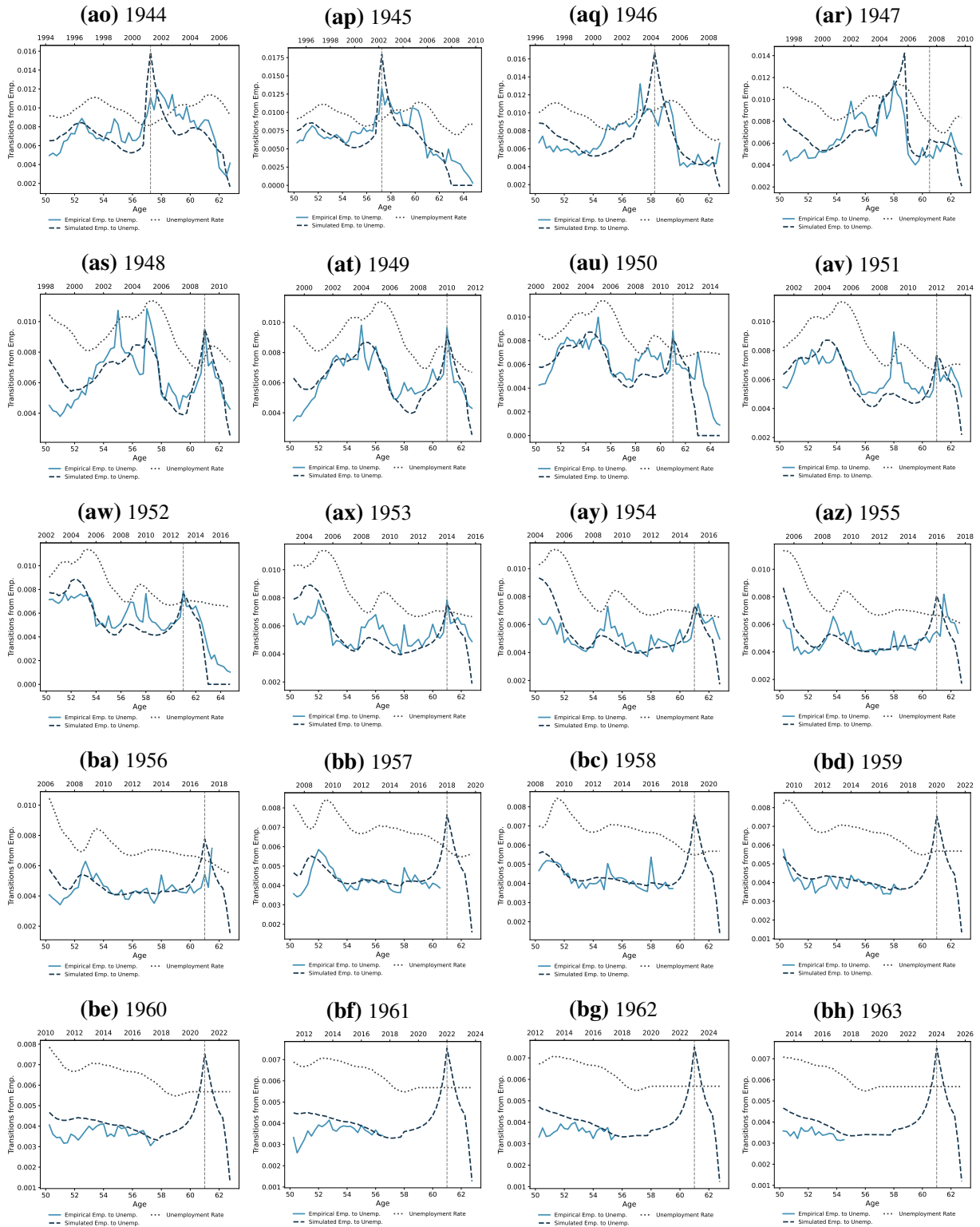


Figure H.9: Empirical and Simulated UI Inflows for all Cohorts (Baseline Model), continued



Notes: These figures compare the model-based transitions from employment to unemployment to their corresponding empirical moments for all cohorts, aggregated to the quarterly level.

Figure H.11: Empirical and Simulated Nonemployment Durations for Cohorts 1924–1943 (Baseline Model)

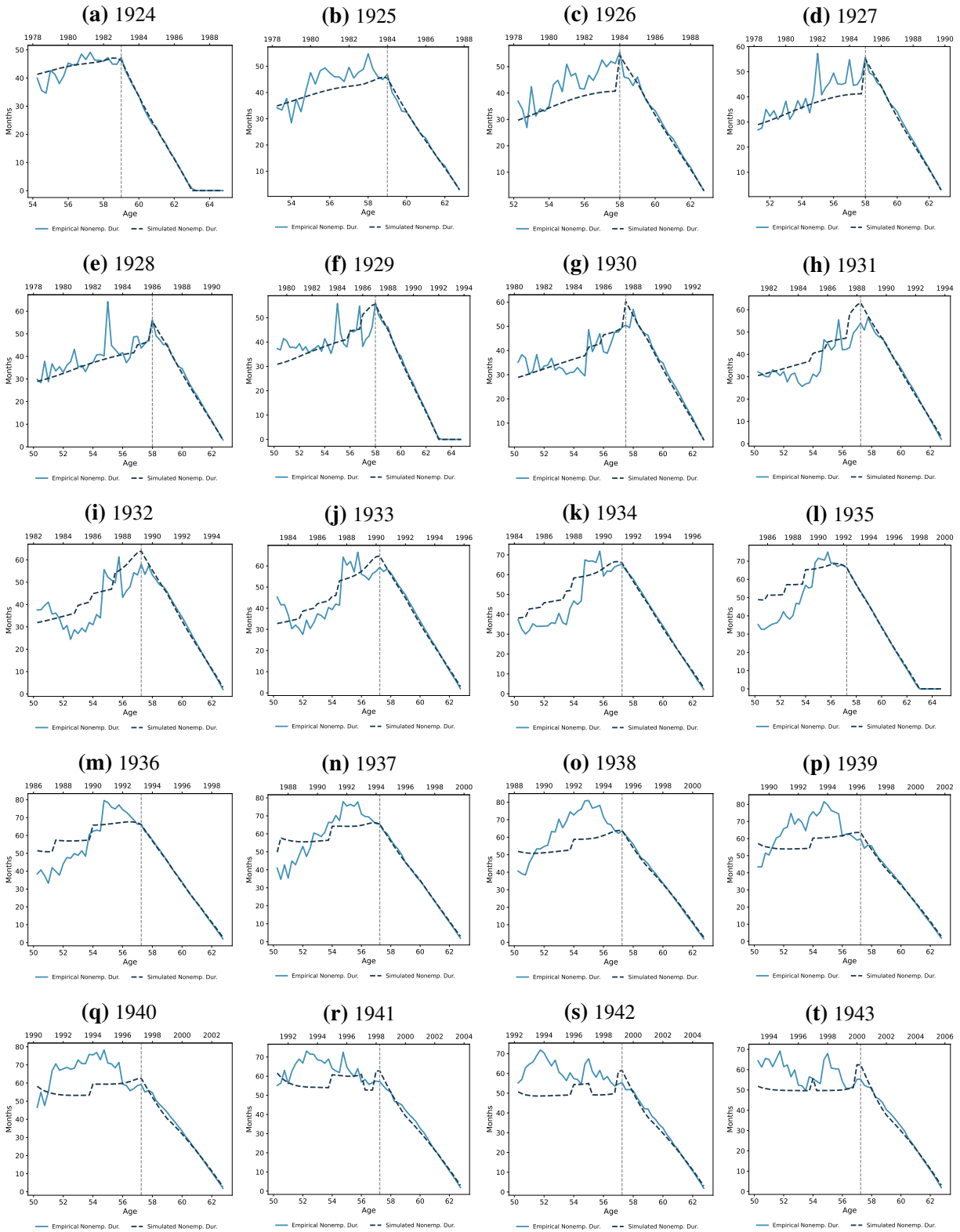
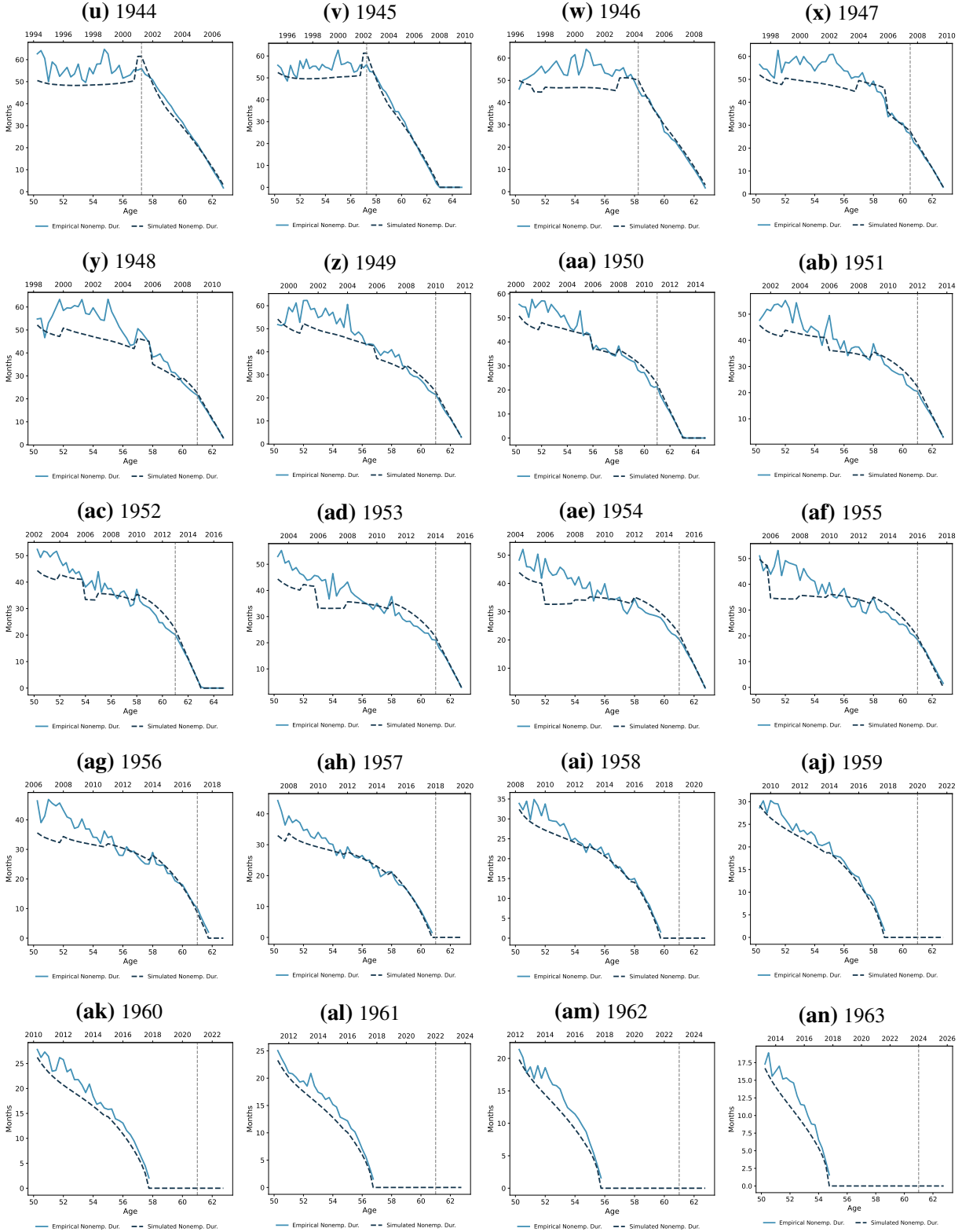
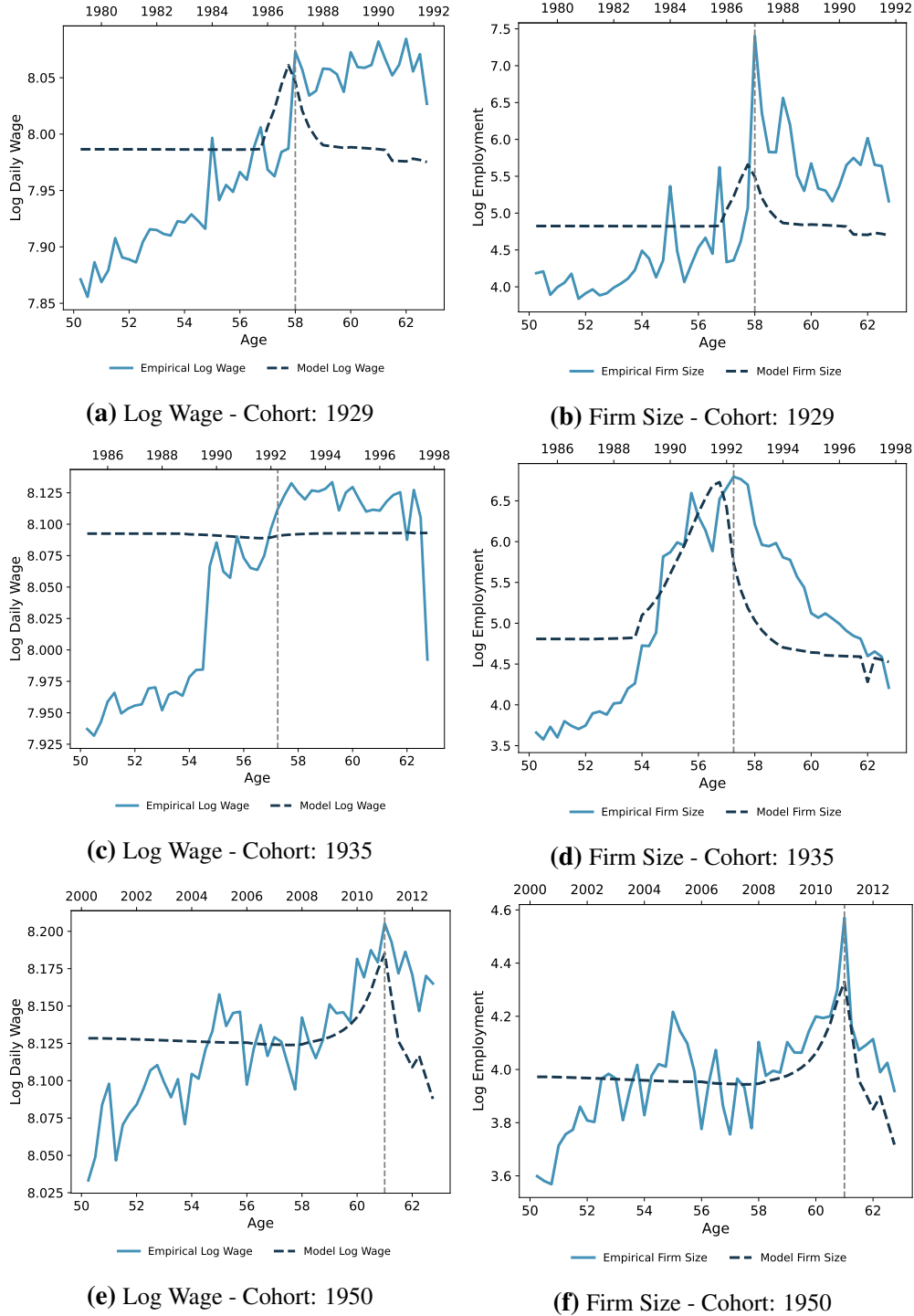


Figure H.12: Empirical and Simulated Nonemployment Durations for Cohorts 1944–1963 (Baseline Model)



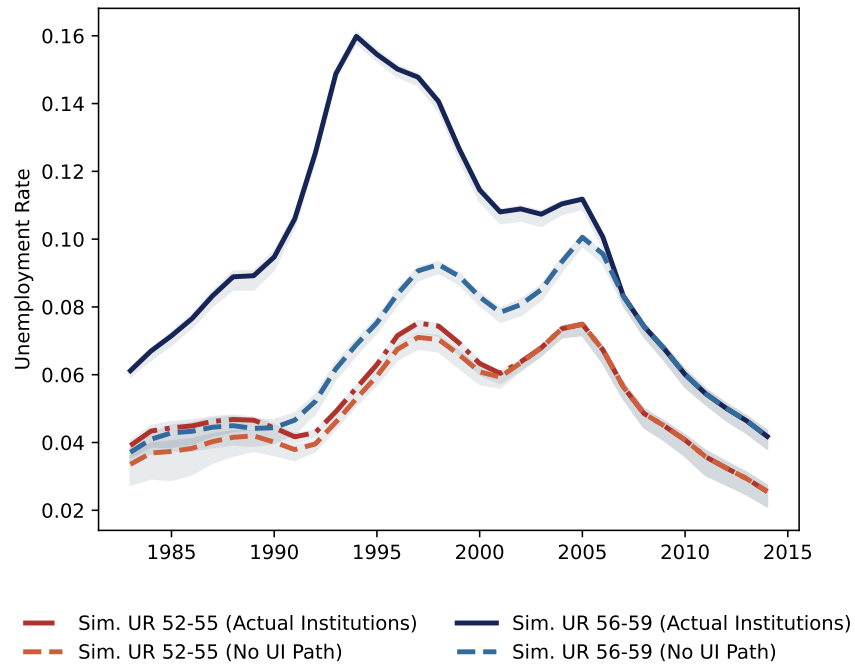
Notes: These figures compare model-based non-employment durations to their corresponding empirical moments for all cohorts, aggregated to the quarterly level.

Figure H.13: Pre-Unemployment Log Wage and Firm Size of UI Entrants by Age



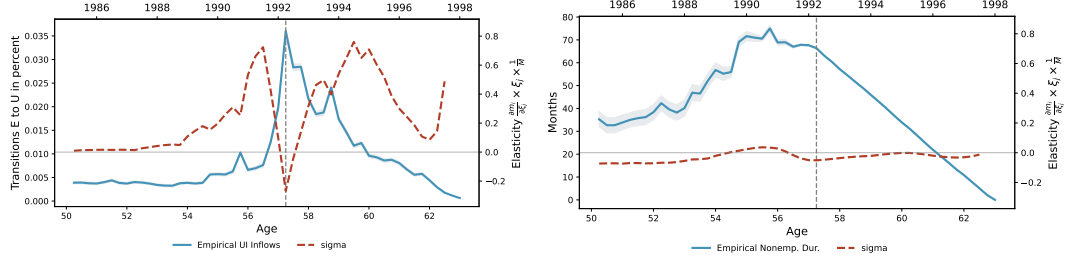
Notes: This figure shows the pre-unemployment log wage and firm size (log of the number of employees) of UI entrants by age of entering UI. The log wage and firm size are measured at age 50 (the beginning of our sample period). The model log wages and firm sizes are computed from the severance pay model (see Section V.B, Robustness) and show the predicted values of a (cohort specific) regression of log wage and log firm size on the log of severance pay.

Figure H.14: Simulated Unemployment Rate by Age Group: Actual vs. No UI Pathway

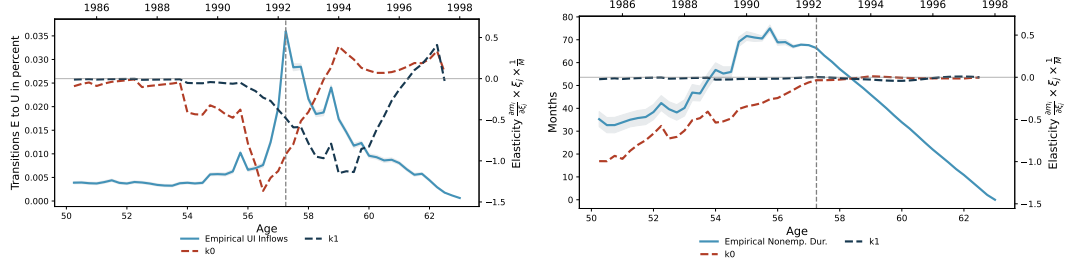


Notes: This figure shows the simulated unemployment rate from the model for two age groups: 52-55 years old and 56-59 years old under two scenarios. The solid lines show the model simulation under the actual institutions, the dashed lines show model simulations for a counterfactual in which the UI pathway does not exist, and hence the earliest possible retirement age was at 63.

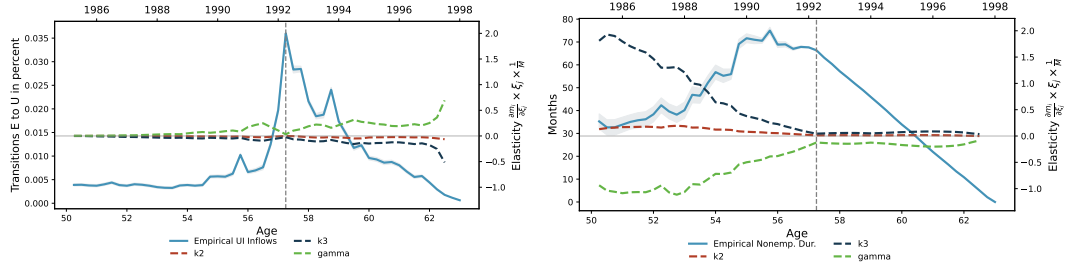
Figure H.15: Influence of Model Parameters on Simulated Moments – 1935 Cohort



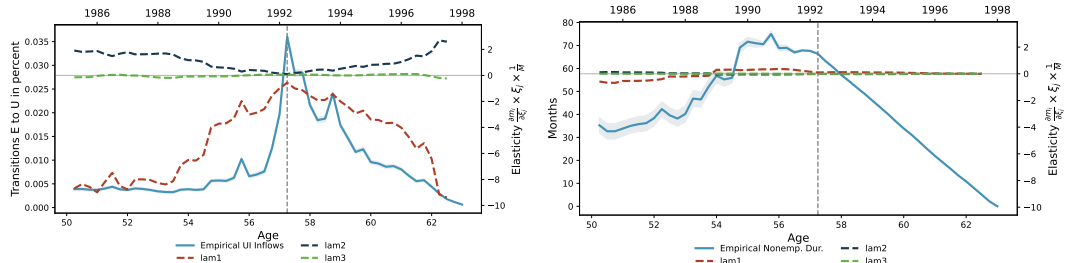
(a) Std. dev. of productivity shock (sigma) (Transitions) **(b)** Std. dev. of productivity shock (sigma) (Durations)



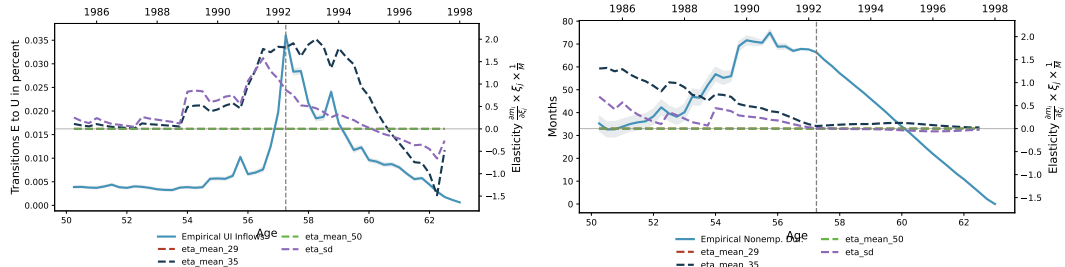
(c) Fixed costs of Job Search (k0) and Entering UI (k1) (Transitions) **(d)** Fixed costs of Job Search (k0) and Entering UI (k1) (Durations)



(e) Search cost trend (k2), slope (k3), and elasticity (gamma) (Transitions) **(f)** Search cost trend (k2), slope (k3), and elasticity (gamma) (Durations)



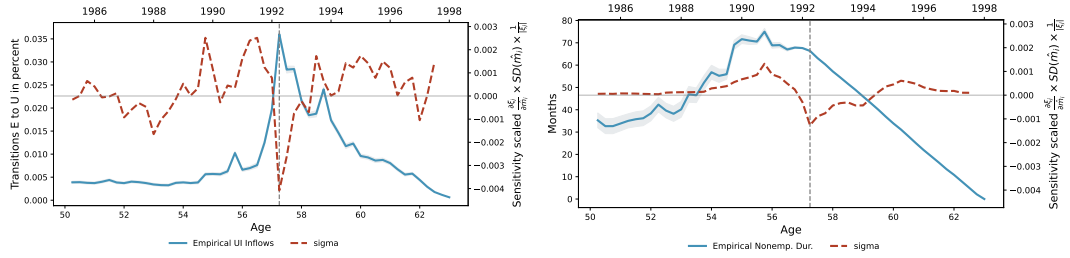
(g) Parameters (lam1, lam2, lam3) (Transitions) **(h)** Parameters (lam1, lam2, lam3) (Durations)



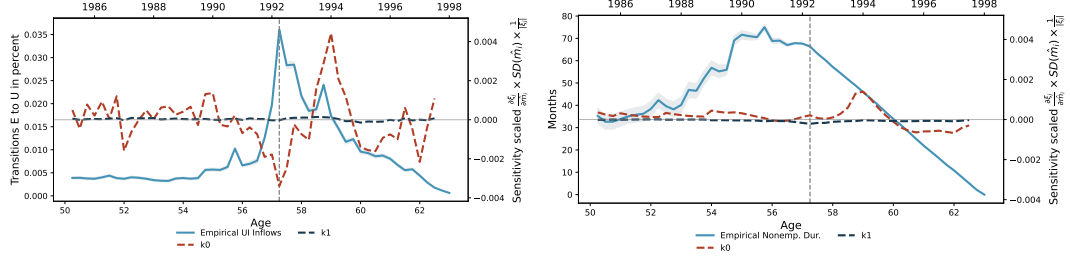
(i) Eta means and standard deviation (Transitions) **(j)** Eta means and standard deviation (Durations)

Notes: These figures show the influence of the model parameters on the simulated moments for the 1935 cohort. The influence is calculated as $\frac{\partial m_i}{\partial \xi_j} \times \frac{|\xi_j|}{m_i}$.

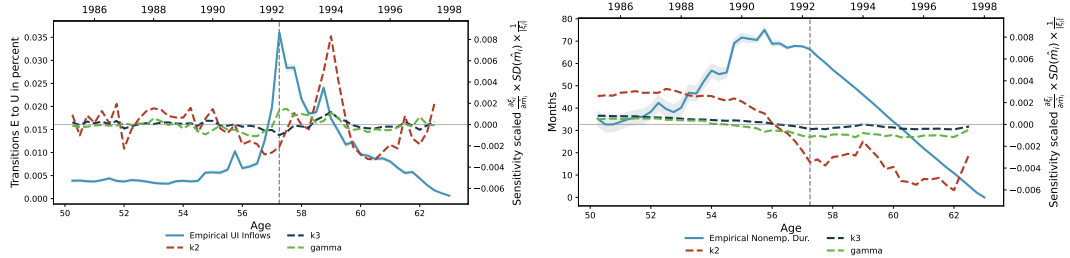
Figure H.16: Sensitivity of Model Parameters to Empirical Moments – 1935 Cohort



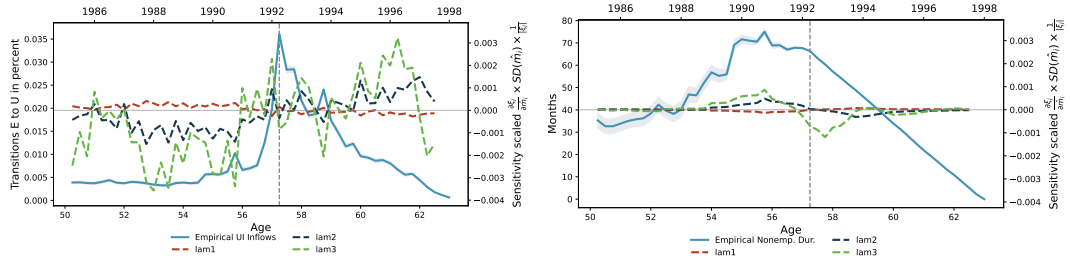
(a) Std. dev. of productivity shock (sigma) (Transitions) **(b)** Std. dev. of productivity shock (sigma) (Durations)



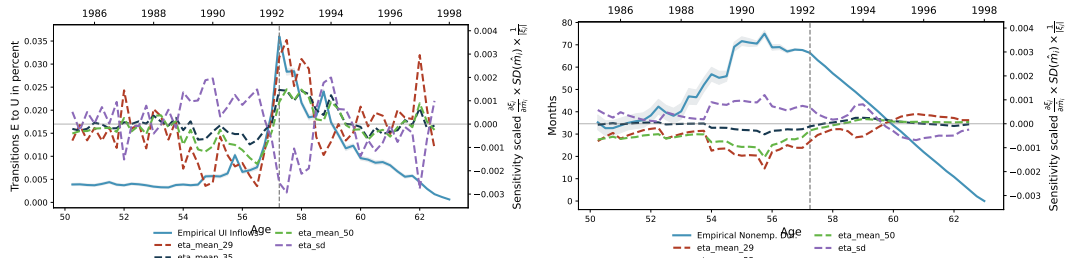
(c) Fixed costs of Job Search (k0) and Entering UI (k1) (Transitions) **(d)** Fixed costs of Job Search (k0) and Entering UI (k1) (Durations)



(e) Search cost trend (k2), slope (k3), and elasticity (gamma) (Transitions) **(f)** Search cost trend (k2), slope (k3), and elasticity (gamma) (Durations)



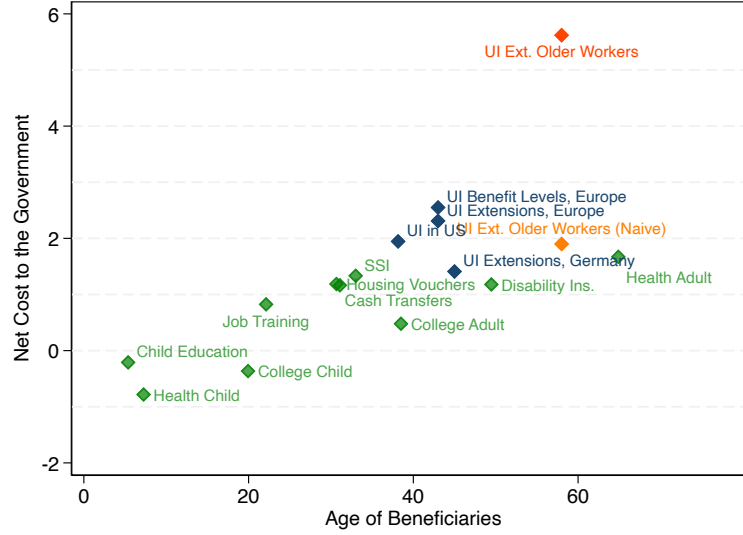
(g) Parameters (lam1, lam2, lam3) (Transitions) **(h)** Parameters (lam1, lam2, lam3) (Durations)



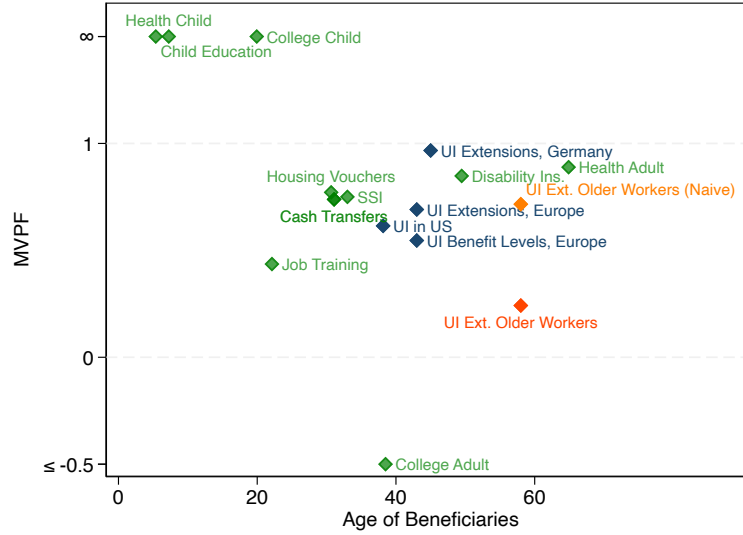
(i) Eta means and standard deviation (Transitions) **(j)** Eta means and standard deviation (Durations)

Notes: These figures show the sensitivity of the model to the parameters to empirical moments for the 1935 cohort (Andrews et al., 2017). The sensitivity is calculated as $\frac{\partial \xi_j}{\partial \hat{m}_i} \times \frac{SD(\hat{m}_i)}{|\xi_j|}$.

Figure H.17: MVPF and Net Cost to the Government of Different UI and Other Expenditures



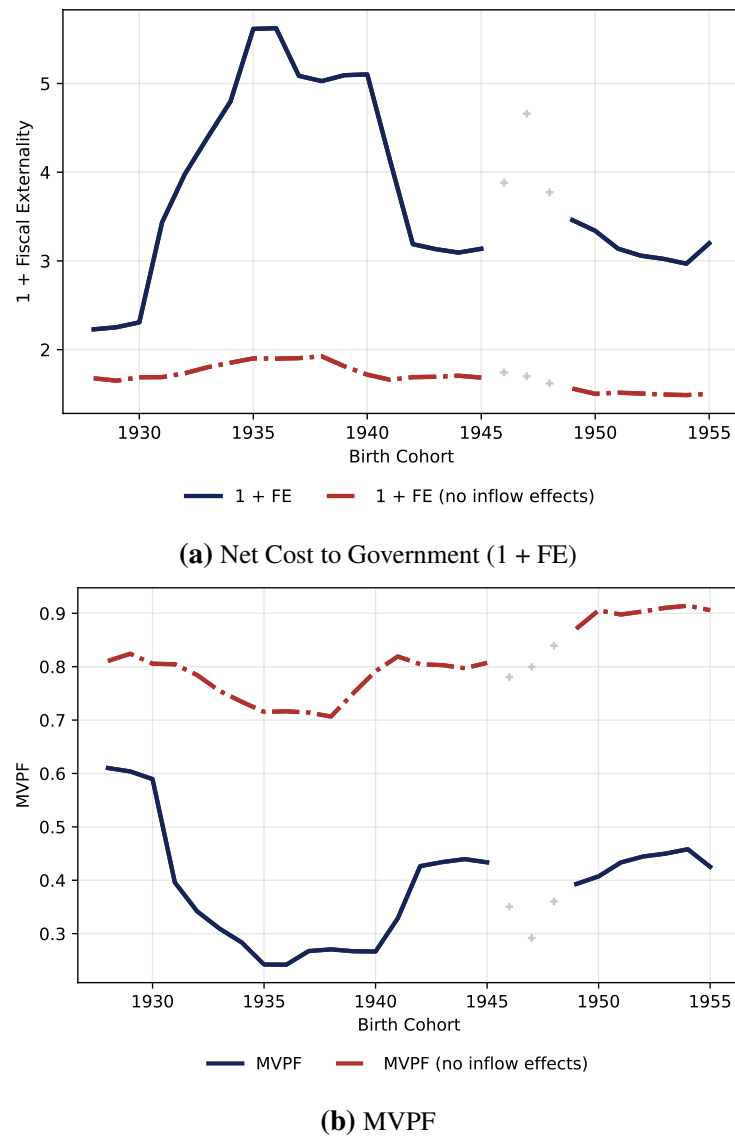
(a) Net Cost



(b) MVPF

Notes: This figure plots the net cost to the government ($= 1 + \text{Fiscal Externality}$) and the associated marginal value of public funds (MVPF) for various policies. The green diamonds are program groups taken directly from [Hendren and Sprung-Keyser \(2020\)](#). The four blue diamonds are UI policies; UI in U.S. is the group-level average of U.S. UI policies as reported in [Hendren and Sprung-Keyser \(2020\)](#). The points for “UI extensions, Europe”, and “UI benefit levels, Europe”, are averages across all PBD and all Benefit policies, respectively, taken from [Le Barbanchon et al. \(2024\)](#). We omit two papers ([Lalive, 2007, 2008](#)) from these averages, where the constant hazard approximation used to calculate the values performs poorly. “The UI Extensions, Germany” point reports the fiscal externality and MVPF associated with Germany’s 3 prime age PBD extensions studied in [Schmieder et al. \(2012\)](#) for workers ages 42, 44, and 49. Finally, in orange and red we report the results from this paper for the effects of extending PBD among older workers (specifically for the 1935 birth cohort), based on our model. The dark red “UI Ext. Older Workers” contains the full model results. The lighter orange “UI Ext. Older Workers (Naive)” shows what we obtain from our model when we allow PBD to only affect UI durations and not inflows.

Figure H.18: The Marginal Value of Public Funds and the Net Cost to Government (1+FE) by Birth Cohort



Notes: Panel (a) shows the net cost to government, i.e., one plus the fiscal externality, of spending an additional Euro to extend the UI benefit duration for each cohort. Panel (b) shows the MVPF, i.e., the ratio of WTP to the net cost to the government. The birth cohorts 1946, 1947, and 1948 are grayed out since they are unusual in that they were affected by two reforms: the increase in the early retirement age via UI as well as a sharp reduction in the maximum potential benefit duration in 2006 (see [Dlugosz et al. \(2014\)](#)), which led to a sharp increase in UI inflows in 2006 which leads to a larger fiscal externality in these cohorts.

I Appendix Tables

Table I.1: Potential Unemployment Insurance Benefit (UIB) Durations as a Function of Age and Months Worked in the Previous 7 Years.

Months Worked in prev. X years	January 1983- December 1984	January 1985- December 1985	January 1986- June 1987	July 1987- March 1997	April 1997* - January 2006	February 2006 - December 2007	January 2008 - Present
12	4	4	4	6	6	6	6
16	4	4	4	8	8	8	8
18	6	6	6	8	8	8	8
20	6	6	6	10	10	10	10
24	8	8	8	12	12	12	12
28	8	8	8	14 (>42)	14 (>45)	12	12
30	10	10	10	14 (>42)	14 (>45)	15 (>55)	15 (>50)
32	10	10	10	16 (>42)	16 (>45)	15 (>55)	15 (>50)
36	12	12	12	18 (>42)	18 (>45)	18 (>55)	18 (>55)
40	12	12	12	20 (>44)	20 (>47)	18 (>55)	18 (>55)
42	12	14 (>49)	14 (>44)	20 (>44)	20 (>47)	18 (>55)	18 (>55)
44	12	14 (>49)	14 (>44)	22 (>44)	22 (>47)	18 (>55)	18 (>55)
48	12	16 (>49)	16 (>44)	24 (>49)	24 (>52)	18 (>55)	24 (>58)
52	12	16 (>49)	16 (>44)	26 (>49)	26 (>52)	18 (>55)	24 (>58)
54	12	18 (>49)	18 (>49)	26 (>49)	26 (>52)	18 (>55)	24 (>58)
56	12	18 (>49)	18 (>49)	28 (>54)	28 (>57)	18 (>55)	24 (>58)
60	12	18 (>49)	20 (>49)	30 (>54)	30 (>57)	18 (>55)	24 (>58)
64	12	18 (>49)	20 (>49)	32 (>54)	32 (>57)	18 (>55)	24 (>58)
66	12	18 (>49)	22 (>54)	32 (>54)	32 (>57)	18 (>55)	24 (>58)
72	12	18 (>49)	24 (>54)	32 (>54)	32 (>57)	18 (>55)	24 (>58)
Rahmenfrist - Min emp dur. for new UI eligibility	12	12	12	12	12	12	12
X - Base Period for P≥12	7	7	7	7	7	5	5
X - Base Period for P<12	4	4	4	3	3	2	2
Replacement Rates on Net Wages in Percent:							
UI (children)	68	68	68	67 [‡]	67	67	67
UI (no children)	63 [†]	63	63	60 [‡]	60	60	60
UA (children)	58	58	58	57 [‡]	57	UIB II	UIB II
UA (no children)	53 [†]	53	53	50 [‡]	50	UIB II	UIB II

Source: Hunt (1995), Bundesgesetzblatt (1983-2015) and Dlugosz et al (2014).

Notes:*The reform in 1997 was phased in gradually: For workers who had worked for more than one year during the three years before April 1997, the old rules applied until March 1999 (See Arntz, Simon Lo, and Wilke 2007).

[†] UI and UA replacement rates were lowered starting in January 1984. Until December 1983, ALG was 68 percent and ALH 58 percent of the previous net wage, irrespective of whether the recipient had children.

[‡] UI and UA were lowered starting in January of 1994.

Table I.2: Retirement age by retirement pathways from 1957 till now

Pathways	Time of implementation	Affected cohorts	SRA		Reform
Standard old-age pension (Years of contribution: 5 [†])	1957 - 2011	<1947 Jan	65	-	
	2012 - 2030	1947 Jan- 1964 Jan	65 to 67	-	2007 Reform
	> 2031	≥1964 Jan	67	-	
			NRA (no penalty)	ERA (earliest possible)	
Old-age pension for long-term insured (Years of contribution:35)	1972 - 1999	1909 Jan - 1936 Dec	63	-	1972 Reform [‡]
	2000 - 2003	1937 Jan - 1938 Dec	63 to 65	63	1992 Reform [§]
	2004 - 2010	1939 Jan - 1947 Dec	65	63	
	2011 - 2030	1949 Jan - 1964 Jan	65 to 67	63	2007 Reform *
Old-age pension due to unemployment or part-time work (at least 52 weeks unemployed after 58½, or 2 years part-time) (Years of contribution: 15(8 in last 10 yrs))	1972 - 1996	< 1937 Jan	60	-	1972 Reform
	1997 - 2006	1937 Jan - 1941 Dec	60 to 65	60	1992/99 Reform
		1942 Jan - 1945 Dec	65	60	
	2006 - 2011	1946 Jan - 1948 Dec	65	60 to 63	1992 Reform
	2012 - 2016	1949 Jan - 1951 Dec	65	63	
	> 2017 Jan	> 1952 Jan	Phased out	-	2007 Reform
Old-age pension for women (Years of contribution: 15 (10 after age 40))	1957 - 2000	<1940 Jan	60	-	
	2000 - 2009	1940 Jan - 1944 Dec	60 to 65	60	1992 Reform
	2010 - 2016	1945 Jan - 1951 Dec	65	60	
	> 2017 Jan	> 1952 Jan	Phased out	-	2007 Reform
Old-age pension for disabled workers (Years of contribution: 35) (Loss of at least 50 percent of earnings capability)	1972 - 1977	1911 - 1917	62	-	1972 Reform
	1978 - 1980	1918 Jan - 1919 Dec	62 to 60	-	1978 Reform
	1981- 2000	1920 Jan - 1940 Dec	60	-	
	2001 - 2006	1941 Jan - 1943 Dec	60 to 63	60	1992 Reform
	2007 - 2011	1944 Jan - 1951 Dec	63	60	
	2012 - 2025	1952 Jan - 1963 Dec	63 to 65	60 to 62	2007 Reform
	> 2026	> 1964 Jan	65	62	
Old-age pension for especially long-term insured (qualifying period of 45 years)	2014-2016	1951- 1953	63	-	
	2016 - 2028	1953 Jan - 1963 Dec	63 to 65	-	2014 Reform
	> 2029 Jan	> 1964 Jan	65	-	
Disability pension : independent of age[◇]	<1985	5 years of contribution			
	> 1985	5 yrs with minimum 3 in last 5 yrs			1984 Reform

Source: Sozialgesetzbuch (SGB) Sechstes Buch (VI), Börsch-Supan and Jürges (2012), Börsch-Supan and Wilke (2006), Giesecke and Kind (2013).

Notes:

[†] Before 1984, the required contribution period was 15 years.

[‡] The 1972 reform made "flexible retirement" after age 63 with full benefits possible for the long-term insured; Moreover, retirement at age 60 with full benefits became possible for women, the unemployed, and older disabled workers.

[§] The 1992 reform introduced actuarial adjustment. Since then, we distinguish ERA and NRA. It also increased NRA to 65 for all pathways except for disabled workers. It increased ERA for the unemployed to 63 (See SGB VI Appendix 19).

* The 2007 reform announced the stepwise increase of SRA between 2012 and 2029 from 65 to 67 for both men and women (see SGB VI 235). For cohorts born in 1952 and after, the old-age pension for women and the old-age pension due to unemployment are abolished.

◇ German statutory pension insurance distinguishes old-age pensions from disability (reduced-earning-capacity) pensions; the latter are converted to regular old-age pensions upon reaching the statutory retirement age.

Table I.3: Intensive Margin Effects of UI Extension on Nonemployment Duration

		No Controls	Controls
Period Jul 1987 - Feb 1999			
Age 42, P: (12-18), ΔP : 6	$\frac{dy}{dP}$	0.092	0.080
		[0.026]**	[0.025]**
N		173,313	173,313
Mean Dep. Var		16.049	16.049
Age 44, P: (18-22), ΔP : 4	$\frac{dy}{dP}$	0.079	0.068
		[0.041]+	[0.039]+
N		170,270	170,270
Mean Dep. Var		17.046	17.046
Age 49, P: (22-26), ΔP : 4	$\frac{dy}{dP}$	0.121	0.103
		[0.068]+	[0.062]
N		107,255	107,255
Mean Dep. Var		18.568	18.568
Age 54, P: (26-32), ΔP : 6	$\frac{dy}{dP}$	0.129	0.126
		[0.053]*	[0.048]**
N		66,720	66,720
Mean Dep. Var		24.331	24.331
Period Mar 1999 - Jan 2006			
Age 45, P: (12-18), ΔP : 6	$\frac{dy}{dP}$	0.024	0.024
		[0.028]	[0.027]
N		156,927	156,927
Mean Dep. Var		15.637	15.637
Age 47, P: (18-22), ΔP : 4	$\frac{dy}{dP}$	0.113	0.104
		[0.044]*	[0.042]*
N		148,285	148,285
Mean Dep. Var		16.794	16.794
Age 52, P: (22-26), ΔP : 4	$\frac{dy}{dP}$	0.128	0.126
		[0.049]**	[0.048]**
N		113,128	113,128
Mean Dep. Var		20.546	20.546
Period Jan 2008 - Dec 2010			
Age 50, P: (12-15), ΔP : 3	$\frac{dy}{dP}$	0.048	0.062
		[0.103]	[0.100]
N		57,116	57,116
Mean Dep. Var		18.539	18.539

Notes: This table shows RD estimates of the effect of a 1 month UI PBD extension at various age cutoffs on non-employment duration in months (capped at 36 months). Estimates are obtained using local polynomial regressions controlling linearly for age (allowing for different slopes on each side of cutoff), using a rectangular kernel and a bandwidth of 2 years on each side of the cutoff, except for the 49 and 54 age cutoffs where we use a bandwidth of one year on the right due to other discontinuities. We exclude the 2 closest months on each side of the cutoff. Standard errors (in brackets) clustered on day level (+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$).

Table I.4: Placebo Outcomes, Men

		Fraction of UI entries per day	Pre UI Wage	Foreign Citizen	Years of Education	Occ. Tenure Last Job	Ind. Tenure Last Job	Times until UI Claim
Period Jul 1987 - Feb 1999								
Age 42, P: (12-18), Δ P: 6	$\frac{dy}{dP}$	-0.002	0.083	0.001	0.007	-0.000	0.011	0.000
		[0.001]	[0.069]	[0.001]*	[0.006]	[0.011]	[0.011]	[0.006]
N		173,313	173,313	173,313	173,313	173,313	173,313	173,313
Mean Dep. Var		1.000	80.712	0.134	12.911	9.955	8.921	1.517
Age 44, P: (18-22), Δ P: 4	$\frac{dy}{dP}$	0.006	0.027	0.000	0.012	-0.008	0.000	0.013
		[0.003]*	[0.107]	[0.001]	[0.010]	[0.018]	[0.019]	[0.009]
N		170,270	170,270	170,270	170,270	170,270	170,270	170,270
Mean Dep. Var		1.000	80.460	0.147	12.851	10.626	9.578	1.526
Age 49, P: (22-26), Δ P: 4	$\frac{dy}{dP}$	-0.001	-0.255	-0.000	-0.001	0.017	0.047	-0.008
		[0.003]	[0.165]	[0.002]	[0.016]	[0.027]	[0.028]	[0.009]
N		107,255	107,255	107,255	107,255	107,255	107,255	107,255
Mean Dep. Var		1.000	81.666	0.175	12.755	12.342	11.258	1.036
Age 54, P: (26-32), Δ P: 6	$\frac{dy}{dP}$	0.005	0.496	-0.002	-0.008	0.023	0.025	0.007
		[0.004]	[0.151]**	[0.001]	[0.010]	[0.018]	[0.020]	[0.006]
N		66,720	66,720	66,720	66,720	66,720	66,720	66,720
Mean Dep. Var		1.000	75.899	0.143	11.566	12.178	11.842	0.906
Period Mar 1999 - Jan 2006								
Age 45, P: (12-18), Δ P: 6	$\frac{dy}{dP}$	0.001	-0.036	0.000	-0.003	0.016	0.019	-0.007
		[0.002]	[0.079]	[0.001]	[0.007]	[0.014]	[0.014]	[0.006]
N		156,927	156,927	156,927	156,927	156,927	156,927	156,927
Mean Dep. Var		1.000	78.133	0.083	13.467	11.736	5.542	1.271
Age 47, P: (18-22), Δ P: 4	$\frac{dy}{dP}$	0.004	-0.129	0.001	0.019	-0.007	-0.010	-0.011
		[0.003]	[0.131]	[0.001]	[0.011]	[0.024]	[0.022]	[0.009]
N		148,285	148,285	148,285	148,285	148,285	148,285	148,285
Mean Dep. Var		1.000	77.292	0.082	13.424	12.750	5.957	1.287
Age 52, P: (22-26), Δ P: 4	$\frac{dy}{dP}$	0.004	-0.088	0.000	-0.012	0.055	0.045	-0.004
		[0.003]	[0.162]	[0.001]	[0.011]	[0.029]+	[0.031]	[0.010]
N		113,128	113,128	113,128	113,128	113,128	113,128	113,128
Mean Dep. Var		1.000	76.409	0.089	13.139	15.562	8.502	1.413
Period Jan 2008 - Dec 2010								
Age 50, P: (12-15), Δ P: 3	$\frac{dy}{dP}$	0.014	0.462	-0.000	0.013	0.029	0.007	-0.007
		[0.005]**	[0.286]	[0.002]	[0.020]	[0.059]	[0.009]	[0.010]
N		57,116	57,116	57,116	57,116	57,116	57,116	57,116
Mean Dep. Var		1.000	75.830	0.089	12.548	13.415	6.020	0.453

Notes: This table shows RD estimates of UI extensions at various cutoffs on different placebo outcomes. For details on the specification see the notes to Table I.3. Standard errors (in brackets) clustered on day level (+ p<0.1, * p<0.05, ** p<0.01).

Table I.5: Robustness for RD-Estimates, Men

	(1) baseline estimate	(2) more controls	(3) exclude 3 months	(4) bw 12 months	(5) kernel triangular
Period Jul 1987 - Feb 1999, Age 42					
$\frac{dy}{dP}$	0.0916** [0.0259]	0.0679** [0.0249]	0.0741* [0.0289]	0.0734 [0.0448]	0.0854** [0.0294]
N	173313	173313	165203	78519	173313
Period Jul 1987 - Feb 1999, Age 44					
$\frac{dy}{dP}$	0.0795+ [0.0412]	0.0522 [0.0378]	0.0507 [0.0444]	0.189** [0.0685]	0.0991* [0.0451]
N	170270	170270	162222	77334	170270
Period Jul 1987 - Feb 1999, Age 49					
$\frac{dy}{dP}$	0.137* [0.0589]	0.0650 [0.0518]	0.0854 [0.0681]	0.180** [0.0692]	0.139* [0.0589]
N	128050	128050	119883	80498	128050
Period Jul 1987 - Feb 1999, Age 54					
$\frac{dy}{dP}$	0.129* [0.0530]	0.0813+ [0.0451]	0.150* [0.0584]	0.180** [0.0624]	0.147** [0.0518]
N	66720	66720	62507	43057	66720
Period Mar 1999 - Jan 2006, Age 45					
$\frac{dy}{dP}$	0.0242 [0.0282]	0.0225 [0.0261]	0.0201 [0.0300]	0.0116 [0.0465]	0.0153 [0.0305]
N	156927	156927	149712	71417	156927
Period Mar 1999 - Jan 2006, Age 47					
$\frac{dy}{dP}$	0.113* [0.0442]	0.0920* [0.0407]	0.0908+ [0.0469]	0.139+ [0.0729]	0.120* [0.0479]
N	148285	148285	141405	67315	148285
Period Mar 1999 - Jan 2006, Age 52					
$\frac{dy}{dP}$	0.128** [0.0491]	0.108* [0.0468]	0.128* [0.0550]	0.0547 [0.0855]	0.0637 [0.0561]
N	113128	113128	107910	51966	113128
Period Jan 2008 - Dec 2010, Age 50					
$\frac{dy}{dP}$	0.0476 [0.103]	0.0895 [0.0931]	0.0289 [0.108]	0.117 [0.166]	0.0150 [0.109]
N	57116	57116	54455	26172	57116

Notes: This table explores robustness of the RD estimates in Table I.3. Column (1) copies the baseline results. Column (2) adds to the baseline controls, in addition to one-digit industry controls, state fixed effects, as well as calendar month and year FE. Column (3) excludes three instead of two months on each side of the cut-off, column (4) uses a bandwidth of 24 months, and column (5) uses a triangular kernel. Standard errors (in brackets) clustered on day level (+ p<0.1, * p<0.05, ** p<0.01).

Table I.6: Description of Moments

Moments	Description	Parameters Primarily Disciplined
E to U inflows by age 50-63 (Quarterly), 1929 cohort	Following everyone in the birth cohort sample who is employed at age 50, we track their labor market state in each quarter since age 50. Quarterly UI inflows are the share of the sample with a transition from E to UI or Nu in the given quarter.	<i>UI inflows around the bridge age:</i>
E to U inflows by age 50-63 (Quarterly), 1935 cohort		$\sigma, k_0, k_1, \bar{\eta}_{1929}, \bar{\eta}_{1935}, \bar{\eta}_{1950}, \eta_{sd}$
E to U inflows by age 50-63 (Quarterly), 1950 cohort		<i>UI inflows at younger ages:</i> $\lambda_1, \lambda_2, \lambda_3$
Average Non-Employment Duration by age 50-63 (Quarterly), 1929 cohort	For the same sample, we construct non-employment durations conditional on entering UI or Nu at the given quarterly age as the number of months out of work before the next employment spell until age 63, winsorized at 36 months	<i>Nonemp. Dur. at younger ages:</i>
Average Non-Employment Duration by age 50-63 (Quarterly), 1935 cohort		k_2, k_3, γ
Average Non-Employment Duration by age 50-63 (Quarterly), 1950 cohort		
$\frac{dNonEmp}{dP}$ at age 52, 1950 cohort	RD estimate of the non-employment duration effects (capped at 36 months) of a 1 month PBD extension, based on the age 52 cutoff for the 1999-2006 period	k_0, k_2, k_3, γ

Notes: This table describes the moments used to estimate our model. The final column describes which parameters the moments primarily discipline. This is based on the absolute values of the influence elasticities in Table I.10.

Table I.7: Model-estimated dD/dP

	In-Sample Cohorts			Out-of-Sample Cohorts		
	1929	1935	1950	1924	1945	1952
dD/dP (target=0.128; age 52 in 1950)						
dD/dP at age 50	0.120	0.144	0.114		0.130	0.110
dD/dP at age 52	0.148	0.168	0.127		0.152	0.123
dD/dP at age 55	0.196	0.182	0.152	0.194	0.183	0.155
dD/dP at age 57	0.183	0.089	0.176	0.205	0.157	0.178
dD/dP at age 59	0.000	0.000	0.179	0.000	0.000	0.182

Notes: This table shows the model's estimated dD/dP at different ages. The targeted empirical moment is 0.128 at age 52 in 1950 and the corresponding estimate, shown in bold, is 0.127.

Table I.8: MVPF when UI and Retirement Interact

Program	Source	WTP	Net Cost to Gvt 1+FE	MVPF
UI in US - Summary	Hendren & Sprung-Keyser (2020)	1.2	1.95	0.61
UI Extensions, Europe, Prime Age	Schmieder and von Wachter (2016) Le Barbanchon, Schmieder, Weber (2024)	1.36	2.31	0.691
UI Extensions, Germany, Prime Age	Schmieder, von Wachter, Bender (2012)	1.36	1.41	0.967
UI Benefit Levels, Europe	Schmieder and von Wachter (2016) Le Barbanchon, Schmieder, Weber (2024)	1.18	2.55	0.546
UI Extensions - Older Workers Naïve - 1935 Cohort	This Paper	1.36	1.9	0.716
UI Extensions - Older Workers Full Model - 1935 Cohort	This Paper	1.36	5.62	0.242

The first row is taken from Hendren and Sprung-Keyser (2020).

The second and fourth rows are based on averaging the values across all the relevant studies covered in Le Barbanchon, Schmieder, and Weber (2024), which in turn uses values from Schmieder and von Wachter (2016). We omit the values corresponding to Lalive (2007) and Lalive (2008), which had very large UI extensions and where the constant hazard approximation used by Schmieder and von Wachter (2016) appears to be problematic.

The third row is taken from Schmieder, von Wachter, Bender (2012).

The last two rows are based on the model simulations in this paper.

For the rows that aggregate multiple studies, WTP, net cost to government, and MVPF are each averaged separately across the underlying study-level estimates; consequently, the reported row averages need not satisfy $MVPF = WTP/(1 + FE)$.

Table I.9: The Marginal Value of Public Funds of UI Extension across Cohorts

Variable	1929	1935	1945	1950	1952
Full Model (intensive and extensive margin responses)					
Revenue Loss dR	321.00	1121.43	612.70	779.72	644.13
Benefit Increase dB	293.39	582.04	409.38	537.84	488.93
Benefit Increase (Mechanical) dB	272.77	303.37	325.84	394.53	370.35
Behavioral Cost BC	341.63	1400.09	696.25	923.03	762.71
Mechanical Cost MC	272.77	303.37	325.84	394.53	370.35
Fiscal Externality FE	1.25	4.62	2.14	2.34	2.06
Net Cost to Government $1 + FE$	2.25	5.62	3.14	3.34	3.06
WTP	1.36	1.36	1.36	1.36	1.36
MVPF	0.60	0.24	0.43	0.41	0.44
Naïve Calculation (only intensive margin response)					
Revenue Loss dR	172.61	263.92	212.49	179.59	169.38
Benefit Increase dB	277.46	312.92	336.44	412.98	388.10
Benefit Increase (Mechanical) dB	272.77	303.37	325.84	394.53	370.35
Behavioral Cost BC	177.30	273.46	223.10	198.04	187.13
Mechanical Cost MC	272.77	303.37	325.84	394.53	370.35
Fiscal Externality FE	0.65	0.90	0.68	0.50	0.51
Net Cost to Government $1 + FE$	1.65	1.90	1.68	1.50	1.51
WTP	1.36	1.36	1.36	1.36	1.36
MVPF	0.82	0.72	0.81	0.91	0.90

Each column reports per-capita cohort totals for a 3-month PBD increase, summed without discounting over ages 50–63. Mechanical benefits apply the longer schedule to baseline UI inflows and constant-hazard survival implied by baseline simulated mean durations; hence $MC = dB|_{mechanical}$ in both panels. The full model allows inflows and durations to adjust; the naïve calculation fixes inflows and uses the RD duration response. Fiscal accounting covers UI outlays and labor-tax revenue, using a monthly wage of EUR 3,000, a tax wedge of 0.49, and a monthly UI benefit of EUR 918 (60 percent of after-tax wages). $WTP = 1.36$ is from Le Barbanchon et al. (2024).

Table I.10: The Influence of Model Parameters on Simulated Moments, $\frac{\partial m_i}{\partial \xi_j} \times \frac{|\xi_j|}{m_i}$

Age Cohort	Transitions						Durations						dD/dP
	Workers Age 52			Workers at Bridge Age			Workers Age 52			Workers at Bridge Age			
	1929	1935	1950	58 1929	57.4 1935	61 1950	1929	1935	1950	58 1929	57.4 1935	61 1950	
Productivity Shocks													
Std. dev. of productivity shock σ	0.0016	0.0154	0.0471	-0.0600	-0.4898	0.3968	-0.0565	-0.0752	0.0221	-0.0564	-0.0501	-0.0229	-0.0435
Job Search Parameters													
Fixed cost of job search k_0	-0.0024	-0.0187	-0.2358	-1.7297	-1.6817	-1.1423	-0.5323	-0.7080	-0.8421	-0.0937	-0.0344	-0.0779	-0.2435
Cost of UI entry k_1	-0.0006	-0.0033	-0.0329	-1.4315	-0.8778	-2.0045	-0.0014	-0.0048	-0.0446	-0.0394	0.0019	-0.0769	-0.0000
Exp. time trend in search cost k_2	0.0010	-0.0001	0.0000	-0.0064	-0.0015	-0.0250	0.0369	0.1827	0.0834	0.0241	0.0122	0.0157	0.2883
Slope parameter of search cost k_3	-0.0071	-0.0190	-0.0169	-0.0818	-0.0435	-0.2573	1.4746	1.5534	0.9660	0.1023	0.0439	0.0768	-0.1645
Elasticity of search cost γ	0.0007	0.0115	0.0117	0.1131	0.0576	0.3355	-0.5252	-1.0836	-0.5526	-0.2091	-0.1206	-0.1643	-1.0160
Exog. job destruction rate													
λ_1	-6.9744	-8.4665	-6.2265	-2.6752	-1.0302	-3.5463	-0.6453	-0.4968	-0.4635	0.3112	0.0955	0.1877	0.2370
λ_2	0.6975	1.7235	1.4000	0.5511	0.1861	0.5911	0.1226	0.0945	0.1236	-0.0636	-0.0138	-0.0356	-0.0503
λ_3	0.3603	-0.0077	0.1808	-0.0218	0.0748	-0.0402	0.0098	-0.0020	-0.0103	-0.0004	-0.0037	0.0026	0.0014
Disutility of work													
$\bar{\eta}_{1929}$	0.0013	-0.0000	-0.0000	2.4937	-0.0000	-0.0000	0.4517	-0.0000	-0.0000	0.1349	-0.0000	-0.0000	-0.0000
$\bar{\eta}_{1935}$	-0.0000	0.0212	-0.0000	-0.0000	3.3159	-0.0000	-0.0000	0.9853	-0.0000	-0.0000	0.0546	-0.0000	-0.0000
$\bar{\eta}_{1950}$	-0.0000	-0.0000	0.2887	-0.0000	-0.0000	1.9108	-0.0000	-0.0000	1.0674	-0.0000	-0.0000	0.1423	0.3136
η_{SD}	0.0146	0.0659	0.4950	4.0442	1.5853	1.5563	0.1791	0.1616	0.8126	0.1341	0.0130	0.0461	-0.1876

Notes: The table shows elements of the Jacobian matrix of $m(\xi)$, where $m(\xi)$ is the mapping from model parameters ξ to simulated moments m . The table scales these elements $m_i(\xi_j)$ as elasticities (preserving the sign): $\frac{\partial m_i}{\partial \xi_j} \times \frac{|\xi_j|}{m_i}$. Cells are shaded as darker blue for larger absolute values of the elasticity.

Table I.11: The Sensitivity of Model Parameters to Empirical Moments, $\frac{\partial \xi_j}{\partial \hat{m}_i} \times \frac{SD(\hat{m}_i)}{|\xi_j|}$

Age Cohort	Transitions						Durations						dD/dP
	Workers Age 52			Workers at Bridge Age			Workers Age 52			Workers at Bridge Age			
	1929	1935	1950	58 1929	57.4 1935	61 1950	1929	1935	1950	58 1929	57.4 1935	61 1950	
Productivity Shocks													
Std. dev. of productivity shock σ	-0.0073	-0.0926	-0.0061	-0.1568	-0.4077	-0.0043	-0.0039	0.0045	-0.0041	-0.0565	-0.1268	-0.0035	0.0250
Job Search Parameters													
Fixed cost of job search k_0	0.0401	-0.1582	0.0287	-0.0312	-0.3393	0.0178	-0.0609	0.0131	0.0314	-0.0356	0.0211	-0.0212	-0.0348
Cost of UI entry k_1	-0.0058	-0.0029	0.0028	-0.0117	-0.0023	-0.0387	0.0033	0.0007	0.0034	-0.0057	-0.0213	-0.0102	0.0220
Exp. time trend in search cost k_2	0.0344	-0.2270	-0.0733	0.0276	-0.2089	-0.3481	-0.0279	0.2905	-0.0327	-0.2108	-0.3470	-0.3083	2.2754
Slope parameter of search cost k_3	0.0127	-0.0379	-0.0153	-0.0319	-0.0995	-0.0548	0.0486	0.0656	0.0293	-0.0302	-0.0434	-0.0344	-0.1926
Elasticity of search cost γ	-0.0158	0.0128	-0.0372	0.0158	0.1331	-0.0853	0.0421	0.0541	0.0045	-0.0566	-0.1146	-0.0600	0.1570
Exog. job destruction rate													
λ_1	0.0036	0.0012	-0.0206	-0.0065	0.0153	-0.0154	-0.0002	0.0009	-0.0000	-0.0054	0.0003	-0.0007	0.0039
λ_2	-0.0258	0.0281	0.1300	0.0295	-0.0345	0.0526	0.0087	-0.0002	-0.0078	0.0222	0.0029	-0.0012	-0.0132
λ_3	0.3208	0.0032	0.3581	0.0980	-0.0841	0.1839	0.0002	-0.0058	-0.0216	0.0218	-0.0706	0.0066	-0.0068
Disutility of work													
$\bar{\eta}_{1929}$	-0.0777	0.1448	-0.0635	0.1138	0.2863	-0.1254	0.0892	-0.0213	-0.0437	0.0678	-0.0738	0.0042	0.1234
$\bar{\eta}_{1935}$	-0.0196	0.0081	-0.0208	-0.0185	0.1464	-0.0428	0.0101	0.0052	-0.0057	-0.0105	-0.0109	-0.0062	0.0280
$\bar{\eta}_{1950}$	-0.0282	0.0261	0.0092	-0.0179	0.0948	0.0908	-0.0049	-0.0460	0.0749	-0.0105	-0.0280	0.0512	0.0767
η_{SD}	0.0606	-0.1519	0.0596	0.0527	-0.2520	0.0961	-0.0763	0.0120	0.0466	-0.0082	0.0681	-0.0123	-0.0906

Notes: The table shows the sensitivity of model parameters ξ_j with respect to empirical moments $\hat{m}_i$. The table scales these elements by multiplying them with the standard error of the moment $SE(\hat{m}_i)$ and dividing by the absolute value of the parameter (to preserve the sign): $\frac{\partial \xi_j}{\partial \hat{m}_i} \times \frac{SE(\hat{m}_i)}{|\xi_j|}$. Cells are shaded in darker green for larger absolute values of the sensitivity.

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